

Agent acceleration

Making LLM agents that operate software finish a task in less time and for less money, without finishing fewer tasks.

Computer-use, web and coding agents, which work a desktop, a browser or a code base one step at a time through a model.

September 2026

PART 1

Where the time and the money go

Why LLM agents that operate software are slow and expensive, and how the two are linked.

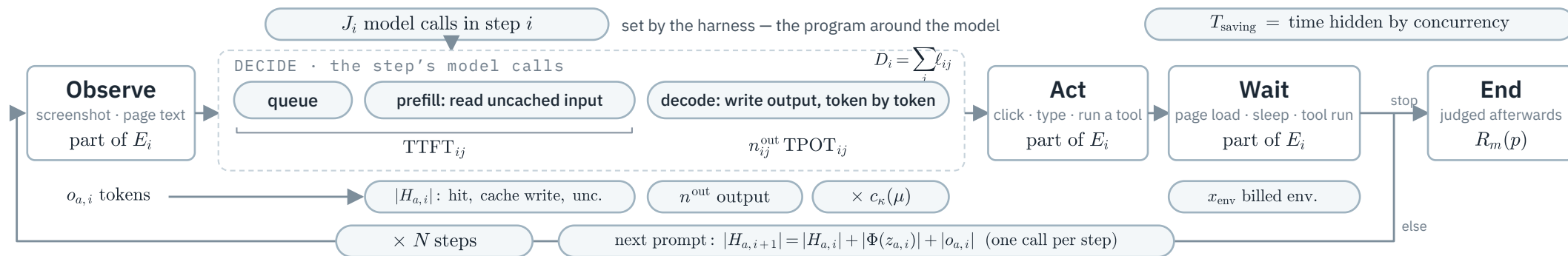
Every number comes from a published measurement of a computer-use, web or coding agent, or from a vendor’s own price page (2024–2026); the conditions of each measurement are stated next to it, and the full references are at the end of the part.

IN THIS PART

02	One attempt of an agent, step by step: the loop and its variables
03–04b	The problem, defined: one formula for time, one for money, the goal, and what is inside their terms
05–10	Where the time goes, term by term
11–12	Where the money goes, term by term, and per success
13–14	How time and money are linked, and the levers
15–17	References · Appendix A0–A5

One attempt of an agent, step by step

One attempt is N steps; each step is some model calls plus everything that is not a model call. Every variable below is defined here once; pages 3–4 add them up.



- i , N step (one observe–decide–act–wait round); steps in the attempt (Hua et al., 2025)
- J_i , j model calls in step i (planner, judge, retries ...); call index (self-defined)
- ℓ_{ij} latency of call j of step i (Li et al., 2026, arXiv · Salesforce Research)
- $TTFT_{ij}$ time to the first generated token: queueing, sending, prefill of the uncached input (Zhong et al., 2024; Artificial Analysis, 2026)
- n_{ij}^{out} , $TPOT_{ij}$ output tokens of the call, thinking included; time per output token after the first (Erol et al., 2026; Zhong et al., 2024)
- D_i , E_i model time of step i ; all its other time (observe, act, wait, harness gaps, back-off) (Hua et al., 2025; Chen et al., 2026, arXiv · Imperial)
- T_{saving} time hidden because some of it ran at the same time; 0 if strictly serial (Feng et al., 2026, arXiv · UC Berkeley; precedent Kim et al., 2024)
- $o_{a,i}$, a tokens the result or screenshot of step i adds to the prompt; agent index (Yao et al., 2023; Yuan et al., 2026)
- $|H_{a,i}|$, Φ , $z_{a,i}$ length of the prompt step i 's call reads (one call per step); chat template; the call's output: thinking, message and tool-call tokens (Yuan et al., 2026)
- n^{hit} , n^w , n^{unc} input tokens read from the cache, written to it (5-min or 1-h), or uncached; $n^w = n^{w5m} + n^{w1h}$ (Anthropic, 2026b; Xu et al., 2026; Zheng et al., 2024; Huang et al., 2026, arXiv · JHU)
- $c_\kappa(\mu)$, κ , μ price per token of billing class κ on serving model μ (Erol et al., 2026)
- x_{env} billed environment usage (e.g. sandbox hours) (Erol et al., 2026)
- $R_m(p)$, m , p probability that one attempt of design m (model + harness) on task p succeeds (Erol et al., 2026)
- reading, writing reading = prefill of input tokens; writing = decoding output tokens (a cache write is storage) (self-defined)

The problem, defined (1/3): time, money and the goal

numerals run left to right: time I–V as on page 5, money I–VI as on page 11

TIME ONE ATTEMPT	$T_{\text{attempt}} = \underbrace{\sum_{i=0}^N}_{\text{I} \cdot \text{steps}} \left[\underbrace{\sum_{j=1}^{J_i}}_{\text{II} \cdot \text{calls}} \underbrace{\ell_{ij}}_{\text{III} \cdot \text{one call}} + \underbrace{E_i}_{\text{IV} \cdot \text{non-model time}} \right] - \underbrace{T_{\text{saving}}}_{\text{V} \cdot \text{overlap}}$	T_{attempt} wall-clock time of one attempt (self-defined) $i=0$ calls outside any step, e.g. an up-front plan (self-defined) Hua et al., 2025; Feng et al., 2026, arXiv · UC Berkeley; Kim et al., 2024
MONEY ONE ATTEMPT	$c_m(p) = \underbrace{\sum_i \sum_j}_{\text{I} \cdot \text{calls}} \underbrace{\sum_{\kappa} n_{ij}^{\kappa}}_{\text{II, III} \cdot \text{tokens read, written}} \underbrace{c_{\kappa}(\mu_{ij})}_{\text{IV} \cdot \text{price per class}} + \underbrace{x_{\text{env}} c_{\text{env}}}_{\text{V} \cdot \text{environment}}$	$c_m(p)$ dollars of one attempt of design m on task p (Erol et al., 2026) c_{env} price per unit of billed environment usage (Erol et al., 2026) κ billing class (Anthropic, 2026b)
GOAL PER SUCCESS	$T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}, \quad v(m, p) = \frac{C_m(p)}{R_m(p)}$ $\min_m (T_{\text{success}}(m, p), v(m, p))^{\top} \quad \text{s. t.} \quad R_m(p) \geq R_0$	T_{success} expected seconds per success (self-defined) $v, C_m(p)$ expected dollars per success; per attempt (Erol et al., 2026) $R_m(p)$ probability one attempt succeeds; dividing by it is money VI (Erol et al., 2026) $\min_m(\cdot, \cdot)^{\top}$ the Pareto set: designs that no other design matches on both and beats on one (Boyd & Vandenberghe, 2004) R_0 minimum acceptable success rate (self-defined)

REFERENCES

Anthropic. (2026b). *Claude API pricing; Prompt caching; Fast mode; Vision* (developer documentation, read 28 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>

Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge University Press. §4.7 Vector optimization (pp. 174–187). <https://web.stanford.edu/~boyd/cvxbook/>

Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

Feng, G., Mao, H., Dutta, P., & Gonzalez, J. E. (2026). Concurrency without model changes: Future-based asynchronous function calling for LLMs (AsyncFC). arXiv:2605.15077. University of California, Berkeley.

Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.00079. Rutgers University; Microsoft; Google DeepMind.

Kim, S., Moon, S., Tabrizi, R., Lee, N., Mahoney, M. W., Keutzer, K., & Gholami, A. (2024). An LLM compiler for parallel function calling. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 24370–24391. arXiv:2312.04511. University of California, Berkeley; International Computer Science Institute; Lawrence Berkeley National Laboratory.

The problem, defined (2/3): inside time III and time V

EXPANDS	FORMULA	SYMBOL	DEFINITION	SOURCE
Time III · one call ℓ_{ij}	$\ell_{ij} \approx \text{TTFT}_{ij} + n_{ij}^{\text{out}} \text{TPOT}_{ij}$ $\text{TTFT}_{ij} \approx t_{ij}^{\text{queue}} + t_{ij}^{\text{net}} + t_{ij}^{\text{prefill}} + t_{ij}^{\text{first}}$ Zhong et al., 2024; Artificial Analysis, 2026	ℓ_{ij}	Latency of call j in step i : time from sending the request to receiving its last output token.	Li et al., 2026, arXiv · Salesforce Research
		TTFT_{ij}	Time to first token: time from sending the request to receiving the first output token.	Zhong et al., 2024; Artificial Analysis, 2026
		TPOT_{ij}	Time per output token: mean time between successive output tokens after the first.	Zhong et al., 2024
		n_{ij}^{out}	Number of output tokens the call generates, thinking tokens included.	Erol et al., 2026
		t_{ij}^{queue}	Time the request waits in the server’s queue before its prefill starts.	Zhong et al., 2024
		t_{ij}^{net}	Network time between the client and the server.	self-defined
		t_{ij}^{prefill}	Time to process the input tokens not read from the cache; this pass also computes the first output token.	Zhong et al., 2024
		t_{ij}^{first}	Time to deliver the first output token once it is computed.	self-defined
Time V · overlap T_{saving}	$T_{\text{saving}} \triangleq \mathcal{S}(\mathcal{M}) + \mathcal{S}(\mathcal{E}) - \mathcal{D}(\mathcal{M} \cup \mathcal{E})$ Feng et al., 2026, arXiv · UC Berkeley; Kim et al., 2024	$\mathcal{M}$	Set of time intervals during which a model call is running.	Feng et al., 2026, arXiv · UC Berkeley
		$\mathcal{E}$	Set of time intervals during which anything else runs: actions, tools, harness code.	Feng et al., 2026, arXiv · UC Berkeley
		$\mathcal{S}(\cdot)$	Sum of the lengths of the intervals in a set; time covered by two intervals counts twice.	Feng et al., 2026, arXiv · UC Berkeley
		$\mathcal{D}(\cdot)$	Length of the union of the intervals in a set; time covered by two intervals counts once.	Feng et al., 2026, arXiv · UC Berkeley
		$\triangleq$	Is defined as.	—

REFERENCES

Artificial Analysis. (2026). *Methodology; Performance benchmarking methodology* (read 28 September 2026). <https://artificialanalysis.ai/methodology>; <https://artificialanalysis.ai/methodology/performance-benchmarking> — third-party measurement definitions.

Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

Feng, G., Mao, H., Dutta, P., & Gonzalez, J. E. (2026). Concurrency without model changes: Future-based asynchronous function calling for LLMs (AsyncFC). arXiv:2605.15077. University of California, Berkeley.

Kim, S., Moon, S., Tabrizi, R., Lee, N., Mahoney, M. W., Keutzer, K., & Gholami, A. (2024). An LLM compiler for parallel function calling. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 24370–24391. arXiv:2312.04511. University of California, Berkeley; International Computer Science Institute; Lawrence Berkeley National Laboratory.

Li, Y., Ye, Q., Choubey, P. K., Zhang, J., & Wu, C.-S. (2026). Speculate with memory. arXiv:2607.12236. Salesforce Research.

Zhong, Y., Liu, S., Chen, J., Hu, J., Zhu, Y., Liu, X., Jin, X., & Zhang, H. (2024). DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving. *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 193–210. <https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin>. arXiv:2401.09670. Peking University; StepFun; University of California, San Diego.

The problem, defined (3/3): input tokens and their bill

EXPANDS	FORMULA	SYMBOL	DEFINITION	SOURCE
Input tokens of a call $ H_{a,i} $ — read in time III (prefill), billed in money II	$ H_{a,i+1} = H_{a,i} + \Phi(z_{a,i}) + o_{a,i} $	$H_{a,i}$	Prompt agent a sends to the model at step i : the attempt’s history so far.	Yuan et al., 2026
		$ \cdot $	Length in tokens.	Yuan et al., 2026
	$\sum_{i=1}^N H_{a,i} = N H_{a,1} + \frac{N(N-1)}{2} \bar{g}$	$z_{a,i}$	Output of the model at step i : its thinking, message and tool call.	Yuan et al., 2026
		Φ	Chat template that writes the output $z_{a,i}$ into the history.	Yuan et al., 2026
	line 1: Yuan et al., 2026; Yao et al., 2023	$o_{a,i}$	Observation returned by the action of step i , e.g. a screenshot or a tool result.	Yuan et al., 2026
	line 2: summed from line 1 (ours)	N	Number of steps in the attempt.	Hua et al., 2025
		g_i	Tokens step i adds to the history: $g_i = \Phi(z_{a,i}) + o_{a,i} $.	self-defined
Money II, III · tokens by class n_{ij}^κ	$K = \{\text{hit, w5m, w1h, unc, out}\}$	n_{ij}^κ	Number of tokens of billing class κ in call j of step i .	Erol et al., 2026; Anthropic, 2026b
	$n_{ij}^{\text{hit}} + n_{ij}^{\text{w5m}} + n_{ij}^{\text{w1h}} + n_{ij}^{\text{unc}} = H_{a,i} $ (one call per step)	K	Set of billing classes.	self-defined
		hit	Input tokens read from the prompt cache.	Anthropic, 2026b; Xu et al., 2026
	Anthropic, 2026b; Zheng et al., 2024	w5m, w1h	Input tokens written to the cache, kept for 5 minutes or 1 hour.	Anthropic, 2026b
		unc	Input tokens neither read from nor written to the cache.	Anthropic, 2026b; Huang et al., 2026, arXiv · JHU
		out	Output tokens.	Anthropic, 2026b

REFERENCES

Anthropic. (2026b). *Claude API pricing; Prompt caching; Fast mode; Vision* (developer documentation, read 28 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>

Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.00079. Rutgers University; Microsoft; Google DeepMind.

Huang, Z., Wang, X., Wang, A., Jurayj, W., Jiménez Gutiérrez, B., Khashabi, D., & Andrews, N. (2026). Better, faster, stronger: Programmatic skill learning best reduces agent cost (SpeedRunner). arXiv:2608.11338v1. Johns Hopkins University.

Xu, B., Xue, Z., Chen, D., Fu, C., Wu, C., Huang, C., ... Zhang, N. (2026). TokenPilot: Cache-efficient context management for LLM agents. *Findings of the Association for Computational Linguistics: EMNLP 2026* (to appear). arXiv:2606.17016v2. Zhejiang University; University of Electronic Science and Technology of China; Xidian University; HomologyAI.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *The Eleventh International Conference on Learning Representations (ICLR 2023)*. arXiv:2210.03629. Princeton University; Google Research.

Yuan, Y., Nayak, A., Kundu, S., & Talati, N. (2026). Agentic AI workload characterization. *Proceedings of the 2026 IEEE International Symposium on Workload Characterization (IISWC 2026)*, Boulder, CO. arXiv:2605.26297v2 (21 September 2026; arXiv title “Agentic AI workload characteristics”). University of Illinois Urbana-Champaign; Gimlet Labs; Intel.

Zheng, L., Yin, L., Xie, Z., Sun, C., Huang, J., Yu, C. H., Cao, S., Kozyrakis, C., Stoica, I., Gonzalez, J. E., Barrett, C., & Sheng, Y. (2024). SGLang: Efficient execution of structured language model programs. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 62557–62583. <https://doi.org/10.52202/079017-2000>. arXiv:2312.07104. Stanford University; University of California, Berkeley; Shanghai Jiao Tong University; Texas A&M University; independent researcher.

Where the time goes: five classes, one per term

Four of the five are terms of the time equation; the fifth is how they combine: the concurrency term T_{saving} is small in the one production trace that reports it, so the terms nearly add. Which class dominates depends on the harness, not on the kind of agent (page 10). Cause-by-cause tables with measurement conditions: Appendix A1.

$$T_{\text{attempt}} \approx \left| \sum_i \left[\sum_{j \leq J_i} (\text{TTFT}_{ij} + n_{ij}^{\text{out}} \text{TPOT}_{ij}) + E_i \right] - T_{\text{saving}} \right|$$

I · STEPS

II · CALLS PER STEP

III · EACH CALL

IV · NON-MODEL TIME

V · LITTLE RUNS CONCURRENTLY

I · TOO MANY STEPS (N)

Hour-scale tasks take hundreds of steps, and every step is a full round trip

- 318 tool calls per task on OSWorld 2.0 — 108 desktop tasks, Claude Opus 4.7, one action per step

(XLANG Lab, 2026, arXiv · HKU)

II · TOO MANY CALLS PER STEP (J_i)

A harness that plans, judges and reflects multiplies every step

- 4–12 planning calls per judging call — GTA1 harness: four parallel planners, up to three retries, one judge

(Abhyankar, Qi & Zhang, 2026)

III · EACH CALL IS SLOW (ℓ_{ij})

Inside a call, waiting, reading and writing are three separate delays

- Writing, one token at a time, is 91–98.6% of model time — local 27–31B models with a warm cache
- About 10 s per model call, the largest phase of a step — UFO2 desktop agent, GPT-4o and o1

(Yuan et al., 2026; Zhang et al., 2026)

IV · NON-MODEL TIME (E_i)

The environment costs time whether or not the model is running

- 6.6 s per browser action vs 4.7 s per model call, median — 151 WebVoyager live-site tasks, GPT-4o

(Wong et al., 2026, arXiv · Princeton)

V · LITTLE RUNS CONCURRENTLY (T_{saving})

Almost nothing overlaps, so task time is close to a sum: fixing one term saves only its own share

- Concurrency within a turn: 1.15 — GitHub Copilot production telemetry

(Liu et al., 2026, arXiv · UIUC)

I–II · Too many steps, too many calls per step

N multiplies every term, and J_i multiplies N . Steps accumulate because tasks are long, each step does one action, navigation-only steps still go through the model, and the agent idles or loops; calls accumulate because the harness plans, judges and reflects, or samples several candidates per step.

I-1, I-2 · LONG TASKS, ONE ACTION PER STEP

Long tasks with one action per step need hundreds of steps; several actions per call cut them by half or more

- 318 steps per task with one action per step; 160.7 steps with several actions per call, which make 597 tool calls — OSWorld 2.0, 108 long tasks, Claude Opus 4.7

(XLANG Lab, 2026, arXiv · HKU)

I-3 · NAVIGATION-ONLY STEPS

Most steps are navigation that needs no thinking, yet each one is a model call

- 66.7% of the steps in the median task are pure navigation — 151 WebVoyager tasks on live websites, three text agents driven by GPT-4o

(Wong et al., 2026, arXiv · Princeton)

I-4 · IDLING AND DEAD LOOPS

When an agent is stuck, every repeated step is billed and timed in full

- One element-locating loop repeated a step 18 times: 27 minutes and \$8.47 at list price without caching — GTA1 harness, OSWorld

(Abhyankar, Qi & Zhang, 2026)

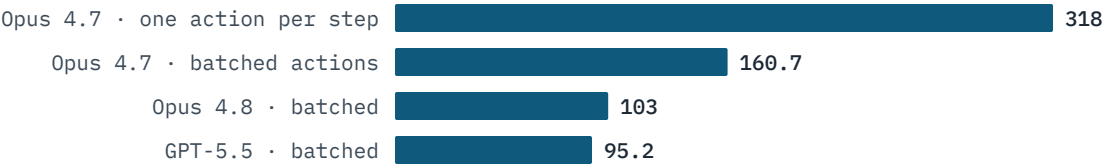
II · PLANNING, JUDGING, SAMPLING

Extra calls per step buy accuracy at a rising token cost per point

- 10 candidates per step instead of 1: 38.8% → 43.2% success for 96K → 920K tokens per task — gpt-oss-120b, 165 WebArena-Lite tasks

(Lee et al., 2026)

Steps per task on OSWorld 2.0, 108 tasks



XLANG Lab, 2026 · mean over tasks

III · Inside one call: first token, then decoding

A call waits in a queue, reads its uncached input, then writes its output one token at a time: $\ell_{ij} \approx \text{TTFT}_{ij} + n_{ij}^{\text{out}} \text{TPOT}_{ij}$. Queueing depends on the provider, reading on the prompt, writing on how much the model writes, thinking included.

III-1 · QUEUEING

The same request can wait far longer depending on the provider's load

- Identical requests take up to 69× longer depending on the time of day — 15 models, 5 providers

(Bian et al., 2025)

III-2 · PREFILL

For screenshot agents, reading the prompt dominates a call, and the prompt grows every step

- Planning, judging and reflection latency is “dominated by the prefill stage” — GTA1 and Agent S2 harnesses, 39 OSWorld tasks

(Abhyankar, Qi & Zhang, 2026)

III-3 · DECODING

Writing is the slowest segment per token, and thinking modes write far more tokens

- 224K vs 37K output tokens per task — Claude Opus 4.8 vs GPT-5.5 on the same 108 OSWorld 2.0 tasks
- Writing is 91–98.6% of model time — local 27–31B models with a warm cache

(XLANG Lab, 2026, arXiv · HKU; Yuan et al., 2026)

HOW TTFT IS MEASURED

Published first-token times lump queueing, network and prefill together

- Serving papers model first-token time as prefill plus queueing
- API measurements start when the request is sent and “include network latency”; output speed excludes the first chunk

(Zhong et al., 2024; Artificial Analysis, 2026)

The prompt grows: every step re-reads the whole history

Each step appends its output and its result, so the tokens read over an attempt grow with N^2 ; caching makes re-reading cheaper and faster, not free. The square matters once N passes a few dozen steps.

HISTORY RE-SENT

Most computer-use agents send the whole history at every step

- “At each step, the prompt sent to the LLM includes the history of all previous steps”; later steps take up to 3× longer — GTA1 harness, 39 OSWorld tasks

(Abhyankar, Qi & Zhang, 2026)

SCREENSHOTS ARE THE BULK

One screenshot adds about a thousand tokens to every later prompt

- A 1280×720 screenshot is $46 \times 26 = 1,196$ tokens on Claude, one token per 28×28-pixel patch (calc.)

(Anthropic, 2026b)

CACHED, STILL PAID

A high cache hit rate lowers the price of re-reading, but the prefix stays the larger part of the bill

- 95.7% hit rate, yet prefix tokens are 59.5% of the list-price cost — 4,265 Claude Code / Codex sessions

(Zhu et al., 2026, arXiv · U. Washington)

WHEN THE SQUARE BITES

Below a few dozen steps the first prompt dominates; long tasks are deep in the square

- the square term overtakes the first-prompt term after $N^* = 2|H_{a,1}|/\bar{g} + 1$ steps: a 6,000-token first prompt and 1,500 tokens added per step give $N^* = 9$ (calc., assumed values); OSWorld 2.0 tasks take about 318 steps

(XLANG Lab, 2026, arXiv · HKU)

Per step, the pressure on prefill is the appended part, not the whole prompt: with a warm cache, “the relevant per-turn prefill pressure is better captured by incremental appends” (Yuan et al., 2026).

IV–V · The time outside the model, and how little of it overlaps

The environment is slow, and today's agents run almost everything in sequence, so T_{saving} is small and the attempt time is close to the full sum. Published “total times” differ in what they count, so the boundary has to be stated.

IV · NON-MODEL TIME

The environment is slow on its own, and the benchmarks add fixed sleeps on top

- OSWorld 2.0 pauses 3 s after every action: × 318 steps (Claude Opus 4.7, one action per step) ≈ 16 min of pure waiting per task (calc.)
- 6.6 s per browser action vs 4.7 s per model call, median — 151 WebVoyager live-site tasks, GPT-4o

(XLANG Lab, 2026, arXiv · HKU; Wong et al., 2026, arXiv · Princeton)

V · CONCURRENCY

Almost nothing overlaps, so fixing one term saves only its own share

- Concurrency within a turn: 1.15 — GitHub Copilot production telemetry

(Liu et al., 2026, arXiv · UIUC)

Where a Claude Code / Codex request’s 4.3 minutes go



Zhu et al., 2026 · 4,265 sessions, 43 developers

WHAT A “TOTAL TIME” COUNTS

Some published times include the person, others exclude them

- 80–92% of a coding session’s elapsed time is the person thinking between turns, outside any one attempt

(Liu et al., 2026, arXiv · UIUC; Zhu et al., 2026, arXiv · U. Washington)

Which term dominates depends on the harness, not on the kind of agent

All five classes exist in every agent; the heaviest term differs, and it flips when the harness changes.

	SCREENSHOT DESKTOP AGENT, MULTI-CALL HARNESS	TEXT WEB AGENT	CODING AGENT WITH A PROMPT CACHE
HEAVIEST	Reading × calls per step × steps. 87–97% of task time is planning, judging and reflection calls (GTA1, Agent S2; 39 OSWorld tasks) (Abhyankar, Qi & Zhang, 2026)	Browser execution and waiting. Per step, median: browser 6.6 s, model 4.7 s (151 WebVoyager live-site tasks, GPT-4o) (Wong et al., 2026, arXiv · Princeton)	Writing, or tool tails. Writing is 91–98.6% of model time (Yuan et al., 2026); in Claude Code / Codex requests tools take 59.8%, the model 41.0% (Zhu et al., 2026, arXiv · U. Washington)
FLIPS WHEN	The harness makes one call per step: over 70% of the time is then the sandbox, because the benchmark sleeps 2–3 s after every action (Chang et al., 2026, arXiv · HKUST; xlang-ai, 2026, code; XLANG Lab, 2026, arXiv · HKU)	The observation grows: 4.8× the input moved the model’s share of step time from 46.9% to 61.6% (Chang et al., 2026, arXiv · HKUST). The site is a fast benchmark sandbox: on five benchmark web apps, model calls are 73% of Browser-Use’s latency, averaged over three models (Winston et al., 2026)	Load evicts the cache: latency up to 7.14× (Kang et al., 2026)
IN ONE LINE	Slow in reading: every step re-reads the growing screenshot history	Slow in the environment — until the observation grows or the site is a fast sandbox, then slow in reading	Reading is cached away; slow in writing and in the tools

1 • The harness sets the dominant term — calls per step, observation size, fixed sleeps, cache on or off. The explanation “agents are slow because inference is slow” is accurate only for multi-call screenshot agents.

2 • N multiplies every term — one step fewer saves a whole step of time and money in all three kinds.

Where the money goes: six classes, one per term

Money comes from tokens read and tokens written, each times a price, plus billed environment usage, which is usually small. Writing costs 5× reading; a cached read costs a small fraction of a normal read (DeepSeek-V4.1-Flash: \$0.006 vs \$0.30 per million tokens); fast mode costs 2× (Anthropic, 2026b; OpenAI, 2026; DeepSeek, 2026).

$$v(m, p) = \left| \begin{array}{c} \mathbb{E} \left[\sum_i \sum_{j \leq J_i} \right] \\ \text{I} \cdot \text{NUMBER OF CALLS} \end{array} \right| \left| \begin{array}{c} \sum_{\kappa} n_{ij}^{\kappa} \\ \text{II, III} \cdot \text{TOKENS BY CLASS} \end{array} \right| \left| \begin{array}{c} c_{\kappa}(\mu_{ij}) \\ \text{IV} \cdot \text{PRICE PER CLASS} \end{array} \right| \left| \begin{array}{c} + x_{\text{env}} c_{\text{env}} \\ \text{V} \cdot \text{ENVIRONMENT} \end{array} \right| \left| \begin{array}{c} \div R_m(p) \\ \text{VI} \cdot \text{FAILURES, RETRIES} \end{array} \right|$$

I · NUMBER OF CALLS

The calls that make the agent slow also make it expensive

- 5–13× the calls of a single-call harness per step — GTA1: planners, retries and a judge (calc.)

(Abhyankar, Qi & Zhang, 2026)

II · TOKENS READ

Reading is the larger bill and grows with the square of the steps

- Even at a 95.7% cache hit rate, prefix tokens are still 59.5% of the bill — 4,265 Claude Code / Codex sessions

(Zhu et al., 2026, arXiv · U. Washington)

III · TOKENS WRITTEN

Writing is the dearer token, thinking included, but the smaller bill

- 31% of an uncached screenshot agent’s bill: \$2.43 counting output only vs \$7.87 counting all tokens — GTA1, 39 OSWorld tasks (calc.)

(Abhyankar, Qi & Zhang, 2026)

IV · UNIT PRICE

The price list sets the constant; fast tiers double it

- Claude Opus 5.5 \$4 / \$20 vs GPT-6 Luna \$0.10 / \$0.50 per million tokens read / written — a 40× spread (calc.)

(Anthropic, 2026b; OpenAI, 2026)

V · THE ENVIRONMENT MACHINE

Published costs are API bills only; a CPU environment adds little

- \$0.04–0.36 for a one-hour CPU environment: 0.5–4.6% of a \$7.87 bill (calc.; an AWS t3.medium VM to an OpenAI 4 GB container; next page)

(Amazon Web Services, 2026; OpenAI, 2026)

VI · FAILURES AND RETRIES

Failed attempts are billed in full

- \$72.4 per attempt ÷ 20.6% completion ≈ \$351 per success — OSWorld 2.0, Claude Opus 4.8, 108 tasks (calc.); cost per success = cost per attempt ÷ success rate

(XLANG Lab, 2026, arXiv · HKU; Erol et al., 2026)

Four accounting conventions for one task, and the cost of the environment machine

The same run’s cost differs by up to 30× across the four conventions in use, and none includes the environment machine.

CONVENTION	THE SAME RUN, COUNTED BOTH WAYS	GAP
OUTPUT ONLY VS ALL TOKENS	GTA1 on 39 OSWorld tasks, o3 list price, no cache: \$2.43 counting output only; \$7.87 counting all tokens (Abhyankar, Qi & Zhang, 2026)	3.2×
PER ATTEMPT VS PER SUCCESS	OSWorld 2.0, Claude Opus 4.8, 108 tasks: \$72.4 per attempt at 20.6% completion → ≈ \$351 (calc.; a benchmark average) (XLANG Lab, 2026, arXiv · HKU); per success = per attempt ÷ success rate (Erol et al., 2026)	4.9×
UNCACHED VS CACHED	A cached read costs a fraction of a normal read (DeepSeek-V4.1-Flash: \$0.006 vs \$0.30 per million tokens; other vendors in A3) (DeepSeek, 2026)	4–50× on the read price only, by vendor (calc.)
STANDARD VS FAST MODE	Claude Opus 5.5 \$4 / \$20 → \$8 / \$40; OpenAI 2× on every listed model; up to 2.5× faster writing (vendor-stated), reading unchanged (Anthropic, 2026b; OpenAI, 2026)	2×

The three that apply to a whole bill compound to about 31× (3.2 × 4.9 × 2, calc.). Screenshot-agent measurements do not report their cache state.

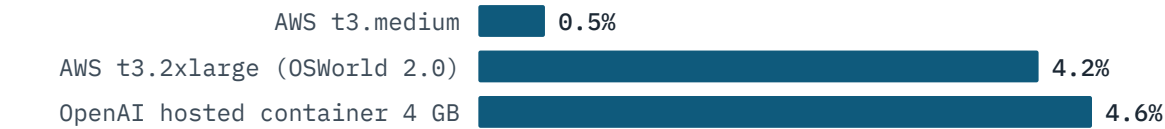
V · THE ENVIRONMENT MACHINE, FROM LIST PRICES

For CPU environments the machine is a small addition; idle time, proxy traffic and GPUs change that

- An H100 sandbox at \$3.95 per hour is already half of a \$7.87 bill, and hourly billing runs while the agent waits — the median AgentSysBench session is active only 20% of its lifetime

vendor price pages, read 2026-09-27; Chang et al., 2026, arXiv · HKUST

A one-hour CPU environment as a share of a \$7.87 API bill (calc.)



AWS and OpenAI price pages · smaller vendors (Browser Use, Browserbase, E2B, Daytona, Modal) in Appendix A3

How slow and expensive are linked: three kinds of relationship

Time and money share the token counts; queueing and environment time reach money only through billed environment usage — and through the cache, which expires while the agent waits.

TIME

$\approx \sum_i \left[\sum_j \left(t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \right] - T_{\text{saving}}$

MONEY

$= \sum_i \sum_j \sum_{\kappa} n_{ij}^{\kappa} c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}}$

in both — same source

time only, or money only through billed environment usage

the price — money for speed or accuracy

$t_{ij}^{\text{queue}}, t_{ij}^{\text{prefill}}$ the queueing and prefill parts of TTFT_{ij} (our split; network and first-token time left out; Appendix A0)

time, money one attempt each; divide by $R_m(p)$ for per success

SAME SOURCE — IN BOTH EQUATIONS

fix it, and time and money fall together

- Steps N , calls per step J_i , reading, writing, cache hits, failures — every one of them lives in the token counts

(Abhyankar, Qi & Zhang, 2026; Anthropic, 2026b)

SLOW BUT NOT EXPENSIVE — NO TOKENS, ONLY MACHINE HOURS

fix it, and only time falls

- Queueing, page loads, fixed sleeps, tool tails — none produces a token; the machine behind them is billed by the hour, usually under 5% of the API bill

MONEY FOR TIME OR FOR ACCURACY — OPPOSITE SIGNS

paying for speed shortens only the writing segment; paying for accuracy costs more tokens per point as accuracy rises

- Fast mode: up to 2.5× faster writing (vendor-stated) at 2× the price, reading unchanged

(Anthropic, 2026b; OpenAI, 2026; accuracy: Appendix A4)

TIME THAT COSTS MONEY — THE CACHE EXPIRES

a slow step or a long pause can turn cheap cache reads into cache writes

- A cache entry lives 5 minutes or 1 hour from the start of the last request that read or wrote it; at most 12.8% of cost could be saved if the cache survived human pauses

(Anthropic, 2026b; Zhu et al., 2026, arXiv · U. Washington)

The levers, and what is still unmeasured

Seven levers follow from the formulas; each moves one term and carries a known risk. Part 2 asks how far the literature has pushed each. Right: gaps in the evidence, not claims of a research gap.

LEVER	TIME	MONEY	THE CATCH
Fewer steps N — several actions per call, skip navigation-only steps	yes	yes, the quadratic term	which steps need no thinking?
Read less $ H_{a,i} $, n^{unc} — trim history, shrink screenshots, cache the prefix	yes	yes	forgets; trimming invalidates the cache
Write less n^{out} — less thinking, smaller model	yes	yes	accuracy may fall
Fewer calls per step J_i — drop judging and reflection	yes	yes	fewer errors caught
Overlap the waiting T_{saving} — pre-load, parallel tools, page-ready events	yes	no	only where the environment dominates
Buy a fast or priority tier TPOT, c_κ	writing only	no — 2× more	reading unchanged; cache dropped
Stop retrying failures $R_m(p)$ — early stop, detect dead loops	yes	yes	stops some attempts that would have succeeded

NOT YET MEASURED

- **Read / write time split and cache hit rate for a frontier API model** — the only split is on local 27–31B models (Yuan et al., 2026); screenshot-agent measurements do not report cache state.
- **How much slower than a person** — no benchmark times agent and human on the same tasks; the closest, AXIS’s small user study, has a UI agent 1.69× slower than manual work on easy tasks (Lu et al., 2025).
- **Whether the benchmark’s fixed sleep sits inside “action” time** in OSWorld-Human — if so, “environment under 3.5%” understates the waiting.
- **How much slowness costs accuracy** — OSWorld 2.0 records the stale-screen failure mode without its share (XLANG Lab, 2026, arXiv · HKU).

References · Part 1 (1 of 3)

author–year tags on the pages refer to these entries · “arXiv · institution” in the text marks a work with no peer-reviewed venue · published version where one exists

1. Abhyankar, R., Qi, Q., & Zhang, Y. (2026). OSWorld-Human: Benchmarking the efficiency of computer-use agents. *Proceedings of Machine Learning and Systems 8 (MLSys 2026)*. arXiv:2506.16042v2. University of California, San Diego.
2. Amazon Web Services. (2026). *Amazon EC2 T3 instances*, on-demand prices, Linux, US East (N. Virginia) (read 27 and 29 September 2026). <https://aws.amazon.com/ec2/instance-types/t3/>
3. Anthropic. (2026a, May 13). *Best practices for computer and browser use with Claude*. <https://claude.com/blog/best-practices-for-computer-and-browser-use-with-claude>
4. Anthropic. (2026b). *Claude API pricing; Prompt caching; Fast mode; Vision* (developer documentation, read 28 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>
5. Artificial Analysis. (2026). *Methodology; Performance benchmarking methodology* (read 28 September 2026). <https://artificialanalysis.ai/methodology>; <https://artificialanalysis.ai/methodology/performance-benchmarking> — third-party measurement definitions.
6. Bian, S., Yan, M., Jayaraman, A., Pekhimenko, G., & Venkataraman, S. (2025). What limits agentic systems efficiency? *NeurIPS 2025 Workshop on Scaling Environments for Agents* (poster). arXiv:2510.16276v1. University of Wisconsin–Madison; University of Toronto; NVIDIA.
7. Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge University Press. §4.7 Vector optimization (pp. 174–187). <https://web.stanford.edu/~boyd/cvxbook/>
8. Chang, C., Zhou, Y., Fu, K., An, D., Feng, T., Lu, H., Yao, S., Guo, P., Yu, Y., Shan, Y., Li, B., Yuan, B., & Wang, W. (2026). From LLM inference to agentic workloads: Characterization and implications for serving systems (AgentSysBench). arXiv:2608.15127. Hong Kong University of Science and Technology; Alibaba Group; ByteDance.
9. Chen, H. M., Guo, J., Luk, W., & Fan, H. (2026). AOSpec: Action and observation co-speculation for low-latency agent serving. arXiv:2608.00881. Imperial College London.
10. Chen, L., Zaharia, M., & Zou, J. (2024). FrugalGPT: How to use large language models while reducing cost and improving performance. *Transactions on Machine Learning Research* (December 2024; Featured Certification). <https://openreview.net/forum?id=cSimKw5p6R>. arXiv:2305.05176. Stanford University.
11. DeepSeek. (2026). *Models & pricing* (API documentation, read 28 September 2026); *DeepSeek-V4.1-Flash release* (10 September 2026). https://api-docs.deepseek.com/quick_start/pricing; <https://api-docs.deepseek.com/news/news260910>
12. Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.
13. Feng, G., Mao, H., Dutta, P., & Gonzalez, J. E. (2026). Concurrency without model changes: Future-based asynchronous function calling for LLMs (AsyncFC). arXiv:2605.15077. University of California, Berkeley.
14. Google. (2026). *Gemini Developer API pricing* (read 28 September 2026). <https://ai.google.dev/gemini-api/docs/pricing>

References · Part 1 (2 of 3)

continued

15. Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.00079. Rutgers University; Microsoft; Google DeepMind.
16. Huang, Z., Wang, X., Wang, A., Jurayj, W., Jiménez Gutiérrez, B., Khashabi, D., & Andrews, N. (2026). Better, faster, stronger: Programmatic skill learning best reduces agent cost (SpeedRunner). arXiv:2608.11338v1. Johns Hopkins University.
17. Kang, H., Li, Z., Yang, X., Xu, W., Chen, Y., Wang, J., Chen, B., Krishna, T., Xu, C., & Arora, S. (2026). ThunderAgent: A fast, simple, and program-aware agentic inference system. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, poster. arXiv:2602.13692. Georgia Institute of Technology; University of Illinois Urbana-Champaign; Carnegie Mellon University; Together AI.
18. Kapoor, S., Stroebl, B., Kirgis, P., Nadgir, N., Siegel, Z. S., Wei, B., ... Narayanan, A. (2026). Holistic Agent Leaderboard: The missing infrastructure for AI agent evaluation. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2510.11977. Princeton University et al.
19. Kapoor, S., Stroebl, B., Siegel, Z. S., Nadgir, N., & Narayanan, A. (2025). AI agents that matter. *Transactions on Machine Learning Research*. arXiv:2407.01502. Princeton University.
20. Kim, S., Moon, S., Tabrizi, R., Lee, N., Mahoney, M. W., Keutzer, K., & Gholami, A. (2024). An LLM compiler for parallel function calling. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 24370–24391. arXiv:2312.04511. University of California, Berkeley; International Computer Science Institute; Lawrence Berkeley National Laboratory.
21. Le Sellier de Chezelles, T., Gasse, M., Lacoste, A., Caccia, M., Drouin, A., Boisvert, L., ... Chapados, N. (2025). The BrowserGym ecosystem for web agent research. *Transactions on Machine Learning Research* (Expert Certification). <https://openreview.net/forum?id=5298fKGmv3>. arXiv:2412.05467v4. ServiceNow Research; Mila; Polytechnique Montréal; Carnegie Mellon University; McGill University; Tel Aviv University; Université de Montréal; iMean AI.
22. Lee, N., Erdogan, L. E., John, C. J., Krishnapillai, S., Mahoney, M. W., Keutzer, K., & Gholami, A. (2026). Agentic test-time scaling for WebAgents. *COLM 2026 Workshop on Efficient Reasoning* (non-archival). arXiv:2602.12276v2. University of California, Berkeley.
23. Li, Y., Ye, Q., Choubey, P. K., Zhang, J., & Wu, C.-S. (2026). Speculate with memory. arXiv:2607.12236. Salesforce Research.
24. Liu, B., Qiu, H., Goiri, Í., Fonseca, R., Bianchini, R., & Choukse, E. (2026). Agentic coding in the wild: Characterizing GitHub Copilot traces at production scale. arXiv:2608.00101. University of Illinois Urbana-Champaign; Microsoft Azure Research.
25. Lu, J., Zhang, Z., Yang, F., Zhang, J., Wang, L., Du, C., Lin, Q., Rajmohan, S., Zhang, D., & Zhang, Q. (2025). AXIS: Efficient human-agent-computer interaction with API-first LLM-based agents. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 7711–7743. <https://doi.org/10.18653/v1/2025.acl-long.381>. Microsoft.
26. Luo, M., Shi, X., Cai, C., Zhang, T., Wong, J., Wang, Y., Wang, C., Huang, Y., Chen, Z., Gonzalez, J. E., & Stoica, I. (2026). Agentix: An efficient serving engine for LLM agents as general programs. *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*, 2443–2459. <https://www.usenix.org/conference/nsdi26/presentation/luo>. arXiv:2502.13965 (as Autellix). University of California, Berkeley; Google DeepMind; Shanghai Jiao Tong University.
27. OpenAI. (2026). *API pricing; API changelog* (fast mode, 30 July 2026; Ultrafast, 13 August 2026) (read 28 September 2026). <https://developers.openai.com/api/docs/pricing>; <https://developers.openai.com/api/docs/changelog>
28. Vendor price pages used on page 12 and in Appendix A3 (all read 27 September 2026): Browser Use, *Pricing* and *API v4: create browser session*; Browserbase, *Pricing* and *Billing plans*; Daytona, *Pricing* and *Billing*; E2B, *Pricing*; Modal, *Pricing* and *Sandbox resources*.

References · Part 1 (3 of 3)

continued

29. Winston, C., Wang, R. Y., Mirhoseini, A., & Kozyrakis, C. (2026). Agent JIT compilation for latency-optimizing web agent planning and scheduling. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, PMLR 306. arXiv:2605.21470. Stanford University.
30. Wong, M., Hsieh, K., Nath, S., & Netravali, R. (2026). Skim: Speculative execution for fast and efficient web agents. arXiv:2605.16565. Princeton University; Microsoft Research.
31. xAI. (2026). *Grok 4.7* model page, SpaceXAI Docs (read 28 September 2026). <https://docs.x.ai/developers/models/grok-4.7>
32. XLANG Lab. (2026). OSWorld 2.0: Benchmarking computer use agents on long-horizon real-world tasks. arXiv:2606.29537. The University of Hong Kong (authored as “XLANG Lab and Collaborators”; the full author list is in the paper’s Appendix A).
33. xlang-ai. (2026). *OSWorld* [Source code], desktop_env/desktop_env.py and run.py at commit b138d348. <https://github.com/xlang-ai/OSWorld>
34. Xu, B., Xue, Z., Chen, D., Fu, C., Wu, C., Huang, C., ... Zhang, N. (2026). TokenPilot: Cache-efficient context management for LLM agents. *Findings of the Association for Computational Linguistics: EMNLP 2026* (to appear). arXiv:2606.17016v2. Zhejiang University; University of Electronic Science and Technology of China; Xidian University; HomologyAI.
35. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *The Eleventh International Conference on Learning Representations (ICLR 2023)*. arXiv:2210.03629. Princeton University; Google Research.
36. Ye, N., Ahuja, A., Liargkovas, G., Lu, Y., Kaffes, K., & Peng, T. (2026). Speculative actions: A lossless framework for faster AI agents. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, oral. arXiv:2510.04371. Columbia University.
37. Yuan, Y., Nayak, A., Kundu, S., & Talati, N. (2026). Agentic AI workload characterization. *Proceedings of the 2026 IEEE International Symposium on Workload Characterization (IISWC 2026)*, Boulder, CO. arXiv:2605.26297v2 (21 September 2026; arXiv title “Agentic AI workload characteristics”). University of Illinois Urbana-Champaign; Gimlet Labs; Intel.
38. Zhang, C., Huang, H., Ni, C., Mu, J., Qin, S., He, S., ... Zhang, D. (2026). UFO2: The desktop AgentOS. *Transactions on Machine Learning Research*. arXiv:2504.14603. Microsoft; ZJU-UIUC Institute; Nanjing University; Peking University.
39. Zheng, L., Yin, L., Xie, Z., Sun, C., Huang, J., Yu, C. H., Cao, S., Kozyrakis, C., Stoica, I., Gonzalez, J. E., Barrett, C., & Sheng, Y. (2024). SGLang: Efficient execution of structured language model programs. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 62557–62583. <https://doi.org/10.52202/079017-2000>. arXiv:2312.07104. Stanford University; University of California, Berkeley; Shanghai Jiao Tong University; Texas A&M University; independent researcher.
40. Zhong, Y., Liu, S., Chen, J., Hu, J., Zhu, Y., Liu, X., Jin, X., & Zhang, H. (2024). DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving. *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 193–210. <https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin>. arXiv:2401.09670. Peking University; StepFun; University of California, San Diego.
41. Zhu, K., Jacob, M., Ma, C., Pan, Y., Wang, S., Krishnamurthy, A., & Kasikci, B. (2026). TraceLab: Characterizing coding agent workloads for LLM serving. arXiv:2606.30560. University of Washington.

A0 · The time of one attempt: each source’s own formula, and why ours differs

the derivation behind pages 3–4 · status: verbatim = as the source writes it; adapted = changed, reason in the last column; our distillation = no source writes it as a f...

FORMULA (PAGES 3–4)	STATUS	THE SOURCE’S OWN FORMULA OR WORDS, IN ITS NOTATION	WHAT CHANGED, AND WHY
$T_{\text{attempt}} \triangleq t_{\text{end}} - t_{\text{start}} = \mathcal{D}(\mathcal{M} \cup \mathcal{E}) = \sum_{i=0}^N (D_i + E_i) - T_{\text{saving}}$	adapted	T_{saving} is “defined as the difference between the serialized baseline and the observed end-to-end asynchronous latency” (App. B.2) (Feng et al., 2026, arXiv · UC Berkeley), combined with the sum below	one attempt, not one task (a task may take several); exact because the non-model intervals cover every instant; $T_{\text{saving}} \geq 0$ (calc.); exact with measured call times, approximate once $\ell \approx \text{TTFT} + n \text{ TPOT}$
$\sum_{i=0}^N (D_i + E_i)$	adapted	“When we do not utilize speculative planning, the time taken to generate and execute the whole plan is $\sum_{i \leq n} (\text{time}(T, s_i) + e(s_i))$ ” (§3) (Hua et al., 2025); the same form with the letters D_i , T_i , $\sum_i (D_i + T_i)$ (§3) (Chen et al., 2026, arXiv · Imperial)	$\text{time}(T, s_i) \rightarrow D_i$ (AOSpec’s letter), widened to all model calls of the step; $e(s_i) \rightarrow E_i$, widened to every non-model interval, both as summed lengths; $n \rightarrow \mathbb{N}$; $i = 0$ holds calls outside any step; preferred to (Kim et al., 2024), whose N counts the function calls of one plan
$D_i = \sum_j \ell_{ij}$	our distillation	No source sums the calls of a step. GTA1 makes “4 parallel calls” per planning step and “between 4 and 12 planning calls” per judging call (Abhyankar, Qi & Zhang, 2026)	parallel calls add here; their overlap is credited to T_{saving} ; the letter ℓ as in (Li et al., 2026, arXiv · Salesforce Research)
$\ell_{ij} \approx \text{TTFT}_{ij} + n_{ij}^{\text{out}} \text{TPOT}_{ij}$	adapted	“The overall request latency equals TTFT plus TPOT times the number of generated tokens in the decoding phase” (fn. 1) (Zhong et al., 2024); API form: Total Response Time = Time to First Token + 100 / Output Speed (Artificial Analysis, 2026)	the decoding-phase tokens are $n^{\text{out}} - 1$ (the first comes with TTFT), hence $\approx$ with n^{out} ; output tokens include thinking; TTFT runs to the first token of any kind; on an API it also includes network time
$\text{TTFT}_{ij} \approx t^{\text{queue}} + t^{\text{net}} + t^{\text{prefill}} (n^{\text{unc}} + n^{\text{w}}; n^{\text{hit}}) + t^{\text{first}}$	adapted	latency “comprises both execution time and queuing delay” (fn. 3); for prefill, the average $\text{Avg_TTFT} = D + \frac{RD^2}{2(1-RD)}$ of an M/D/1 queue (§3.1, Eq. 1) (Zhong et al., 2024); a program’s latency “comprises three components”, the first the “queuing time” of its calls (Luo et al., 2026); TTFT “includes network latency” (Artificial Analysis, 2026)	network time and t^{first} added to DistServe’s two terms, and its M/D/1 average becomes one call’s time; DistServe’s prefill already produces the first token, so t^{first} is only emitting it (≈ 0 on the server); the split used on page 13; without a cache, DistServe App. A.2 models prefill time from the batch’s token counts; the shape with a cached prefix, our second argument, is given by no verified source

A0 · Time hidden by concurrency, and the growing prompt

the derivation behind pages 4 and 4b · verbatim / adapted / our distillation · calc. = our algebra

FORMULA (PAGES 4, 4B)	STATUS	THE SOURCE’S OWN FORMULA OR WORDS, IN ITS NOTATION	WHAT CHANGED, AND WHY
$T_{\text{saving}} \triangleq \mathcal{S}(\mathcal{M}) + \mathcal{S}(\mathcal{E}) - \mathcal{D}(\mathcal{M} \cup \mathcal{E})$	adapted	the same formula, continued $= \Delta_{F \parallel F} + \Delta_{D \parallel E}$, with $\mathcal{M}, \mathcal{E}$ “the sequence of time intervals for model decoding and function execution” (App. B.2) (Feng et al., 2026, arXiv · UC Berkeley); precedent for the effect: parallel tasks’ latency “is determined by the slowest task”, $T^C = \sum_i T_P^C(P_i) + \max_k T_E(E_k)$ against the serial $\sum_i (T_P^R(P_i) + T_E(E_i))$ (App. E.1, Eqs. 1–2) (Kim et al., 2024)	intervals widened to whole calls and all non-model time, so values are not comparable with AsyncFC’s; the split becomes (calc.) $[\mathcal{S}(\mathcal{M}) - \mathcal{D}(\mathcal{M})] + [\mathcal{S}(\mathcal{E}) - \mathcal{D}(\mathcal{E})] + [\mathcal{D}(\mathcal{M}) + \mathcal{D}(\mathcal{E}) - \mathcal{D}(\mathcal{M} \cup \mathcal{E})]$, each ≥ 0 (e.g. a 2-s tool call during a model call gives 2 s, calc.); it can include extra work that concurrency causes, so it is not the saving against a separate serial run (speed-ups use $T_{\text{seq}}/T_{\text{attempt}}$, Part 2)
$ H_{a,i+1} = H_{a,i} + \Phi(z_{a,i}) + o_{a,i} $	adapted	$H_{a,i+1} = H_{a,i} \parallel \Phi(\theta_{a,i}, m_{a,i}, u_{a,i}) \parallel o_{a,i}$, $C_{a,i} = H_{a,i} $ (Eqs. 1–3, v2 §II-C) (Yuan et al., 2026); the full history as context: $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ (§2) (Yao et al., 2023)	lengths of the concatenation; $\Phi(z_{a,i})$ written with Yuan’s own $z = (\theta, m, u)$ of its Eq. 2 (thinking, message, tool-call tokens), because its m would clash with the design m and its $C_{a,i}$ with the cost $C_m(p)$; exact with one call per step and full history; in general add $h_i^+ - h_i^-$ (other calls’ output; compaction)
$\sum_{i=1}^N H_{a,i} = N H_{a,1} + \frac{N(N-1)}{2} \bar{g}$	our distillation (calc.)	no source writes the sum; “at each step, the prompt sent to the LLM includes the history of all previous steps” (Abhyankar, Qi & Zhang, 2026)	$g_i = \Phi(z_{a,i}) + o_{a,i} $, $\bar{g} = \sum_i (N-i) g_i / \frac{N(N-1)}{2}$ makes it exact; over a random step count $\mathbb{E}[\sum H] \approx \mathbb{E}[N] H_{a,1} + \frac{\bar{g}}{2} (\mathbb{E}[N]^2 + \text{Var}[N] - \mathbb{E}[N])$; crossover $N^* = 2 H_{a,1} / \bar{g} + 1$

A0 · Money, per success, and the goal

the derivation behind page 3 · verbatim / adapted / our distillation · calc. = our algebra

FORMULA (PAGE 3)	STATUS	THE SOURCE’S OWN FORMULA OR WORDS, IN ITS NOTATION	WHAT CHANGED, AND WHY
$c_m(p) = \sum_{i,j} \sum_{\kappa} n_{ij}^{\kappa} c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}}$	adapted	$c_m(p) = n_{\text{in}}(m,p) c_{\text{in}}(m) + n_{\text{out}}(m,p) c_{\text{out}}(m)$ (Eq. 13): input and output tokens times their prices; other components go into $C_m(p) = w^{\top} x_m(p)$, w the unit prices and $x_m(p)$ the quantities per attempt (App. D.1) (Erol et al., 2026)	input split into the billed classes (cache hit, 5-min and 1-h cache write, uncached); prices inside the sum because one step can mix models; environment usage added as App. D.1 allows; the class index is κ because x is Cost-of-Pass’s quantity. Closest written three-class form: (Xu et al., 2026)
$v(m,p) = C_m(p)/R_m(p)$	verbatim	“the expected number of attempts to obtain the first correct solution is $1/R_m(p)$ ”, assuming independent trials (§2.2, Eq. 2) (Erol et al., 2026)	none; assumes unlimited independent retries and the verifier’s time and cost in every attempt; $C/R = \mathbb{E}[c \mid \text{success}] + \frac{1-R}{R} \mathbb{E}[c \mid \text{fail}]$ (calc.), the second term is money VI; with at most K retries the ratio is still C/R (calc.); with cache-warm retries $C_m^{(1)}(p) + (1/R_m(p) - 1) C_m^{(2+)}(p)$ (calc.)
$T_{\text{success}}(m,p) = \mathbb{E}[T_{\text{attempt}}]/R_m(p)$	adapted	“alternative units per attempt (FLOPs, time, latency, energy) may matter more than dollar cost” (App. D.1) (Erol et al., 2026)	the same derivation in seconds, named T_{success} by us; serial retries only (parallel votes cost “roughly the latency of a single vote”); correlated failures break independence
$\min_m (T_{\text{success}}(m,p), v(m,p))^{\top}$ s. t. $R_m(p) \geq R_0$	our distillation	$V_p(\mathcal{M}) = \min_{m \in \mathcal{M}} v(m,p)$ (Eq. 3), with a human expert as fallback, $\min(V_p(\mathcal{M}), v(\text{expert}, p))$ (Eqs. 4–5) (Erol et al., 2026); “the new goal of jointly optimizing the two metrics” (arXiv v1) (Kapoor et al., 2025); budgeted dual: maximise quality with expected cost within a budget (Chen, Zaharia & Zou, 2024)	scalar min $\rightarrow$ min of a vector, read as the Pareto set: “minimize (with respect to K)” (4.56), Pareto optimal (§4.7.3–4.7.5) (Boyd & Vandenberghe, 2004), because seconds and dollars are not one scalar; Cost-of-Pass never says Pareto; the human is the reference point; the floor R_0 is our choice, after App. C.8’s remedy (i), which leaves impractical baselines off the frontier set; over a task mix $\sum_p C_m(p)/\sum_p R_m(p)$, unlike Eq. 8’s mean of V_p

A0 · Every symbol, its unit, and where it comes from

origin of each symbol: “as in” a source (its own symbol and meaning), “adapted from” a source (renamed or widened), or self-defined

SYMBOL	MEANING	UNIT	ORIGIN
i, N, j, J_i, a	step (i = 0: calls outside any step); steps; model call; calls in step i; agent	—; steps; —; calls; —	i as in Hua et al., 2025 §3 (their s_i , n steps) · a as in Yuan et al., 2026 §II-C (whose i counts LLM invocations) · N adapted from Hua et al.’s n · j, J_i self-defined (calls ≠ steps)
$D_i, E_i, T_{\text{attempt}}, T_{\text{success}}, T_{\text{saving}}, T_{\text{seq}}$	model time; non-model time of step i; wall-clock of one attempt; expected seconds per success; time hidden by concurrency; a separate serial run	s	D: the quantity of Hua et al., 2025 time(T, s_i) and the letter of Chen et al., 2026, arXiv · Imperial, widened to all calls of a step · E adapted from Hua et al.’s e(s_i) (AOSpec’s T_i) · T_attempt self-defined · T_success self-defined, the time analogue of Cost-of-Pass’s v · T_saving adapted from Feng et al., 2026, arXiv · UC Berkeley App. B.2 (intervals widened; precedent for the effect, not the symbol: Kim et al., 2024 App. E.1, Eqs. 1–2) · T_seq as in Ye et al., 2026 §3.1.2
$\mathcal{S}, \mathcal{D}, \mathcal{M}, \mathcal{E}, \triangleq$	summed length; length of a union; model-call and non-model intervals; defined as	s	adapted from Feng et al., 2026, arXiv · UC Berkeley App. B.2 (intervals widened)
$\ell_{ij}, \text{TTFT}_{ij}, \text{TPOT}_{ij}$	call latency; time to first token; time per output token	s; s; s/token	ℓ as in Li et al., 2026, arXiv · Salesforce Research §2.1 (ℓ_LLM) · TPOT as in Zhong et al., 2024 §1 · TTFT adapted from Zhong et al., 2024 §1 (“the duration of the prefill phase”) and §3.1 Eq. 1 (plus queuing delay), widened by network time as in Artificial Analysis, 2026 · indices self-defined
$t^{\text{queue}}, t^{\text{net}}, t^{\text{prefill}}, t^{\text{first}}$	parts of TTFT: queueing, network, prefill of uncached input, emitting the first token	s	t^queue, t^prefill adapted from Zhong et al., 2024 §3.1 Eq. 1 (queueing delay; prefill execution time) · t^net, t^first self-defined (A0 · 1/5)
$n^{\text{in}}, n^{\text{hit}}, n^{\text{w}}, n^{\text{w5m}}, n^{\text{w1h}}, n^{\text{unc}}, n^{\text{out}}$	tokens: all input (= hit + w + unc); cache read; cache write (= w5m + w1h); uncached input; output	tokens	n as in Erol et al., 2026 (n_in, n_out) · classes as billed by Anthropic, 2026b (usage fields) · hit as in Xu et al., 2026 App. A.2 (cached prompt tokens as cache hits: Zheng et al., 2024 §3) · unc adapted from Huang et al., 2026, arXiv · JHU App. A.3 (“uncached”) · w5m, w1h self-defined
$\kappa, K, \mu_{ij}, c_{\kappa}(\mu)$	billing class; the set {hit, w5m, w1h, unc, out}; serving model; price per token of class κ on μ	—; —; —; \$/token	c adapted from Erol et al., 2026 Eq. 13 (c_in, c_out) · κ, K, μ self-defined
$x_{\text{env}}, c_{\text{env}}, w, x_m(p)$	billed environment usage; its price; price vector; quantity vector per attempt	h; \$/h; \$/unit; units	as in Erol et al., 2026 App. D.1 · env subscript self-defined
$H_{a,i}, \Phi, z_{a,i}, o_{a,i}, g_i, \bar{g}, N^*$	context before step i of agent a; chat template; the step’s output; its tool result; tokens added; weighted mean; crossover	tokens; —; tokens; tokens; tokens; tokens; steps	H, Φ, o, z as in Yuan et al., 2026 Eqs. 1–3 (o also Yao et al., 2023 §2), z = (θ, m, u) = (thinking, message, tool call) in its Eq. 2; writing Φ(z) for Φ(θ, m, u) is our shorthand · g, $\bar{g}$, N* self-defined (calc.)
$m, p, c_m(p), C_m(p), R_m(p), v(m,p), R_0, \mathbb{E}, \text{Var}$	design (model + harness); task; dollars of one attempt; expected; success probability; dollars per success; success floor; expectation; variance	—; —; \$; \$; prob.; \$; prob.	as in Erol et al., 2026 §2.2, Eq. 2, App. B · R_0 self-defined · $\mathbb{E}$, Var standard

A0 · What a published number covers: map it onto the terms before comparing

the measurement boundary · a figure is comparable only with what it includes stated

KIND OF FIGURE	WHAT IT INCLUDES	WHAT IT LEAVES OUT
first-token time, output speed (Artificial Analysis, 2026)	one synthetic call: queueing, network and prefill lumped in TTFT; speed after the first chunk	the step, the tools, the environment
request response time (Zhu et al., 2026, arXiv · U. Washington)	model generation plus tool execution	the person thinking between turns
stopwatch time per task (Abhyankar, Qi & Zhang, 2026)	the whole attempt, T_{attempt}	— (39 tasks, one run each)
time saved (Feng et al., 2026, arXiv · UC Berkeley)	decode and function-execution intervals of the concurrent trace	queueing, prefill, harness gaps
cost with two classes (input, output) (Erol et al., 2026)	all input at the base price	cache reads (overstates cost) and cache writes (understates it)
cost with three classes (hit, miss, output) (Xu et al., 2026)	cache reads at their price	cache writes; environment usage
cache hit rate	denominators differ: cached / cacheable input (Artificial Analysis, 2026), / all input (Xu et al., 2026), per incoming call (Luo et al., 2026)	ours: $n^{\text{hit}}/n^{\text{in}}$
almost every published figure	per attempt	per success: divide by $R_m(p)$; lossless speed papers report no task success

A1 · Slowness, every cause — I. steps and II. calls per step

the complete table behind pages 5–6 · three kinds of agent: screenshot desktop agent, text web agent, coding agent with a prompt cache

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
I-1	The task is long	Hour-scale tasks take hundreds of steps	OSWorld 2.0, 108 long tasks: Claude Opus 4.7 with maximum thinking, one action per step, 500-step cap: 318 tool calls per task on average, i.e. 318 steps; with several actions per call, Claude Opus 4.8 takes 103 steps (481.8 calls) and GPT-5.5 95.2 steps (149.8 calls)	(XLANG Lab, 2026, arXiv · HKU)	screenshot, text
I-2	One action per step	Every click is a whole step: one observation, one model call, one wait	OSWorld 2.0: with one action per step Claude Opus 4.7 needs 318 steps; with batched actions the same model needs 160.7 and Claude Opus 4.8 needs 103	(XLANG Lab, 2026, arXiv · HKU)	screenshot, text
I-3	Navigation-only steps still call the model	Paging, scrolling and opening the next page need no thinking but cost a call each	Skim’s profile of three text web agents (Browser-Use, AgentOccam, WebVoyager; GPT-4o) on 151 WebVoyager live-site tasks: 66.7% of the steps in the median task are pure navigation	(Wong et al., 2026, arXiv · Princeton)	text
I-4	Idling and dead loops	Element not found, repeated retries; steps rise without progress	OSWorld-Human’s records of the GTA1 harness (o3 plans and judges, GTA1-7B locates elements) on 39 OSWorld desktop tasks: in failed tasks that exceeded 50 steps, 66% of steps were wasted repeats; one element-locating loop repeated the same step 18 times — 27 minutes and \$8.47 at list price without caching	(Abhyankar, Qi & Zhang, 2026)	screenshot
I-5	Acting on a stale screen	The screenshot is taken before the page changes; the agent clicks on the old layout and must repair	OSWorld 2.0 records “stale interface state” as a failure mode; no share reported	(XLANG Lab, 2026, arXiv · HKU)	screenshot
II-1	Multi-call harness	One step = plan + judge + reflect, several calls in series	OSWorld-Human, two harnesses: GTA1 runs 4 parallel planners per step, retries up to 3 rounds, then one judging call picks — 4–12 planning calls per judging call; Agent S2 adds a reflection call per step, so planning is 53% and reflection 34% of task time	(Abhyankar, Qi & Zhang, 2026)	screenshot
II-2	Sampling for accuracy	5–20 candidates per step, then pick one	Same model and harness (gpt-oss-120b, ReAct) on 165 WebArena-Lite tasks: 1 → 10 candidates per step raises success from 38.8% to 43.2% and tokens per task from 96K to 920K — 9.6× the tokens for 4.4 points	(Lee et al., 2026)	text
II-3	Coding agents call repeatedly within a turn	The model is called back and forth inside one user turn	GitHub Copilot coding agent production telemetry (first week of June 2026, 13.5 million sessions): 6.6 model calls per user turn on average	(Liu et al., 2026, arXiv · UIUC)	coding

A1 · Slowness, every cause — III. each call is slow

the complete table behind pages 7–8

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
III-1	Queueing	The request waits at the provider; unbilled, but counted as model time	Client-side measurement of 15 models across 5 providers: requests of the same length differ in latency by up to 69× depending on when they are sent	(Bian et al., 2025)	all
III-2	Reading (the prompt is re-processed every call)	Every call re-reads the whole history; more screenshots, slower	OSWorld-Human, multi-call harnesses: later steps up to 3× slower because the prompt at step i holds the $i - 1$ earlier screenshots — “dominated by prefill”. One screenshot is 1,000–1,800 tokens. AgentSysBench’s WebArena agent (Kimi-K2.6): observation switched from a single format to accessibility tree + HTML + screenshot → input 4.8×, model share of time 46.9% → 61.6%. Under load the cache is evicted and re-read: SWE-Agent / OpenHands running GLM-4.6 on 8 H100s for SWE-bench Lite, request latency up to 7.14× as concurrency rises	(Abhyankar, Qi & Zhang, 2026; Anthropic, 2026a; Chang et al., 2026, arXiv · HKUST; Kang et al., 2026)	screenshot; text with rich observations; coding under load
III-3	Writing (one token at a time)	Token-by-token generation, strictly serial	Locally served ReAct agents (Qwen3.6-27B / Gemma4-31B, vLLM, 2 H100s) on five benchmarks: with the cache warm, generation is 91–98.6% of model time. Windows desktop agent UFO2 (GPT-4o / o1 API): about 10 s per model call, the largest item per step in every configuration	(Yuan et al., 2026; Zhang et al., 2026)	coding; single-call screenshot
III-4	Thinking modes and larger models	Thinking writes more tokens; larger models are slower per token	OSWorld 2.0, same 108 tasks: Claude Opus 4.8 writes 224K output tokens per task, GPT-5.5 37K. Holistic Agent Leaderboard, 9 benchmarks, 21,730 runs: raising reasoning effort lowered accuracy in 21 of 36 pairs	(XLANG Lab, 2026, arXiv · HKU; Kapoor et al., 2026)	all

A1 · Slowness, every cause — IV. environment, V. serial

the complete table behind page 9 (non-model time and concurrency)

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
IV-1	Producing the observation	Extracting the accessibility tree or detecting elements takes time	OSWorld desktop applications: 3–26 s to generate one accessibility tree; the screenshot itself is under 2% of task time. UFO2: OmniParser element detection adds about 1 s per step	(Abhyankar, Qi & Zhang, 2026; Zhang et al., 2026)	screenshot
IV-2	Browser execution and page load	After a click the agent waits for the page	151 WebVoyager live-site tasks: per step, median browser action 6.6 s vs model call 4.7 s. WebArena with GenericAgent + Claude 3.5 Sonnet: mean step 12.2 s, 7.6 s of it in the environment (5,493 steps; arXiv v4, App. G, Table 7)	(Wong et al., 2026, arXiv · Princeton; Le Sellier de Chezelles et al., 2025)	text
IV-3	Fixed sleeps	The agent cannot tell when the page is ready, so it sleeps a fixed number of seconds	OSWorld’s environment code sleeps 2 s after every action; OSWorld 2.0 prescribes 3 s, so 318 steps are about 16 minutes of pure waiting (calc.). AgentSysBench: a single-call GUI agent (ReAct, Kimi-K2.6 API) spends over 70% of its OSWorld execution time in the desktop sandbox	(xlang-ai, 2026, code; XLANG Lab, 2026, arXiv · HKU; Chang et al., 2026, arXiv · HKUST)	screenshot (caused by the benchmark harness)
IV-4	Tool tails	Tests and builds take minutes	TraceLab, 4,265 Claude Code / Codex sessions from 43 developers (September 2025 – June 2026): tool calls over 1 minute are 4% of calls but 85% of tool time; a request averages 4.3 minutes — tools 2.5 (59.8%), model 1.7 (41.0%)	(Zhu et al., 2026, arXiv · U. Washington)	coding
V	Serial structure	The four stages never overlap, so task time is the sum of every term, not the largest one; fixing one term saves only its share	Concurrency within a turn is 1.15 in Copilot telemetry. Separately, 80–92% of a coding session’s elapsed time is the human thinking between turns — strip it before reading any total	(Liu et al., 2026, arXiv · UIUC; Zhu et al., 2026, arXiv · U. Washington)	all

A2 · Where the arrows land, with every measurement condition

the complete table behind page 10

	SCREENSHOT DESKTOP AGENT, MULTI-CALL HARNESS	TEXT WEB AGENT	CODING AGENT WITH A PROMPT CACHE
heaviest term	Reading × calls per step × steps. OSWorld-Human timed two multi-call harnesses (GTA1, Agent S2) step by step on 39 OSWorld desktop tasks: 87–97% of task time in planning, judging and reflection calls; screenshots and actions together under 3.5% (Abhyankar, Qi & Zhang, 2026)	Browser execution plus waiting. Skim measured three text web agents (GPT-4o) on 151 WebVoyager live-site tasks: per step, median browser action 6.6 s and model call 4.7 s (Wong et al., 2026, arXiv · Princeton)	Writing, or tool tails, depending on the tools. With the context cache on, token-by-token generation is 91–98.6% of model time (locally served models) (Yuan et al., 2026); in Claude Code / Codex records a request spends 59.8% in tools and 41.0% in the model (Zhu et al., 2026, arXiv · U. Washington); in GitHub Copilot, whose tools are light, agent time is 13.7% model and 2% tools (Liu et al., 2026, arXiv · UIUC)
second	Writing: with thinking on, more reasoning tokens per step, and generation is one token at a time	Reading, once the observation grows: AgentSysBench switched a WebArena agent from one observation format to accessibility tree + HTML + screenshot — input 4.8×, model share 46.9% → 61.6% (Chang et al., 2026, arXiv · HKUST); the Browser-Use harness spends 73% of its latency in model calls (Winston et al., 2026)	Reading, when load evicts the cache and the whole context is re-read: ThunderAgent measured request latency up to 7.14× under cache thrash (Kang et al., 2026)
negligible	Producing the observation and executing the action: under 3.5% together (Abhyankar, Qi & Zhang, 2026)	Nothing: model and browser are close; which is larger depends on observation size and model speed	Navigation-only steps: a coding agent has no pages to turn
counter-example	A single-call agent is different: AgentSysBench’s one-call GUI agent (Kimi-K2.6) spends over 70% of its OSWorld time in the desktop sandbox (Chang et al., 2026, arXiv · HKUST), because the harness sleeps after every action — 2 s in OSWorld’s code (xlang-ai, 2026, code), 3 s in OSWorld 2.0 (XLANG Lab, 2026, arXiv · HKU)	Browser-Use: 73% of latency in model calls, not the browser (Winston et al., 2026)	Two production records point opposite ways: heavy tools in Claude Code / Codex (tools 59.8% > model 41.0%) (Zhu et al., 2026, arXiv · U. Washington); light tools in Copilot (model 13.7% > tools 2%) (Liu et al., 2026, arXiv · UIUC). The difference is the kind of tool (tests and builds vs file reads), not a contradiction
in one line	Slow in reading: every step re-reads the screenshot history, more as the task goes on	Slow in the environment (browser and page waits); once the observation grows, slow in reading again	Reading is cached away; slow in writing and in the tools

A3 · Cost, every cause — II. tokens read and III. tokens written

the complete table behind pages 8 and 11 · prices are list prices from the vendors’ own pages, September 2026

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE
II-1	History re-sent	Step i reads everything from the $i - 1$ earlier steps; total reading $\approx N^2$	“Cost grows quadratically with the number of steps” — OSWorld-Human, real runs	(Abhyankar, Qi & Zhang, 2026)
II-2	Large screenshots	1,000–1,800 tokens per image; 100 images fill a 200K context	Anthropic’s engineering guidance for computer and browser use	(Anthropic, 2026a)
II-3	Rich observation formats	Accessibility tree + HTML + screenshot together: 4.8× the input	AgentSysBench, WebArena agent, Kimi-K2.6	(Chang et al., 2026, arXiv · HKUST)
II-4	Sampling	Several candidates per step, mostly read tokens	gpt-oss-120b + ReAct on 165 WebArena-Lite tasks: 1 → 10 candidates per step, 96K → 920K tokens per task	(Lee et al., 2026)
II-5	The cache is not a cure	However high the hit rate, a long history keeps reading dominant; a miss re-reads everything	TraceLab, 4,265 Claude Code / Codex sessions: prefix-cache hit rate 95.7%, prefix tokens still 59.5% of list-price cost; on a miss the re-read is 3.8× the genuinely new content	(Zhu et al., 2026, arXiv · U. Washington)
III-1	Thinking modes	Thinking tokens are billed as output; when they cannot be switched off they are a fixed overhead	OSWorld 2.0, same tasks: Claude Opus 4.8 224K output tokens per task, GPT-5.5 37K; Claude Opus 5.5: “thinking cannot be disabled and is billed as output”	(XLANG Lab, 2026, arXiv · HKU; Anthropic, 2026b)
III-2	But writing is not always the larger bill	In an uncached screenshot agent, output was 31% of the total	OSWorld-Human, GTA1: the paper’s \$2.43 counts output only; Table 3’s input and output give \$7.87	(Abhyankar, Qi & Zhang, 2026)

A3 · Cost, every cause — I. calls, IV. unit price, VI. failures

the complete table behind page 11 (I calls, IV unit price, VI failures)

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE
I	Number of calls	$N \cdot J_i$ scales both token bills	GTA1: 4 parallel planners per step, up to 3 retry rounds, one judging call → 4–12 planning calls per judging call, i.e. 5–13× the calls of a single-call harness (calc.). Agentic test-time scaling: 5–20 candidates per step multiply the calls by the candidate count. The same extra calls cost time and money at once	(Abhyankar, Qi & Zhang, 2026; Lee et al., 2026)
IV-1	Model tier	Claude Opus 5.5 \$4 / \$20 vs GPT-6 Luna \$0.10 / \$0.50 per million tokens read / written	Vendor price pages, 27 September 2026	(Anthropic, 2026b; OpenAI, 2026)
IV-2	Fast mode	2× on both vendors; switching speed tiers invalidates the cache	Anthropic: Claude Opus 5.5 \$8 / \$40, up to 2.5× output speed (vendor-stated), first-token wait unchanged. OpenAI: 2× on all listed models; “up to 2.5×” stated only for GPT-5.6 Sol	(Anthropic, 2026b; OpenAI, 2026)
IV-3	Cache write and read	Latest model of each vendor, per million tokens, normal read → cached read (ratios calc.): DeepSeek-V4.1-Flash \$0.30 → \$0.006 (0.02×; peak rate, both halve off-peak) Claude Opus 5.5 \$4 → \$0.20 (0.05×; other Claude models 0.1×, Fable 5.1 0.025×; cache writes 1.25× for 5 min, 2× for 1 hour) GPT-6 Sol \$2 → \$0.20 and GPT-6 Luna \$0.10 → \$0.01 (0.1×; cache writes 1.25×) Gemini 3.8 Flash \$0.75 → \$0.075 (0.1×; plus \$0.50 per million tokens per hour of storage; introductory to 31 Dec 2026) Grok 4.7 \$2 → \$0.50 (0.25×)	Vendor price pages, read 28 September 2026	(DeepSeek, 2026; Anthropic, 2026b; OpenAI, 2026; Google, 2026; xAI, 2026)
IV-4	Long-context surcharge	OpenAI: 2× the read price above 272K tokens	Vendor price page	(OpenAI, 2026)
VI-1	Failed attempts are billed	Cost per success = cost per attempt ÷ success rate	OSWorld 2.0: the best agent (Claude Opus 4.8) costs about \$72.4 per attempt at 20.6% completion → about \$351 per success (calc.). An accounting conversion, not the real price of retrying one task until it succeeds	(XLANG Lab, 2026, arXiv · HKU)
VI-2	Idle steps are billed in full	Every step of a dead loop is a full read and write	OSWorld-Human, GTA1 on 39 OSWorld tasks: one element-locating loop repeated the same step 18 times — 27 minutes, \$8.47 at list price without caching; in failed tasks over 50 steps, 66% of steps were such repeats	(Abhyankar, Qi & Zhang, 2026)
VI-3	Evaluation itself is too expensive to repeat	Single runs, no repeats	Holistic Agent Leaderboard: 9 benchmarks, 21,730 runs, one run per configuration, about \$40,000 in total; Claude Opus 4.1 not run on Online-Mind2Web because the estimate was \$20,000	(Kapoor et al., 2026)

A3 · Cost — V. the environment machine, and the four conventions

the complete tables behind page 12 · AWS rows: Amazon Web Services, 2026 (read 27 and 29 Sep 2026) · OpenAI container: OpenAI, 2026 · other vendors: vendor price pages, ...

ENVIRONMENT	PRICE (VENDOR PAGE)	1-HOUR TASK	15-MIN TASK	SHARE OF A \$7.87 API BILL	SHARE OF \$72.4
Browser Use cloud browser	\$0.02 per browser-hour, billed by the minute, 1-minute minimum; traffic extra (residential proxy \$5/GB, direct \$0.20/GB); sessions capped at 240 minutes	\$0.020	\$0.005	0.25%	0.03%
AWS t3.medium (2 vCPU / 4 GiB, Linux, us-east-1)	\$0.0418 per hour on the T3 product page; another AWS page says \$0.0416 (0.5% apart)	\$0.042	\$0.010	0.5%	0.06%
Browserbase cloud browser	\$20 per month for 100 hours, then \$0.12 per hour; \$99 per month for 500 hours, then \$0.10; billed by the minute	\$0.12	\$0.03	1.5%	0.17%
E2B / Daytona sandbox (2 vCPU / 4 GiB)	E2B \$0.000014 per vCPU-second + \$0.0000045 per GiB-second; Daytona \$0.0504 per vCPU-hour + \$0.0162 per GiB-hour — both \$0.1656 per hour	\$0.166	\$0.041	2.1%	0.23%
Modal sandbox (1 physical core = 2 vCPU, 4 GiB)	\$0.00003942 per core-second + \$0.00000667 per GiB-second → \$0.238 per hour; billed on the larger of requested and used	\$0.238	\$0.060	3.0%	0.33%
AWS t3.2xlarge (8 vCPU / 32 GiB, OSWorld 2.0's default instance)	\$0.3341 per hour	\$0.334	\$0.084	4.2%	0.46%
OpenAI hosted container (Hosted Shell / Code Interpreter, 4 GB)	\$0.12 per 20-minute session → \$0.36 per hour; 1 GB \$0.03, 16 GB \$0.48, 64 GB \$1.92 per 20 minutes	\$0.36	\$0.12	4.6%	0.50%

The four accounting conventions (page 12), in one line each: output only vs all tokens — 3.2× (OSWorld-Human, \$2.43 vs \$7.87); per attempt vs per success — 4.9× (OSWorld 2.0, \$72.4 ÷ 20.6%); uncached vs cached — 4–50× on the read price by vendor (calc.), far less on the bill (TraceLab: 95.7% hits, prefix still 59.5% of cost); standard vs fast mode — 2× (both vendors), buying only writing speed.

A4 · Each cause’s effect on time and on money

the complete table behind pages 13–14

CAUSE	EFFECT ON TIME	EFFECT ON MONEY	RELATIONSHIP
Many steps (N)	Roughly linear: one more step is one more call and one more wait (later steps somewhat slower, up to 3×)	Write tokens linear in N ; read tokens quadratic, because every step re-reads the whole history	same source — double the steps: about 2× the time, about 4× the read bill (uncached)
Many calls per step (J_i)	linear	linear	same source
Reading a lot (history, screenshots, rich observations)	Reading time, longer every step	Read-token bill	same source — the same tokens cost time and money
Writing a lot (thinking)	Generation time, roughly 12 ms per token	Write-token bill at 5× the read price	same source , and the dearest time: writing is the slowest and the most expensive token
Cache hits	Saves reading time	Read price 0.02–0.25× of a normal read, by vendor (A3)	same source , but fragile: editing the history or switching to fast mode invalidates it
Queueing	slow	free	slow but not expensive ; escaping it means a fast or priority tier at 2×
Environment waits (page loads, fixed sleeps)	slow	No API cost; machine time by the hour — CPU sandboxes \$0.02–0.36 per hour, usually under 5% of the API bill	slow but not expensive
Tool tails (tests, builds)	slow	No API cost	slow but not expensive
Failures, idling, retries	slow	expensive	same source , and it turns “per attempt” into “per success”
Fast mode	Writing up to 2.5× faster (vendor-stated); reading unchanged	2× the price	money for time
Bigger models, more thinking	slower	dearer	money and time for accuracy , convex: marginal tokens per point 101K → 575K on one model; 6× / 9× / 9.6× tokens for a few points; more thinking not always more accurate (Lee et al., 2026; XLANG Lab, 2026, arXiv · HKU; Kapoor et al., 2026)
Smaller models	faster	cheaper	but if accuracy falls, steps and retries rise — possibly slower and dearer again (inference, no direct evidence)

A5 · The gaps in the evidence, and what each one blocks

the complete table behind page 14

GAP	WHAT EXISTS TODAY	CONSEQUENCE FOR WHAT CAN BE CLAIMED
Read / write time split and cache hit rate for a frontier API model	No measurement reports both; the only read / write split is on locally served 27–31B models (Yuan et al., 2026); screenshot-agent measurements do not report cache state	How much of “87–97% in the model” is reading, how much writing, and what remains once the cache is on — unknown
How much slower an agent is than a person	None of 17 benchmarks measures agent and human elapsed time on the same tasks; the closest is AXIS’s user study, in which a UI agent was 1.69× slower than manual work on easy tasks (small sample) (Lu et al., 2025)	“X times slower than a person” cannot go on a slide
Environment dollars	Benchmarks report instance types; AgentSysBench reports shares; products publish nothing. Computable from instance type × duration × list price: CPU environments usually 0.3–5% of the API bill (page 12)	Every benchmark cost is an API bill; the omission is small for CPU environments, not for GPU sandboxes or long idle sessions
Whether OSWorld-Human’s “action” includes the 2-s sleep after each step	Not stated (xlang-ai, 2026, code); if it does, “actions under 2%” implies more than 100 s per step (calc.)	“Environment under 3.5%” may understate the waiting
The rebound from a smaller model	No direct evidence on how many extra steps and retries a drop in accuracy causes	The last row of the relationship table stays marked “inference”
How much slowness hurts accuracy	Only OSWorld 2.0 records the stale-state failure mode, without its share of failures (XLANG Lab, 2026, arXiv · HKU)	“It exists” can be said; “how large” cannot

PART 2

Acceleration, term by term

What published work has done to each term of the Part 1 formulas, and what each change costs elsewhere.

We classify; we do not propose methods. Each work sits on the term it changes, next to the terms that grow in exchange and one measured effect with its conditions. Every number was re-read in its source on 28 September 2026; calc. marks our arithmetic, vendor a vendor’s own statement. Full references at the end of the part.

IN THIS PART

02	The map: which term each family of work changes
03–04	Fewer model calls, fewer steps
05–07	A faster call: queueing, prefill, decoding
08	Less time outside the model
09–10	Overlap: parallel calls, and speculation with its safety
11–13	The growing prompt, the bill by class, a cheaper model per call
14	Success and cost per success, set-up included
15–21	References · Appendix B0–B10

The map: each family of work targets one term or a pair, and most grow another

Each work sits on the term it changes: it shrinks a term, removes a structure, or adds overlap T_{saving} ; most also grow another term. Speed-up = baseline time ÷ new time on the same tasks; each row says what was timed.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} (t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{ TPOT}_{ij}) + E_i \right] - T_{\text{saving}}$$

MONEY

$$v(m, p) = \mathbb{E} \left[\sum_i \sum_j \sum_{\kappa} n_{ij}^{\kappa} c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}} \right] \div R_m(p)$$

TERM (PART 1)	WHAT THE WORK DOES TO IT	FAMILY OF WORK	PAGE
J_i	→ 0 on steps run as code or replayed from a recording	compile or replay the loop	03
N	fewer steps: more work per call, reusable skills	bigger actions, skills	04
$t^{\text{queue}}, t^{\text{prefill}}$	shorter on self-hosted serving: whole programs scheduled, cache kept across tool waits	agent-aware serving	05
$ o_{a,i} \rightarrow t^{\text{prefill}}$	less of each page shown to the model	observation reduction	06
$n^{\text{out}}, \text{TPOT}$	less thinking; faster decoding, at a price	decoding	07
E_i	fewer or cheaper environment actions	environment	08
T_{saving}	added: independent calls run at once, or the next step guessed and run early	parallel calls; speculation	09–10
$\bar{g}, N^2$	a flatter slope; or the square removed	context management	11
n^{κ}	input moved from uncached to cache hit	caching	12
$c_{\kappa}(\mu_{ij})$	a cheaper model or tier for most calls	routing, small models, tiers	13
$R_m(p)$, set-up	higher success; a one-time set-up spread over the tasks	cost per success	14

Compiling or replaying the loop removes most model calls; where the actions still run, time falls less

term: $J_i \rightarrow 0$ on steps run as code or replayed · E_i unchanged on replay; JIT-Planner’s cached site tools also shorten it · a set-up cost sits outside the formula (page ...

Code or a recorded trajectory takes the model out of most steps, so $D_i \rightarrow 0$ there; the actions in E_i still run unless the code replaces them too. Against Synapse, a planning agent, replay cut median tokens by 96.5% but median time by 59.6% (calc., EchoPath).

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \rightarrow 0 \left(t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \right] - T_{\text{saving}}$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
JIT-Planner (Winston et al., 2026)	Parallel calls draft code plans over cached, checked site tools; the cheapest valid one runs	Task time 150.1 → 15.4 s (9.7×), success 61 → 90% — 37 tasks on 5 web apps (18 author-written), 3 runs, GPT-4.1, vs Browser-Use, same model	Set-up of 25–90 + 25–45 min per app, excluded
MobileGPT (Lee et al., 2024)	Sub-tasks learned once are recalled from app memory; the model fills only what memory lacks	Repeat runs with new parameters: 70.6–98.6% of actions from memory, latency –62.5%, cost –68.8% — 80 tasks, 8 Android apps, GPT-4-Turbo, vs deriving every action	An offline explorer runs 10–15 min per app before any task (\$10.78 in all); GPT-3.5 fills slots
AutoDroid-V2 (Wen et al., 2025)	A local model writes one script per task, not one call per step	Model time 669.2 → 46.3 s per task (screen actions excluded), success 43.9 → 54.4% — 158 DroidTask tasks, same 8B model	\$82.42 of GPT-4o calls per app, offline (calc.)
EchoPath (Zhao et al., 2026, arXiv · JHU)	Validated trajectories kept as callable memories and replayed	Median tokens 586,386 → 20,370, median time 315.7 → 127.5 s, success 91.8 → 91.2% — OSWorld-Verified, 159 replayable tasks, Codex GPT-5.5, vs Synapse	First pass ≈ 572k tokens and ≈ 4.5 min per task, not counted

Vendor tools replay recorded steps and call a model only to repair a failure; none publishes a repair rate with a defined sample (B1).

Fewer steps: more work per call cut steps by 38–46%; skills cut fewer and cost prompt tokens

term: $N \downarrow$ · actions per step grow with batching, fall with API calls (page 8) · skills add module calls ($J_i \uparrow$) and lengthen the prompt $|H_{\{a,i\}}|$

One call that issues an API call or a batch of actions cut N by 38–46% (calc.); the actions still run, and only **AXIS timed the task.** Skills cut 11–21% of steps (calc.) and raise success; a plain agent given 15 steps instead of 10 used 31% fewer tokens (calc.) and did better.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \downarrow \left[\sum_{j=1}^{J_i} \uparrow \left(t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} \uparrow + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \right] - T_{\text{saving}}$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
AXIS (Lu et al., 2025a)	Application API calls instead of UI sequences	Steps 3.2 → 2.0, task time 59.5 → 29.9 s, success 52 → 84% — 50 Word tasks, GPT-4o, vs the UFO agent	One application; the API must exist
OSWorld 2.0, batched (XLANG Lab, 2026, arXiv · HKU)	One call emits several actions	Steps 190.5 → 103, success 18.5 → 20.6%, ~\$76.1 → ~\$72.4 per task — 108 tasks, Claude Opus 4.8, one run	Tool calls 190.5 → 481.8; no wall-clock
ASI (Wang et al., 2025b)	Verified Python skills induced from the agent’s own successes	Steps 5.6 → 5.0 (~10.7%, calc.), success 32.7 → 40.4% — 812 WebArena tasks, Claude 3.5 Sonnet	Induction calls not counted; no time or \$
WALT (Prabhu et al., 2026)	Site tools learned by exploration	Steps 8.9 → 7.0, success 57.5 → 64.1% — 234 VisualWebArena tasks, GPT-5 planner, GPT-5-mini executor	Exploration cost not quantified
Plain agent, 15 steps (Hajimiri et al., 2026)	No skill or memory module; a pruned page tree	Plain 44.78% at 73.6K tokens per task vs ASI 41.02% at 107.3K — 655 WebArena tasks (calc.), Gemini 3 Flash, 3 runs	Skills add tokens and calls

Headroom: grouping actions by screenshot needs 1.35× (Chrome) to 2.93× (Calc) fewer steps in human reference runs (calc.) (Abhyankar, Qi & Zhang, 2026).

Serving the agent as one program cuts queueing and re-reading, on self-hosted models only

terms: t^{queue} and $t^{prefill}$ ↓ under load · self-hosted serving only; an API user cannot apply these

Schedulers that treat a whole agent run as the unit, and keep its cache through tool waits, give 1.2–15× more throughput or load than vLLM-based baselines. All are measured under load on self-hosted models; none reports task success.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \left(\overset{\text{targeted}}{t_{ij}^{\text{queue}} \downarrow} + \overset{\text{in exchange}}{t_{ij}^{\text{prefill}} \downarrow} + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \right] - T_{\text{saving}}$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
Agentix (Luo et al., 2026)	Program-level preemptive scheduling; cache-aware routing	Program throughput up to 15× vLLM and 2–5× vLLM with prefix caching, at equal latency — Llama-3.1 8B/70B and Falcon-180B on A100s	No task quality: scheduling only
InferCept (Abhyankar et al., 2024)	During a tool call, keeps, swaps or recomputes the cache, whichever wastes least	1.6–2× the load of vLLM at similar per-token latency (1.25× for 13B on one GPU) — six tool-augmented workloads	Baseline spent 37–40% of forward time recomputing
ThunderAgent (Kang et al., 2026)	The program and its tool sandboxes are the scheduling unit	Steps per minute 1.24–3.58× vLLM (calc. from its figure) — GLM-4.6 and Qwen-3 on 8×H100	Throughput, not task time

Showing the model less of the page cuts prefill; a reader call can cost more time than it saves

term: $|o_{\{a,i\}}| \downarrow \rightarrow$ the actor's t^{prefill} and $n^{\text{unc}} \downarrow \cdot J_i + 1$ when a model does the reducing; that reader reads the full page · success can fall

Pruning the page before the model reads it shortens the actor’s prompt at that step, and later prompts only where old pages stay in the history; a separate reader call adds a model call per step. A pruning program without a model halved step time and kept 84% of the successes.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \left(t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \right] - T_{\text{saving}}$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
FocusAgent (Kerboua et al., 2026)	A small model picks the relevant page-tree lines for the actor	Input cost −19%, success 53.6 → 51.5% — 330 WorkArena L1 episodes; model time per step 2.5 → 10.1 s on 33 tasks, one seed — GPT-4.1 actor, GPT-4.1-mini reader	One reader call per step; the time excludes the browser
Evolved pruning program (Enomoto et al., 2026, arXiv · NEC)	An offline-evolved program keeps ~20% of the HTML; no model at run time	Step time 65.7 → 30.2 s (2.2×), 84% of successes kept — 33 WorkArena L1 tasks, Qwen3.5-122B, 2 runs	Success falls; timed per step
Aguvis (Xu et al., 2025)	A screenshot instead of HTML, read by a self-hosted 72B model	Input tokens per step 1,196 vs ~4,000 (−70%), success 22.1 → 27.1% — 104 Mind2Web-Live tasks, vs GPT-4o on HTML	The model changes too
AgentOccam (counter) (Yang et al., 2025)	Simplified page and pruned history, aimed at accuracy	Success 16.5 → 43.1%, observation tokens per step +33%, steps +45% (calc.) — 812 WebArena tasks, GPT-4-Turbo	Prefill grows

Less thinking and faster decoding shorten the writing; neither touches the environment

terms: $n^{\text{out}} \downarrow$ (thinking included) · TPOT $\downarrow$ at a higher price per token

Choosing the reasoning effort per step cut reasoning tokens by 45% at equal success; fast tiers are stated by their vendors to be up to 2.5× faster for twice the price. Both shorten D_i only.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \uparrow (t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \downarrow \text{TPOT}_{ij} \downarrow) + E_i \right] - T_{\text{saving}}$$

MONEY

$$v(m, p) = \mathbb{E} \left[\sum_i \sum_j \sum_{\kappa} n_{ij}^{\kappa} (c_{\kappa}(\mu_{ij}) \uparrow + x_{\text{env}} c_{\text{env}}) \right] \div R_m(p)$$

targeted in exchange unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
ARES (Yang et al., 2026a, arXiv · UC Santa Barbara)	A 1.7B router picks the reasoning effort for each step	Reasoning tokens per task 21,424 → 11,723 (−45%), success 45.0 → 46.5%; low effort throughout −7.6 points (calc.) — WebArena, ~129 tasks (calc.; count not printed, B9), gpt-oss-20b	A router call per step, cost not reported; no wall-clock
Overthinking (Cuadron et al., 2025, arXiv · UC Berkeley)	Two low-effort runs; keep the one that overthinks least	\$1,400 → \$800 for the whole run, solved 29.1 → 27.3% — SWE-bench Verified (count not printed, B9), o1	Two attempts per task
GUI-G1 (Zhou et al., 2025)	RL-trained grounding without a thinking section	Output tokens 107–114 → 37–39 per example, ScreenSpot 87.5 → 90.3% — InfGUI-R1-3B (accuracy from its own paper) vs GUI-G1-3B, two differently trained models	Grounding only; no timing
Agent-X (Chung et al., 2026)	Draft tokens looked up from the prompt’s tool-use examples, verified by the model	Decoding 1.73× faster; task time 1.43× faster from this alone (1.61× with its prompt cache) — 1,022 TinyAgent tool-calling examples, 7B model on a Mac mini	On-device only
Fast tiers (vendor) (Anthropic, 2026b; OpenAI, 2026)	The same model, served faster	Anthropic: up to 2.5× output tokens per second; the gain is in output speed, not first-token time. OpenAI: up to 2.5× faster, ‘speed’ undefined	Price 2×; switching speed invalidates Anthropic’s message cache

NOT YET MEASURED · independent task-level timing of fast tiers · the net effect of per-step effort changes, which invalidate Anthropic’s message cache · more rows in Appendix B4 (2/2)

Time outside the model: on self-hosted WebArena it exceeded model time, and few works shrink it

term: $E_i \downarrow$ – fewer environment actions, cached tool results · x_{env} priced by the hour

With one agent on self-hosted WebArena, the mean step took 12.2 s, 7.6 s of it outside the model (Claude 3.5 Sonnet, 5,493 steps, on 8-CPU VMs the authors say limited speed; WorkArena L2: 4.9 of 10.7 s) (Le Sellier de Chezelles et al., 2025); Browser-Use on five web apps spent 73% of its time in model calls (Winston et al., 2026). The works that shrink E_i skip the browser or cache tool results; batching actions moves the other way.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \left(t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{TPOT}_{ij} \right) + E_i \downarrow \right] - T_{\text{saving}}$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
Skim (Wong et al., 2026, arXiv · Princeton)	Fetches a synthesized URL over HTTP instead of driving the browser; a verifier falls back to the agent	Median cost 1.9× lower, latency −33.4%, accuracy kept — WebVoyager and WebShop, “300+” tasks (count not stated; condition (iii) pending, B9), GPT-4o agents	Read-only tasks; site profiling not counted
ToolCaching (Zhai et al., 2026, arXiv · Southeast U.)	Caches tool results, with learned admission	Latency 16.2 → 10.7 s vs no cache at a 0.514 hit ratio — 500 Movie Recommendation queries, LLMCompiler, DeepSeek V3	Accuracy not reported
AXIS (Lu et al., 2025a)	API calls replace clicks	UI actions 103 → 48 over 50 Word tasks	The API must exist
OSWorld 2.0 (counter) (XLANG Lab, 2026, arXiv · HKU)	More actions per call	Tool calls ×2.5 (calc.); a 3-s pause follows each action, whether after each batched call is not stated; no wall-clock	N falls, actions per step grow

Billed environment usage $x_{\text{env}}c_{\text{env}}$: a managed cloud browser for agents is billed per second at \$0.0895 per vCPU-hour used and \$0.00945 per GB-hour of peak memory, with no CPU charge while it waits on I/O (vendor price) (Amazon Web Services, 2026).

Overlap (1): running independent calls and tools together saved 17–73% of task time; one workload got slower

term: T_{saving} raised where it is small today · calls, turns and tokens fall when independent calls share a turn (LLMCompiler, W&D, TPS-Bench)

Running independent calls or tools together creates T_{saving} ; the longest dependent chain sets the limit. Speed-up $\leq (T_{\text{LLM}} + T_{\text{tool}}) / \max(T_{\text{LLM}}, T_{\text{cp}})$: decoding, tool and critical-path tool time (Feng et al., 2026, arXiv · UC Berkeley); for parallel calls the slowest one sets the time (Kim et al., 2024); at most 2× when every tool call waits for the one before ($T_{\text{cp}} = T_{\text{tool}}$; calc.).

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \uparrow \text{ or } \downarrow (t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{ TPOT}_{ij}) + E_i \right] - T_{\text{saving}} \uparrow$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
LLMCompiler (Kim et al., 2024)	A planner builds a dependency graph; independent calls run at once	Task time 20.47 → 5.47 s (3.74×), accuracy 72.47 → 77.13%, input tokens 20,000 → 2,800 — 500 Movie Recommendation questions, gpt-3.5-turbo	WebShop got slower: 5.98 → 10.48 s
W&D (Lin et al., 2026)	Three or more tool calls per turn	Time per task 1,522.6 → 904.2 s (−40.6%), \$102.5 → \$65.7 per 100 tasks, 66 → 68% — first 100 BrowseComp tasks, GPT-5, one run	Call counts forced; \$ include search
AsyncFC (Feng et al., 2026, arXiv · UC Berkeley)	Tool calls return futures; the model keeps decoding until it needs a result	1.44× vs the plain agent (1.21× with parallel calls alone), resolved 47.6 → 44.3% — 300 SWE-bench Lite tasks, GPT-5.2, tool latency doubled	Resolution −3.3 points
TPS-Bench (Xu et al., 2026b)	RL training teaches a small model to issue independent tool calls in the same turn	Time per task 42.0 → 34.8 s (−17%, calc.), completion 26.75 → 35.17%, turns 2.4 → 1.9, output tokens 2.2k → 1.6k — 100 TPS-Bench-Hard tasks, Qwen3-1.7B before and after RL	Parallelism inferred from fewer turns, not measured; no \$

Overlap (2): guessing the next step saved 5–45% of task time, always paid in extra calls

term: T_saving added by guessing · calls, tokens and GPU hours grow · safe only for effects that can wait for a commit

A guesser (a cheaper model or setting, or the model itself) runs the predicted next action early: a right guess hides a wait, a wrong one is discarded but paid for. Steps with side effects need the rule on the right.

TIME

$$T_{\text{attempt}} \approx \sum_{i=0}^N \left[\sum_{j=1}^{J_i} \uparrow (t_{ij}^{\text{queue}} + t_{ij}^{\text{prefill}} + n_{ij}^{\text{out}} \text{TPOT}_{ij}) + E_i \right] - T_{\text{saving}} \uparrow$$

MONEY

$$v(m, p) = \mathbb{E} \left[\sum_i \sum_j \uparrow \sum_{\kappa} n_{ij}^{\kappa} \uparrow c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}} \right] \div R_m(p)$$

targeted in exchange unchanged

WORK	MEASURED EFFECT, AND ITS CONDITIONS	PAID IN
Speculative Actions (Ye et al., 2026)	Time −19.5%, guesses 54.7% right — chess, 5 × 30 steps, GPT-5 high effort guessed by GPT-5 low	Guess calls; ≤ 50% saving in its model
ISP (Hua et al., 2025)	Total time 182.70 → 105.42 s (−42.3%), \$0.2160 → \$0.2973 per plan — 117 OpenAGI tasks, GPT-4-turbo	Baseline spread 2.3× its mean
SMC (Liu et al., 2026c)	τ² Telecom, 2,285 tasks (calc.): 27.60 → 22.47 s (−18.6%), outcomes unchanged; AppWorld: 355.7 → 195.9 s (−44.9%; −40.4% already from one-step guessing), −2 of 168 — Qwen3.5-27B, 4B drafter	3 GPUs instead of 1
DSP (Guan et al., 2026)	Time −37.09% at +62.99% cost with a debate target (−25.26% at +69.36% with ReAct) — 312 OpenAGI tasks, GPT-4.1-mini; cost vs both agents in sequence	Guess calls; cost vs one agent not reported
Ghost Tool Calls (Mohammadi et al., 2026a, arXiv · MPI-SWS)	Naive speculation: p50 −5.4%, p99 11.56 → 14.22 s; tool calls seen by the provider 4.03 vs 1.03 per task — 30 author-built tasks × 3 seeds	Abandoned calls reveal intent

WHEN THE STEP HAS SIDE EFFECTS

Hold irreversible effects until commit; compensating them afterwards leaks

- Held until commit: 0 of 500 aborted sends leaked; compensation 400, checkpoint-and-replay 200, mislabelled effects 300 — one author-built workflow, 5 abort causes × 100 (Mohammadi et al., 2026b)
- Resuming from a checkpoint repeated or skipped an effect in 90 and 93 of 96 author-built workflows; a case study paid an invoice twice (Wu et al., 2026, arXiv · SUSTech)
- Speculate only where a wrong guess can be undone; for deleting records or placing orders, naive speculation is harmful (Ye et al., 2026)

The growing prompt: masking old observations halved the bill; only compiled plans are shown to remove the square

terms: $\bar{g} \downarrow$ (mask, trim, evict) or the N^2 structure removed (a plan instead of the history) · edits to old context turn some cache hits into misses or cache writes

Placeholders remove each old observation once it is 10 turns old: a smaller effective $\bar{g}$, a flatter slope of the N^2 term but the same shape. A compiled plan removes the square: the prompt holds the plan, not the history.

PROMPT

$$\sum_{i=1}^N |H_{a,i}| = N |H_{a,1}| + \frac{N(N-1)}{2} \text{ removed by a plan } \bar{g} \downarrow$$

MONEY

$$v(m,p) = \mathbb{E} \left[\sum_i \sum_j \sum_{\kappa} \left(n_{ij}^{\kappa} \text{ hit} \rightarrow \text{miss, write} \right) c_{\kappa}(\mu_{ij}) + x_{\text{env}} c_{\text{env}} \right] \div R_m(p)$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
Observation masking (Lindenbauer et al., 2025; Yang et al., 2024)	Observations older than 10 turns become a placeholder (SWE-agent’s method, there 5)	Cost per task \$1.29 → \$0.61, solved 53.4 → 54.8% — 500 SWE-bench Verified tasks, Qwen3-Coder-480B; cost = tokens × list price	Model summaries instead lengthened runs by ~15%
AgentDiet (Xiao et al., 2026)	A cheaper model deletes useless or expired content two steps back	Input tokens −39.9 to −59.7%, cost −21.1 to −35.9% with the reducer, success −1 to +2 points — 200 SWE-bench Verified + 300 Multi-SWE-bench Flash tasks, Claude 4 Sonnet, Gemini 2.5 Pro, one run	A reducer call per step
TokenPilot eviction (Xu et al., 2026a)	Evicts stale segments every 3 turns	Cost \$4.22 → \$2.79 while cache-hit tokens fell 26.7M → 8.6M — PinchBench continuous sessions, GPT-5.4-mini	Fewer tokens, fewer of them cached
A plan, not the history (Kim et al., 2024)	The prompt carries a compiled plan instead of every observation	Input tokens per question 20,000 → 2,800 — 500 Movie Recommendation questions, gpt-3.5-turbo	Only on compiled steps

The bill by class: a cached stable prefix cut bills 41–80% vs no cache; with trimming and eviction, up to 87%

term: $n^{\text{unc}} \rightarrow n^{\text{hit}}$ (input read from the cache) · cache writes n^{w} grow · the price $c_{\kappa}(\mu)$ unchanged

A cache hit costs 2–25% of an uncached input token (by vendor), so moving input into the hit class cuts that input’s price by 75–98% (calc.) without changing the model. A change of speed, of tools or of old context turns hits back into writes and misses.

MONEY

$$v(m, p) = \mathbb{E} \left[\sum_i \sum_j \sum_{\kappa} \left(\overset{\text{targeted}}{n_{ij}^{\kappa} \text{ unc} \rightarrow \text{hit}} \overset{\text{in exchange}}{c_{\kappa}(\mu_{ij}) =} + x_{\text{env}} c_{\text{env}} \right) \right] \div R_m(p)$$

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
Don’t Break the Cache (Lumer et al., 2026, arXiv · PwC)	Measures caching the stable prefix and leaving tool results out	Cost –41 to –80%, first-token time –6 to –31% vs a forced no-cache baseline — DeepResearch Bench (web-search agent), 40 sessions per condition, 4 models, 10K-token system prompt	Caching the whole context made GPT-4o’s first token 8.8% slower
TokenPilot (Xu et al., 2026a)	Stable placeholders for changing fields, trimming, eviction	Cost \$8.31 → \$3.22 (123 PinchBench tasks, one per session), \$81.52 → \$10.58 (Claw-Eval, continuous); hit rate 38.7 → 79.2% — GPT-5.4-mini	Score change –2.6 to +2.1 points
TraceLab counterfactual (Zhu et al., 2026, arXiv · U. Washington)	Keeps the cache alive through the user’s think time	At most –12.8% of the priced cost (paper; 15.8% on the live dashboard) — coding-agent traces	An estimate, not a measurement

LATEST MODEL (VENDOR PRICE PAGES, READ 28 SEP 2026)	CACHE HIT ÷ UNCACHED INPUT	CACHE WRITE
DeepSeek-V4.1-Flash	0.02×	not listed
Claude Opus 5.5	0.05×	1.25× (5 min), 2× (1 h)
GPT-6 Sol, GPT-6 Luna	0.1×	1.25×
Gemini 3.8 Flash	0.1×	storage \$0.50 per million tokens per hour
Grok 4.7	0.25×	not listed

A cheaper model for most calls cut dollars per task by 50–97%, usually for a few points of success

term: $c_k(\mu_{ij}) \downarrow$ – a smaller model or cheaper tier for most calls · N and J_i can grow · $R_m(p)$ often falls

Sending most steps to a small model, and only hard ones to a large one, cut dollars per task by 50–97% (calc.), usually for 1–4 points of success and sometimes with more steps. Tiers trade the price per token against queueing and speed.

MONEY

$$v(m, p) = \mathbb{E} \left[\sum_i \uparrow \sum_j \sum_{\kappa} n_{ij}^{\kappa} c_{\kappa}(\mu_{ij}) \downarrow + x_{\text{env}} c_{\text{env}} \right] \div R_m(p) \downarrow$$

targeted

in exchange

unchanged

WORK	WHAT IT CHANGES	MEASURED EFFECT, AND ITS CONDITIONS	WHAT GROWS, OR IS LEFT OUT
StepWise (Wei et al., 2026)	An 8B model by default; the large model when the small one is stuck	\$0.881 → \$0.224 per task (−74.6%), success 58.1 → 55.4%, time per request 6.4 → 4.1 s — OSWorld (count not printed, B9), EvoCUA-8B + Claude Sonnet 4.5 vs Sonnet 4.5 alone	Steps 25.4 → 26.2
WebRouter (Li et al., 2026b)	A router picks one of three models for each step	\$0.98 → \$0.12 per task, accuracy 86.1 → 82.3%, ~14% slower — 5 WebVoyager sites, ≥ 46 tasks each, vs GPT-4o	Steps 7.63 → 8.38
Fara-7B (Awadallah et al., 2025, arXiv · Microsoft Research)	A 7B screenshot agent instead of a frontier computer-use model	\$0.913 → \$0.025 per task, success 70.9 → 73.5% — WebVoyager (count not printed, B9), 3 runs; but Online-Mind2Web 42.9 → 34.1%	7B priced at \$0.20 per million tokens, no caching
Agentic plan caching (Zhang et al., 2025)	Cached plan templates, adapted by a small model	Cost −50.31% on average, at 96.61% of the best accuracy — 5 workloads; latency −27.28% on one FinanceBench run	3.99 s to build each cache entry

Tier prices (vendor, read 28 Sep 2026): OpenAI Flex and Batch 0.5×, Fast 2×; Google Priority 1.8× (Preview); Microsoft Foundry priority 1.75–2.5× (calc.) (OpenAI, 2026; Google, 2026; Microsoft, 2026a).

Cost per success moves with the success rate, and a set-up cost must be spread over the tasks

terms: $R_m(p)$ in $v(m,p) = C_m(p) / R_m(p)$ · plus a one-time set-up cost that Part 1’s per-attempt formula leaves out

A dearer attempt can give a cheaper success: Beyond Browsing’s hybrid agent costs \$1.4 per attempt against \$1.2 for API calls alone, but \$3.60 against \$4.11 per success (calc.). Few papers report the ratio, so the figures here are our calculations. Set-up costs are left out of most headlines; SpeedRunner spreads them over its runs (B8).

PER TASK
SET-UP
SPREAD
adapted

$$v_n(m,p) = \frac{C_{\text{setup}}}{n} + \frac{C_m(p)}{R_m(p)}$$

C_{setup} : one-time cost of building the code, tools or memories; n : tasks it serves; the second term is Part 1’s $v(m,p)$, each task retried until it succeeds, so per task = per success. Against a baseline of cost per success v_{base} it pays off once $n > C_{\text{setup}} / (v_{\text{base}} - v(m,p))$, which needs $v(m,p) < v_{\text{base}}$ (calc.).
fixed + variable cost: Kapoor et al., 2025, §3 (in words); one large tool-making call spread over n cheap calls, $O(nc + C)$ with C, c the cost of one large- and one small-model call: Cai et al., 2024, Table 2 · per success: Erol et al., 2026, Eq. 2

WORK	COST PER SUCCESS (CALC. UNLESS STATED)	WHY IT MOVED
AXIS (Lu et al., 2025a)	\$0.77 → \$0.24	cheaper and more successful
W&D (Lin et al., 2026)	\$1.55 → \$0.97 per correct answer	cheaper per attempt
OSWorld 2.0, batched (XLANG Lab, 2026, arXiv · HKU)	\$411 → \$351	Opus 4.8; success 18.5 → 20.6%
Beyond Browsing (Song et al., 2025)	browsing ≈ \$0.68 · API only ≈ \$4.11 · hybrid ≈ \$3.60	dearer attempt, cheaper success (hybrid vs API)
BATS (Liu et al., 2026a)	\$0.79 → \$4.47	success 12.6 → 24.6%, cost ×11
AI Agents That Matter (Kapoor et al., 2025)	93.2% for \$2.45 vs 88.0% for \$134.50 (run totals, stated)	simpler agent, better and cheaper

SET-UP COSTS LEFT OUT OF THE HEADLINES

Each needs n tasks to pay off

- JIT-Planner: 25–90 min of tool building plus 25–45 min of traces per app (Winston et al., 2026)
- EET (Agentless, GPT-5-mini): \$4.3 to build its experience base once, against a 500-task SWE-bench Verified run cut from \$13.77 to \$6.18 — repaid within that run; with Mini-SWE-Agent the \$7.1 set-up exceeds one run’s \$3.11 saving (calc.) (Guo et al., 2026)
- AutoDroid-V2: \$82.42 of GPT-4o calls per app (calc.) (Wen et al., 2025)

References · Part 2 (1 of 7)

author-year tags on the Part 2 pages refer to these entries · a, b, c separate works whose first authors share a surname and year within Part 2 · published version where...

1. Abhyankar, R., He, Z., Srivatsa, V., Zhang, H., & Zhang, Y. (2024). InferCept: Efficient intercept support for augmented large language model inference. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 81–95. arXiv:2402.01869. University of California, San Diego.

2. Abhyankar, R., Qi, Q., & Zhang, Y. (2026). OSWorld-Human: Benchmarking the efficiency of computer-use agents. *Proceedings of Machine Learning and Systems 8 (MLSys 2026)*. arXiv:2506.16042v2. University of California, San Diego.

3. Amazon Web Services. (2026). *Amazon Bedrock AgentCore pricing*, Browser Tool (read 29 September 2026). <https://aws.amazon.com/bedrock/agentcore/pricing/> — vendor pricing.

4. Anthropic. (2026a, May 13). *Best practices for computer and browser use with Claude*. <https://claude.com/blog/best-practices-for-computer-and-browser-use-with-claude>

5. Anthropic. (2026b). *Claude API pricing; Prompt caching; Fast mode; Vision* (developer documentation, read 28 September 2026). <https://platform.claude.com/docs/en/about-claude/pricing>; <https://platform.claude.com/docs/en/build-with-claude/prompt-caching>; <https://platform.claude.com/docs/en/build-with-claude/fast-mode>; <https://platform.claude.com/docs/en/build-with-claude/vision>

6. Automation Anywhere. (2026). *Automator AI* [Product page, undated, read 28 September 2026]; *Generative Recorder and resilient automation / April 2024* [Community Product Club recap, 13 May 2024]. <https://www.automationanywhere.com/products/automator-ai>; <https://community.automationanywhere.com/generative-recorder-85080/generative-recorder-and-resilient-automation-april-2024-88211> — vendor marketing (the product page's 60% claims); the recap's “nearly 50%” is vendor-stated, undefined.

7. Awadallah, A., Lara, Y., Magazine, R., Mozannar, H., Nambi, A., Pandya, Y., Rajeswaran, A., Rosset, C., Taymanov, A., Vineet, V., Whitehead, S., & Zhao, A. (2025). Fara-7B: An efficient agentic model for computer use. arXiv:2511.19663v1. Microsoft (Microsoft Research AI Frontiers; no affiliation line printed).

8. Bai, H., Lv, W., Zheng, H., Lu, Y., & Shu, J. (2026). SPORK: Self-speculative forking to accelerate agentic LLM inference. arXiv:2607.03333v1. Tsinghua University; Meituan.

9. Browserbase. (2026a). *How caching works in Stagehand (and where it breaks)* [Blog post, by S. Arif, 24 February 2026]; *Caching actions*, Stagehand v3 and v4 documentation (undated) (read 28 September 2026). <https://www.browserbase.com/blog/stagehand-caching>; <https://docs.stagehand.dev/v3/best-practices/caching>; <https://docs.stagehand.dev/v4/best-practices/caching> — the docs describe the mechanism; the blog's speed-up figures are vendor marketing.

10. Browserbase. (2026b). *Pricing* (read 28 September 2026). <https://www.browserbase.com/pricing> — vendor pricing.

11. Cai, T., Wang, X., Ma, T., Chen, X., & Zhou, D. (2024). Large language models as tool makers (LATM). *The Twelfth International Conference on Learning Representations (ICLR 2024)*. arXiv:2305.17126. Google DeepMind; Princeton University; Stanford University.

12. Cerebras. (2026, August 13). *Accelerating GPT-5.6 Sol Ultrafast* [Blog post, by J. Er]. <https://www.cerebras.ai/blog/accelerating-gpt-5-6-sol-ultrafast-with-openai> — vendor and hardware-partner marketing.

13. Chen, B., Shu, C., Shareghi, E., Collier, N., Narasimhan, K., & Yao, S. (2023). FireAct: Toward language agent fine-tuning. arXiv:2310.05915v1. System2 Research; University of Cambridge; Monash University; Princeton University.

14. Chen, G., Zhou, X., Shao, R., Lyu, Y., Zhou, K., Wang, S., Li, W., Li, Y., Qi, Z., & Nie, L. (2025). Less is more: Empowering GUI agent with context-aware simplification (SimpAgent). *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2025)*, 5901–5911. <https://doi.org/10.1109/ICCV51701.2025.00558>. arXiv:2507.03730. Harbin Institute of Technology (Shenzhen); Huawei Noah's Ark Lab.

15. Chen, L., Zaharia, M., & Zou, J. (2024). FrugalGPT: How to use large language models while reducing cost and improving performance. *Transactions on Machine Learning Research* (December 2024; Featured Certification). <https://openreview.net/forum?id=cSimKw5p6R>. arXiv:2305.05176. Stanford University.

16. Choi, D., Park, K., Song, W., Dingliwal, S., Jayanthi, S. M., Shin, J., & Galstyan, A. (2026). IdleSpec: Exploiting idle time via speculative planning for LLM agents. arXiv:2605.22154v1. KAIST; Amazon AGI; Together AI.

17. Chung, J., Shin, B., Kim, J., & Rhu, M. (2026). Agent-X: Full pipeline acceleration of on-device AI agents. *Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services (MobiSys 2026)*. <https://doi.org/10.1145/3745756.3809195>. arXiv:2605.10380. KAIST.

18. Cuadron, A., Li, D., Ma, W., Wang, X., Wang, Y., Zhuang, S., ... Gonzalez, J. E. (2025). The danger of overthinking: Examining the reasoning-action dilemma in agentic tasks. arXiv:2502.08235v1. University of California, Berkeley; ETH Zurich; University of Illinois Urbana-Champaign; Carnegie Mellon University.

19. DeepSeek. (2026). *Models & pricing* (API documentation, read 28 September 2026); *DeepSeek-V4.1-Flash release* (10 September 2026). https://api-docs.deepseek.com/quick_start/pricing; <https://api-docs.deepseek.com/news/news260910>

References · Part 2 (2 of 7)

continued

20. Dihan, M. L., Hashem, T., Ali, M. E., & Parvez, M. R. (2025). WebOperator: Action-aware tree search for autonomous agents in web environment. *arXiv:2512.12692v1*. Bangladesh University of Engineering and Technology; Monash University; Qatar Computing Research Institute.

21. Dong, Y., He, J., Liu, S., Hou, Y., Du, D., Xu, Z., Yu, S., Yang, B., Xia, Y., & Chen, H. (2026). DeltaBox: Scaling stateful AI agents with millisecond-level sandbox checkpoint/rollback. *arXiv:2605.22781v2*. Shanghai Jiao Tong University (IPADS); Huawei Technologies.

22. Enomoto, M., Obara, R., Zhang, H., & Oyamada, M. (2026). Revisiting observation reduction for web agents: Comprehensive evaluation with a lightweight framework. *arXiv:2605.29397v1*. NEC Corporation.

23. Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. *arXiv:2504.13359v2*. Stanford University.

24. Farzaneh, A., & Simeone, O. (2026). Think short, defer smart, act, and repeat: Calibrated reasoning and uncertainty-aware deferral for edge LLM agents. *arXiv:2607.26865v3*. Northeastern University London.

25. Feng, G., Mao, H., Dutta, P., & Gonzalez, J. E. (2026). Concurrency without model changes: Future-based asynchronous function calling for LLMs (AsyncFC). *arXiv:2605.15077v1*. University of California, Berkeley.

26. Gim, I., Lee, S., & Zhong, L. (2024). Asynchronous LLM function calling (AsyncLM). *arXiv:2412.07017v1*. Yale University.

27. Google. (2026). *Gemini Developer API pricing; Priority inference* (Gemini API documentation, updated 24 and 23 September 2026; read 28 September 2026). <https://ai.google.dev/gemini-api/docs/pricing>; <https://ai.google.dev/gemini-api/docs/priority-inference>

28. Guan, Y., Lan, Q., Sun, F., Ding, D., Acharya, D., Wang, C., Wang, W. Y., & Hua, W. (2026). Dynamic speculative agent planning. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. *arXiv:2509.01920*. Johns Hopkins University; University of Alberta; University of British Columbia; Avey Research Center; Google DeepMind; University of California, Santa Barbara.

29. Guo, Y., Xiao, Y., Zhang, J. M., Harman, M., Lou, Y., Liu, Y., & Chen, Z. (2026). EET: Experience-driven early termination for cost-efficient software engineering agents. *Findings of the Association for Computational Linguistics: ACL 2026*, 33008–33024. <https://doi.org/10.18653/v1/2026.findings-acl.1652>. *arXiv:2601.05777*. Nanyang Technological University; King’s College London; University College London; University of Illinois Urbana-Champaign; Tsinghua University.

30. Hajimiri, S., Aminbeidokhti, M., Dolz, J., Ben Ayed, I., Laradji, I. H., Gella, S., & Gontier, N. (2026). Are online skill and memory modules always worth their tokens? A budget-constrained study of web agents. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)* (to appear). *arXiv:2606.15017v2*. ServiceNow AI Research; ÉTS Montréal; University of British Columbia; McGill University.

31. Hua, W., Wan, M., Vadrevu, S., Nadel, R., Zhang, Y., & Wang, C. (2025). Interactive speculative planning: Enhance agent efficiency through co-design of system and user interface (ISP). *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. *arXiv:2410.00079*. Rutgers University; Microsoft; Google DeepMind.

32. Huang, K.-H., Qiu, H., Dai, Y., Xiong, C., & Wu, C.-S. (2026a). GUI-KV: Efficient GUI agents via KV cache with spatio-temporal awareness. *Transactions on Machine Learning Research*. *arXiv:2510.00536*. Salesforce AI Research; University of California, Los Angeles.

33. Huang, Y., Srivatsa, V., Asch, A., Patwa, H. T., & Zhang, Y. (2026b). TClone: Low-latency forking of live GUI environments for computer-use agents. *arXiv:2605.17320v1*. University of California, San Diego; GenseeAI.

34. Huang, Z., Wang, X., Wang, A., Jurayj, W., Jiménez Gutiérrez, B., Khashabi, D., & Andrews, N. (2026c). Better, faster, stronger: Programmatic skill learning best reduces agent cost (SpeedRunner). *arXiv:2608.11338v1*. Johns Hopkins University.

35. Hyperbrowser. (2026). *Action caching* (HyperAgent documentation, undated, read 28 September 2026); *HyperAgent* [Source code], README.md on the main branch. <https://www.hyperbrowser.ai/docs/hyperagent/action-cache>; <https://github.com/hyperbrowserai/HyperAgent> — vendor documentation; “near-instant replay” is vendor marketing.

36. Jia, J., & Li, Q. (2026). AutoTool: Efficient tool selection for large language model agents. *Proceedings of the AAAI Conference on Artificial Intelligence, 40(37)*, 31265–31273. <https://doi.org/10.1609/aaai.v40i37.40389>. *arXiv:2511.14650*. Huazhong University of Science and Technology.

37. Jiang, W., Zhuang, Y., Song, C., Yang, X., Zhou, J. T., & Zhang, C. (2025). AppAgentX: Evolving GUI agents as proficient smartphone users. *arXiv:2503.02268v3*. Westlake University; Henan University; Southeast University; A*STAR (IHPC; CFAR).

38. Kang, H., Li, Z., Yang, X., Xu, W., Chen, Y., Wang, J., Chen, B., Krishna, T., Xu, C., & Arora, S. (2026). ThunderAgent: A fast, simple, and program-aware agentic inference system. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, poster. *arXiv:2602.13692*. Georgia Institute of Technology; University of Illinois Urbana-Champaign; Carnegie Mellon University; Together AI; independent researcher.

References · Part 2 (3 of 7)

continued

39. Kapoor, S., Stroebel, B., Siegel, Z. S., Nadgir, N., & Narayanan, A. (2025). AI agents that matter. *Transactions on Machine Learning Research*. arXiv:2407.01502. Princeton University.

40. Kerboua, I., Omidi Shayegan, S., Lù, X. H., Boisvert, L., Thakkar, M., Caccia, M., Espinas, J., Aussem, A., Eglin, V., & Lacoste, A. (2026). FocusAgent: Simple yet effective ways of trimming the large context of web agents. *Transactions on Machine Learning Research*. arXiv:2510.03204. Esker; INSA Lyon; ServiceNow Research; Mila; McGill University; Polytechnique Montréal; Université Lyon 1 (LIRIS).

41. Kerboua, I., Omidi Shayegan, S., Thakkar, M., Lù, X. H., Caccia, M., Eglin, V., Aussem, A., Espinas, J., & Lacoste, A. (2025). LineRetriever: Planning-aware observation reduction for web agents. arXiv:2507.00210v1. INSA Lyon; Esker; ServiceNow Research; Mila; McGill University; Université Lyon 1 (LIRIS).

42. Kim, H., Wu, Y., & Tambe, T. (2026). AgenticCache: Cache-driven asynchronous planning for embodied AI agents. *Proceedings of Machine Learning and Systems 8 (MLSys 2026)*. arXiv:2604.24039. Seoul National University; Stanford University.

43. Kim, S., Moon, S., Tabrizi, R., Lee, N., Mahoney, M. W., Keutzer, K., & Gholami, A. (2024). An LLM compiler for parallel function calling. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, PMLR 235, 24370–24391. arXiv:2312.04511. University of California, Berkeley; International Computer Science Institute; Lawrence Berkeley National Laboratory.

44. Lai, H., Liu, X., Zhao, Y., Xu, H., Zhang, H., Jing, B., Ren, Y., Yao, S., Dong, Y., & Tang, J. (2026). ComputerRL: Scaling end-to-end online reinforcement learning for computer use agents. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2508.14040. Tsinghua University; Z.AI; University of Chinese Academy of Sciences.

45. Le Sellier de Chezelles, T., Gasse, M., Lacoste, A., Caccia, M., Drouin, A., Boisvert, L., ... Chapados, N. (2025). The BrowserGym ecosystem for web agent research. *Transactions on Machine Learning Research* (Expert Certification). <https://openreview.net/forum?id=5298fKGmv3>. arXiv:2412.05467v4. ServiceNow Research; Mila; Polytechnique Montréal; Carnegie Mellon University; McGill University; Tel Aviv University; Université de Montréal; iMean AI.

46. Lee, S., Choi, J., Lee, J., Wasi, M. H., Choi, H., Ko, S., Oh, S., & Shin, I. (2024). MobileGPT: Augmenting LLM with human-like app memory for mobile task automation. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom '24)*, 1119–1133. <https://doi.org/10.1145/3636534.3690682>. arXiv:2312.03003v3. KAIST; Simon Fraser University; Korea University.

47. Li, G., Wu, R., Tan, H., & Chen, G. (2024). A plan reuse mechanism for LLM-driven agent. *Journal of Computer Research and Development*, 61(11), 2706–2720. <https://doi.org/10.7544/issn1000-1239.202440380>. English version: Li, G., Wu, R., & Tan, H. (2025), arXiv:2512.21309v2. University of Science and Technology of China.

48. Li, H., He, R., Mang, Q., Zhang, Q., Mao, H., Chen, X., Zhou, H., Zhang, H., Cheung, A., Gonzalez, J., & Stoica, I. (2026a). Continuum: Efficient and robust multi-turn LLM agent scheduling with KV cache time-to-live. arXiv:2511.02230v7. University of California, Berkeley; Stanford University; Tsinghua University.

49. Li, K., Meng, Z., Lin, H., Luo, Z., Tian, Y., Ma, J., Huang, Z., & Chua, T.-S. (2025). ScreenSpot-Pro: GUI grounding for professional high-resolution computer use. *Proceedings of the 33rd ACM International Conference on Multimedia (MM 2025)*, 8778–8786. <https://doi.org/10.1145/3746027.3755688>. arXiv:2504.07981. National University of Singapore; East China Normal University; Hong Kong Baptist University.

50. Li, T., Hu, J., Wang, Y., Liu, J., & Liu, X. (2026b). WebRouter: Query-specific router via variational information bottleneck for cost-sensitive web agent. *ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing*, 4086–4090. <https://doi.org/10.1109/ICASSP55912.2026.11464950>. arXiv:2510.11221. Nanjing University of Aeronautics and Astronautics; Hong Kong Baptist University; Beihang University; Pengcheng Laboratory.

51. Li, Y., Ye, Q., Choubey, P. K., Zhang, J., & Wu, C.-S. (2026c). Speculate with memory: Lossless acceleration for LLM agents. arXiv:2607.12236v1. Salesforce Research.

52. Lin, C., Han, Z., Zhang, C., Yang, Y., Yang, F., Chen, C., & Qiu, L. (2024). Parrot: Efficient serving of LLM-based applications with semantic variable. *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*. Shanghai Jiao Tong University; Microsoft Research.

53. Lin, X., Liew, J. H., Savarese, S., & Li, J. (2026). W&D: Scaling parallel tool calling for efficient deep research agents. *ICLR 2026 Workshop on Agents in the Wild: Safety, Security, and Beyond* (non-archival). <https://openreview.net/forum?id=JYM4xKpRgN>. arXiv:2602.07359v1. Salesforce AI Research.

54. Lindenbauer, T., Slinko, I., Felder, L., Bogomolov, E., & Zharov, Y. (2025). The complexity trap: Simple observation masking is as efficient as LLM summarization for agent context management. Paper presented at the 4th Deep Learning for Code Workshop (DL4C): Deep Learning for Code in the Agentic Era, NeurIPS 2025. <https://openreview.net/forum?id=OHVzruJl5k>. arXiv:2508.21433v3. JetBrains Research; Technical University of Munich.

55. Liu, T., Wang, Z., Miao, J., Hsu, I.-H., Yan, J., Chen, J., ... Lee, C.-Y. (2026a). Budget-aware tool use enables effective agent scaling (Budget Tracker, BATS). *Third Conference on Language Modeling (COLM 2026)*. arXiv:2511.17006. University of California, Santa Barbara; Google Cloud AI Research; Google DeepMind; New York University.

56. Liu, Y., Huang, Y., Yao, J., Feng, S., Gu, Z., Du, K., Li, H., Cheng, Y., Jiang, J., Lu, S., Musuvathi, M., & Choukse, E. (2026b). DroidSpeak: KV cache sharing across fine-tuned model variants. *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*, 319–338. arXiv:2411.02820 (under another title). University of Chicago; Microsoft.

57. Liu, Z., Kundu, S., & Beerel, P. A. (2026c). Speculative macro commit for faster tool-using agents. *2026 IEEE International Workshop on Machine Learning for Signal Processing (MLSP 2026)*, Atlanta, GA (poster). arXiv:2609.03236v1. University of Southern California; Intel Labs.

References • Part 2 (4 of 7)

continued

58. Lu, J., Zhang, Z., Yang, F., Zhang, J., Wang, L., Du, C., Lin, Q., Rajmohan, S., Zhang, D., & Zhang, Q. (2025a). AXIS: Efficient human-agent-computer interaction with API-first LLM-based agents. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 7711–7743. <https://doi.org/10.18653/v1/2025.acl-long.381>. Peking University; Nanjing University; Microsoft.
59. Lu, Q., Ding, L., Cao, S., Liu, X., Zhang, K., Zhang, J., & Tao, D. (2025b). Runaway is ashamed, but helpful: On the early-exit behavior of large language model-based agents in embodied environments. *Findings of the Association for Computational Linguistics: EMNLP 2025*, 24014–24027. <https://doi.org/10.18653/v1/2025.findings-emnlp.1304>. arXiv:2505.17616. Southeast University; The University of Sydney; Harbin Institute of Technology (Shenzhen); Nanyang Technological University.
60. Lumer, E., Nizar, F., Jangiti, A., Frank, K., Gulati, A., Phadate, M., & Subbiah, V. K. (2026). Don’t break the cache: An evaluation of prompt caching for long-horizon agentic tasks. arXiv:2601.06007v2. PricewaterhouseCoopers (PwC U.S.).
61. Luo, M., Shi, X., Cai, C., Zhang, T., Wong, J., Wang, Y., Wang, C., Huang, Y., Chen, Z., Gonzalez, J. E., & Stoica, I. (2026). Agentix: An efficient serving engine for LLM agents as general programs. *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*, 2443–2459. <https://www.usenix.org/conference/nsdi26/presentation/luo>. arXiv:2502.13965 (as Autellix). University of California, Berkeley; Google DeepMind; Shanghai Jiao Tong University.
62. Microsoft. (2026a). *Enable priority processing for Microsoft Foundry models* (Microsoft Learn, 22 September 2026); *Azure OpenAI pricing* (prices taken from the page’s data attributes, US East) (read 28 September 2026). <https://learn.microsoft.com/en-us/azure/foundry/openai/concepts/priority-processing>; <https://azure.microsoft.com/en-us/pricing/details/azure-openai/>
63. Microsoft. (2026b). *Self-healing (preview)* (Power Automate documentation, Microsoft Learn, 30 June 2026, updated 11 July 2026); *FAQ for Repair with Copilot at runtime in Power Automate desktop* (16 January 2026) (read 28 September 2026). <https://learn.microsoft.com/en-us/power-automate/desktop-flows/self-healing>; <https://learn.microsoft.com/en-us/power-automate/faqs-repair-copilot> — vendor documentation.
64. Mohammadi, B., Klein, L., Arora, A., & Bindschaedler, L. (2026a). Ghost tool calls: Issue-time privacy for speculative agent tools. arXiv:2606.02483v1. Max Planck Institute for Software Systems; EPFL; Aarhus University.
65. Mohammadi, B., Potamitis, N., Klein, L., Arora, A., & Bindschaedler, L. (2026b). Atomix: Timely, transactional tool use for reliable agentic workflows. *ICLR 2026 Workshop on Agents in the Wild: Safety, Security, and Beyond; ICLR 2026 Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy* (non-archival). arXiv:2602.14849v2. Max Planck Institute for Software Systems; Aarhus University; EPFL.
66. Nichols, D., Singhanian, P., Jekel, C., Bhatele, A., & Menon, H. (2025). Optimizing agentic language model inference via speculative tool calls. arXiv:2512.15834v1. Lawrence Livermore National Laboratory; University of Maryland.
67. Ong, I., Almahairi, A., Wu, V., Chiang, W.-L., Wu, T., Gonzalez, J. E., Kadous, M. W., & Stoica, I. (2025). RouteLLM: Learning to route LLMs from preference data. *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2406.18665. University of California, Berkeley; Anyscale; Canva.
68. OpenAI. (2026). *API pricing; API changelog* (Priority processing, 27 June 2025; Fast mode, 30 July and 5 August 2026; Ultrafast, 13 August 2026); *Fast mode, Flex processing, Batch API and Prompt caching* guides (developer documentation, read 28 September 2026); *Prompt Caching 201* (OpenAI Cookbook, 18 February 2026). <https://developers.openai.com/api/docs/pricing>; <https://developers.openai.com/api/docs/changelog>; <https://developers.openai.com/api/docs/guides/fast-mode>; <https://developers.openai.com/api/docs/guides/flex-processing>; <https://developers.openai.com/api/docs/guides/batch>; <https://developers.openai.com/api/docs/guides/prompt-caching>; https://developers.openai.com/cookbook/examples/prompt_caching_201
69. Pan, Z., Patel, A., Shen, Y., Hu, Z., Guan, Y., Li, W.-L., Qin, L., Wang, Y., & Ding, Y. (2025). KVFlow: Efficient prefix caching for accelerating LLM-based multi-agent workflows. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 139912–139931. <https://doi.org/10.52202/085713-4208>. arXiv:2507.07400. University of California, San Diego; Amazon Web Services.
70. Prabhu, V., Dai, Y., Fernandez, M., Ramakrishnan, K., Gu, J., Luo, Y., Savarese, S., Xiong, C., Li, J., Chen, Z., & Xu, R. (2026). WALT: Web agents that learn tools. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2510.01524. Salesforce AI Research.
71. Qin, Z., Yue, S., Hua, X., Fu, Y., & Ren, J. (2026). Executable agentic memory for GUI agent. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. arXiv:2605.12294. Tsinghua University; Sun Yat-sen University.
72. Shen, J., Bai, H., Zhang, L., Zhou, Y., Setlur, A., Tong, S., Caples, D., Jiang, N., Zhang, T., Talwalkar, A., & Kumar, A. (2025). Thinking vs. doing: Improving agent reasoning by scaling test-time interaction. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 187840–187881. <https://doi.org/10.52202/085713-5647>. arXiv:2506.07976 (under another title). Carnegie Mellon University; Scribe; University of Illinois Urbana-Champaign; University of Toronto; University of California, Berkeley; The AGI Company; New York University.
73. Shi, C., Chen, Y., Chen, Y., Miao, X., & Cui, B. (2026a). Multi-segment attention: Enabling efficient KV-cache management for faster large language model serving (AsymCache). arXiv:2606.02964v1. Peking University.
74. Shi, H., Wang, W., Fang, W., Liang, Y., Jin, T., Zhao, P., Liu, G., Chen, S., & Wang, Y. (2026b). CUA-Universe: A scalable and dynamic environment for hybrid GUI+CLI agents. arXiv:2609.05374v1. Shanghai Jiao Tong University; Zhejiang University.
75. Skyvern. (2026). *Code caching; Cost control* (developer documentation, undated, read 28 September 2026). <https://www.skyvern.com/docs/developers/features/code-caching>; <https://skyvern.mintlify.app/developers/optimization/cost-control> — vendor documentation.
76. Song, L., Dai, Y., Prabhu, V., Zhang, J., Shi, T., Li, L., Li, J., Savarese, S., Chen, Z., Zhao, J., Xu, R., & Xiong, C. (2026). CoAct-1: Computer-using multi-agent system with coding actions. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2508.03923. University of Southern California; Salesforce; University of Washington.

References · Part 2 (5 of 7)

continued

77. Song, Y., Xu, F. F., Zhou, S., & Neubig, G. (2025). Beyond browsing: API-based web agents. *Findings of the Association for Computational Linguistics: ACL 2025*, 11066–11085. <https://doi.org/10.18653/v1/2025.findings-acl.577>. arXiv:2410.16464. Carnegie Mellon University.

78. Srivatsa, V., He, Z., Abhyankar, R., Li, D., & Zhang, Y. (2025). Preble: Efficient distributed prompt scheduling for LLM serving. *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2407.00023. University of California, San Diego.

79. Tang, F., Shen, Y., Zhang, H., Chen, S., Hou, G., Zhang, W., Zhang, W., Song, K., Lu, W., & Zhuang, Y. (2025). Think twice, click once: Enhancing GUI grounding via fast and slow systems (FOCUS). arXiv:2503.06470v1. Zhejiang University; Microsoft Research Asia.

80. UiPath. (2026). *Healing Agent user guide: What is Healing Agent?; Recovery strategies; Licensing; Frequently asked questions* (Automation Cloud documentation, undated, read 28 September 2026). <https://docs.uipath.com/agents/automation-cloud/latest/user-guide-ha/what-is-healing-agent>; <https://docs.uipath.com/agents/automation-cloud/latest/user-guide-ha/deterministic-recovery-strategies>; <https://docs.uipath.com/agents/automation-cloud/latest/user-guide-ha/licensing>; <https://docs.uipath.com/agents/automation-cloud/latest/user-guide-ha/frequently-asked-questions> — vendor documentation.

81. Wadlom, N., Shen, J., & Lu, Y. (2026). Efficient LLM serving for agentic workflows: A data systems perspective (Helium). *Proceedings of the ACM on Management of Data*, 4(3), 1–29 (SIGMOD 2026). <https://doi.org/10.1145/3802046>. arXiv:2603.16104v1 (extended version). National University of Singapore.

82. Wang, N., Hu, X., Liu, P., Zhu, H., Hou, Y., Huang, H., ... Zhou, W. (2025a). Efficient agents: Building effective agents while reducing cost. arXiv:2508.02694v1. OPPO (OPPO AI Agent Team).

83. Wang, Z. Z., Gandhi, A., Neubig, G., & Fried, D. (2025b). Inducing programmatic skills for agentic tasks (ASI). *Second Conference on Language Modeling (COLM 2025)*. arXiv:2504.06821. Carnegie Mellon University.

84. Wang, Z. Z., Mao, J., Fried, D., & Neubig, G. (2025c). Agent workflow memory. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, PMLR 267, 63897–63911. arXiv:2409.07429. Carnegie Mellon University; Massachusetts Institute of Technology.

85. Wei, J., Ni, K., Zhao, Y., Gan, G., & Cohan, A. (2026). Step-level optimization for efficient computer-use agents (StepWise). *COLM 2026 Workshop on Efficient Reasoning* (non-archival). <https://openreview.net/forum?id=AQZh4nx7Jx>. arXiv:2604.27151v1. Yale University; University of North Carolina at Chapel Hill.

86. Wen, H., Tian, S., Pavlov, B., Du, W., Li, Y., Chang, G., Zhao, S., Liu, J., Liu, Y., Zhang, Y.-Q., & Li, Y. (2025). AutoDroid-V2: Boosting SLM-based GUI agents via code generation. *Proceedings of the 23rd Annual International Conference on Mobile Systems, Applications and Services (MobiSys 2025)*, 223–235. <https://doi.org/10.1145/3711875.3729134>. arXiv:2412.18116. Tsinghua University (AIR); Shanghai AI Laboratory; Beijing Academy of Artificial Intelligence.

87. Winston, C., Wang, R. Y., Mirhoseini, A., & Kozyrakis, C. (2026). Agent JIT compilation for latency-optimizing web agent planning and scheduling. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, PMLR 306. arXiv:2605.21470. Stanford University.

88. Wong, M., Hsieh, K., Nath, S., & Netravali, R. (2026). Skim: Speculative execution for fast and efficient web agents. arXiv:2605.16565v2. Princeton University; Microsoft Research.

89. Wu, G., Li, D., Jiang, K., Niu, J., Wang, C., & Zhang, Y. (2026). Safe to resume? Breaking execution continuity of agent execution via rollback. arXiv:2608.29381v1. Southern University of Science and Technology; City University of Hong Kong.

90. xAI. (2026). *Grok 4.7* model page, SpaceXAI Docs (read 28 September 2026). <https://docs.x.ai/developers/models/grok-4.7>

91. Xia, H., Li, Y., Du, C., Song, M., & Li, W. (2026). ToolSpec: Accelerating tool calling via schema-aware and retrieval-augmented speculative decoding. arXiv:2604.13519v2. The Hong Kong Polytechnic University; Peking University.

92. Xiao, Y.-A., Gao, P., Peng, C., & Xiong, Y. (2026). Reducing cost of LLM agents with trajectory reduction (AgentDiet). *Proceedings of the ACM on Software Engineering*, 3(FSE), Article FSE056. <https://doi.org/10.1145/3797084>. arXiv:2509.23586. Peking University; ByteDance.

93. XLANG Lab. (2026). OSWorld 2.0: Benchmarking computer use agents on long-horizon real-world tasks. arXiv:2606.29537v2. The University of Hong Kong (authored as “XLANG Lab and Collaborators”; the full author list is in the paper’s Appendix A).

94. Xu, B., Xue, Z., Chen, D., Fu, C., Wu, C., Huang, C., Jiang, C., Fang, J., Deng, X., Chen, Y., Yao, Y., Wang, X., Shang, J., Yu, G., & Zhang, N. (2026a). TokenPilot: Cache-efficient context management for LLM agents. *Findings of the Association for Computational Linguistics: EMNLP 2026*. arXiv:2606.17016v2. Zhejiang University; University of Electronic Science and Technology of China; Xidian University; HomologyAI.

95. Xu, H., Huang, X., Liu, Y., & Deng, Z. (2026b). TPS-Bench: Evaluating AI agents’ tool planning & scheduling abilities in compounding tasks. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 34949–34961. <https://doi.org/10.18653/v1/2026.acl-long.1614>. arXiv:2511.01527. Shanghai Jiao Tong University.

References · Part 2 (6 of 7)

continued

96. Xu, Y., Wang, Z., Wang, J., Lu, D., Xie, T., Saha, A., Sahoo, D., Yu, T., & Xiong, C. (2025). Aguviz: Unified pure vision agents for autonomous GUI interaction. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, PMLR 267, 69772–69805. <https://proceedings.mlr.press/v267/xu25ae.html>. arXiv:2412.04454. The University of Hong Kong; Salesforce Research.

97. Yang, J., Hou, B., Wei, W., Bao, Y., & Chang, S. (2026a). Ares: Adaptive reasoning effort selection for efficient LLM agents. arXiv:2603.07915v1. University of California, Santa Barbara; Accenture.

98. Yang, J., Jimenez, C. E., Wettig, A., Lieret, K., Yao, S., Narasimhan, K., & Press, O. (2024). SWE-agent: Agent-computer interfaces enable automated software engineering. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 50528–50652. <https://doi.org/10.52202/079017-1601>. arXiv:2405.15793v3. Princeton University.

99. Yang, K., Liu, Y., Chaudhary, S., Fakoor, R., Chaudhari, P., Karypis, G., & Rangwala, H. (2025). AgentOccam: A simple yet strong baseline for LLM-based web agents. *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.13825. University of Illinois Urbana-Champaign; Amazon.

100. Yang, Y., Jin, C., Zhao, J., Wu, J., Zhou, Y., Wang, Z., Wang, Z., Zhou, M., & Metaxas, D. N. (2026b). Act more, decide less: Skill-guided adaptive action chunking for long-horizon LLM agents (SPACE). *Findings of the Association for Computational Linguistics: EMNLP 2026* (to appear). arXiv:2609.02042v1. Rutgers University; University of Toronto; The Hong Kong Polytechnic University; Amazon; Microsoft.

101. Yao, J., Li, H., Liu, Y., Ray, S., Cheng, Y., Zhang, Q., Du, K., Lu, S., & Jiang, J. (2025). CacheBlend: Fast large language model serving for RAG with cached knowledge fusion. *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys 2025)*, 94–109. <https://doi.org/10.1145/3689031.3696098>. arXiv:2405.16444. University of Chicago; The Chinese University of Hong Kong, Shenzhen; Stanford University; Microsoft Research.

102. Ye, H., Gao, Z., Ma, M., Wang, Q., Fu, Y., Chung, M.-Y., Lin, Y., Liu, Z., Zhang, J., Zhuo, D., & Chen, Y. (2025). KVCOMM: Online cross-context KV-cache communication for efficient LLM-based multi-agent systems. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. <https://doi.org/10.52202/085713-0605>. arXiv:2510.12872. Duke University; Massachusetts Institute of Technology; NVIDIA.

103. Ye, N., Ahuja, A., Liargkovas, G., Lu, Y., Kaffes, K., & Peng, T. (2026). Speculative actions: A lossless framework for faster AI agents. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, oral. arXiv:2510.04371. Columbia University.

104. Yu, Z., Li, Q., Gao, Z., Xiang, L., Shi, C., Liu, Y., Wu, H., Wei, Y., Fei, Y., Fu, Y., & He, Z. (2026). Beyond sequential interaction: Benchmarking parallel execution and coordination for GUI agents (ParaGUIBench). arXiv:2607.22689v1. Beijing University of Posts and Telecommunications; BIGAI; China University of Geosciences (Beijing); Beijing Institute of Technology.

105. Zhai, Y., Shen, D., Luo, J., & Yang, B. (2026). ToolCaching: Towards efficient caching for LLM tool-calling. arXiv:2601.15335v1. Southeast University.

106. Zhang, C., Huang, H., Ni, C., Mu, J., Qin, S., He, S., ... Zhang, D. (2026a). UFO2: The desktop AgentOS. *Transactions on Machine Learning Research*. arXiv:2504.14603. Microsoft; ZJU-UIUC Institute; Nanjing University; Peking University.

107. Zhang, C., Xia, M., Zhang, X., Madrigal, D., Mallick, A., Kessler, S., Rühle, V., & Rajmohan, S. (2026b). Budget-aware agentic routing via boundary-guided training (BoPO). arXiv:2602.21227v1. University of Cambridge; Microsoft (M365 Research).

108. Zhang, J., Chen, K., Lu, Z., Zhou, E., Yu, Q., & Zhang, J. (2026c). Prune4Web: DOM tree pruning programming for web agent. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(41), 34710–34718. <https://doi.org/10.1609/aaai.v40i41.40772>. arXiv:2511.21398. Beihang University.

109. Zhang, L., Gao, L., & Xu, M. (2026d). Does chain-of-thought reasoning help mobile GUI agents? An empirical study. *Findings of the Association for Computational Linguistics: ACL 2026*, 7981–7996. <https://doi.org/10.18653/v1/2026.findings-acl.392>. Beijing University of Posts and Telecommunications; Tsinghua University.

110. Zhang, Q., Wornow, M., & Olukotun, K. (2025). Agentic plan caching: Test-time memory for fast and cost-efficient LLM agents. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. <https://doi.org/10.52202/085713-3451>. arXiv:2506.14852 (v2 adds a fourth author, G. Wan). Stanford University.

111. Zhao, Y., Shanmugham, A., Roy, S., & Xu, Y. (2026). EchoPath: Execution-level replayable memory for GUI agents. arXiv:2609.16635v1. Johns Hopkins University; Amazon AGI.

112. Zheng, L., Yin, L., Xie, Z., Sun, C., Huang, J., Yu, C. H., Cao, S., Kozyrakis, C., Stoica, I., Gonzalez, J. E., Barrett, C., & Sheng, Y. (2024). SGLang: Efficient execution of structured language model programs. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 62557–62583. <https://doi.org/10.52202/079017-2000>. arXiv:2312.07104. Stanford University; University of California, Berkeley; Shanghai Jiao Tong University; Texas A&M University; independent researcher.

113. Zhong, H., Faisal, F., França, L., Leesatapornwongsa, T., Szekeres, A., Rong, K., & Nath, S. (2026). ActionEngine: From reactive to programmatic GUI agents via state machine memory. arXiv:2602.20502v1 (24 February 2026; v2 of 28 September 2026 is retitled ‘...Programmatic web agents...’ and reports new results). Georgia Institute of Technology; Microsoft Research.

114. Zhou, Y., Dai, S., Wang, S., Zhou, K., Jia, Q., & Xu, J. (2025). GUI-G1: Understanding R1-Zero-like training for visual grounding in GUI agents. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. <https://doi.org/10.52202/085713-3201>. arXiv:2505.15810. Renmin University of China; Huawei Noah’s Ark Lab.

References · Part 2 (7 of 7)

continued

115. Zhu, K., Jacob, M., Ma, C., Pan, Y., Wang, S., Krishnamurthy, A., & Kasikci, B. (2026). TraceLab: Characterizing coding agent workloads for LLM serving. arXiv:2606.30560v2. University of Washington; Wuhan University of Technology; Shanghai Jiao Tong University. Dashboard: <https://tracelab.cs.washington.edu> (read 28 September 2026).

B0 · Source formulas for overlap: how much concurrency can save

verbatim from each source, in its own symbols; the right-hand column maps them onto Part 1 · behind pages 9–10

SOURCE, LOCATION	THE FORMULA IN THE SOURCE’S OWN SYMBOLS	ITS SYMBOLS	WHAT IT BOUNDS, IN PART 1’S TERMS
AsyncFC, §4 Eq. 1 (= App. B Eq. 2) (Feng et al., 2026, arXiv · UC Berkeley)	$R = \frac{T_{\text{LLM}} + T_{\text{tool}}}{\max(T_{\text{LLM}}, T_{\text{cp}})}$	T_{LLM} total decoding time; T_{tool} total function time; T_{cp} function time on the critical path of the dependency graph	the largest speed-up that T_{saving} can give; decoding sits in D_i , functions in E_i ; its R is not the success probability $R_m(p)$ (page 9 writes “speed-up”)
LLMCompiler, App. E.1 (Kim et al., 2024)	$T^R = \sum_{i=1}^N (T_P^R(P_i) + T_E(E_i))$ $T^C = \sum_{i=1}^N T_P^C(P_i) + \max_k T_E(E_k)$ $\gamma = T^R / T^C$	N planned tasks (not our steps); P_i planning and E_i execution of task i (not our E_i); R: ReAct, C: compiled	serial against parallel execution; $\gamma_{\text{max}} \approx N$ when execution dominates and all calls take equally long, ≈ 1 when planning dominates
Speculative tool calls, Eq. 2 and Lemma 1 (Nichols et al., 2025, arXiv · LLNL)	$S_{\text{spec}} = \frac{G + T}{\alpha \max\{G, g + T\} + (1 - \alpha)(G + T)}$ $S_{\text{max}} = \frac{G + T}{\max\{G, g + T\}} < 2$	G mean generation time of the main model; g of the speculator; T mean tool time; α acceptance rate	one model call overlapped with one tool call: under 2× per step, whatever the guesser
SPORK, EQ1 and App. A (Bai et al., 2026, arXiv · Tsinghua)	$\text{Ratio} = \frac{T_{\text{base}}}{T_{\text{base}}^* - \alpha t_{\text{overlap}} + T_{\text{oh}}}$ $S_{\text{max}} = \frac{1}{1 - f_{\text{tool}}}$	$T_{\text{base}} = T_{\text{dec}} + T_{\text{tool}}$; $T_{\text{base}}^* = T_{\text{base}} - T_{D3}$, less the decoding saved by its third design (D3: a rejected probe’s prefix reused as draft tokens); α accepted share of tool-call turns; t_{overlap} overlap per accepted turn; T_{oh} probe overhead; $f_{\text{tool}} = T_{\text{tool}} / T_{\text{base}}$	break-even: $\alpha t_{\text{overlap}} \geq T_{\text{oh}}$ (without its third design); the ceiling is Amdahl’s law on the tool share
ParaGUI, §III-C (Yu et al., 2026, arXiv · BUPT)	$L_{\text{parallel}} = \sum_{r=1}^R \max_k s_{r,k}$ $S = L_{\text{serial}} / L_{\text{parallel}}, \quad P = L_{\text{total}} / L_{\text{parallel}}$	$s_{r,k}$ GUI steps of worker k in round r; R rounds (not $R_m(p)$); L_{serial} steps of a single-agent serial run; L_{total} all worker steps	steps on the critical path against total work: P = 2.13 means 2.13 desktops busy on average — machine hours x_{env} for time

B0 · Source formulas for speculation, cache retention and a reader call

verbatim from each source, in its own symbols · behind pages 5, 6 and 10

SOURCE, LOCATION	THE FORMULA IN THE SOURCE’S OWN SYMBOLS	ITS SYMBOLS	WHAT IT BOUNDS, IN PART 1’S TERMS
Speculative Actions, Prop. 1 and Thm 3 (Ye et al., 2026)	$\frac{\mathbb{E}[T_s]}{\mathbb{E}[T_{\text{seq}}]} \rightarrow 1 - \frac{p(k)}{1 + p(k)} \cdot \frac{\alpha}{\alpha + \beta}$ $\frac{\mathbb{E}[M_{\text{spec}} - M_{\text{seq}}]}{\mathbb{E}[M_{\text{seq}}]} \rightarrow \tilde{k} - (\tilde{k} + \frac{\alpha}{\alpha + \beta}) \frac{p(k)}{1 + p(k)}$	$p(k) = 1 - (1 - p)^k$, k guesses of accuracy p; speculator and API latencies exponential with rates α, β ; T_s, T_{seq} time with and without speculation; $\rightarrow$ the limit as the number of steps grows; $\tilde{k}$ distinct guessed actions; M money	time saved and money added from the same guess accuracy; in the limit the time ratio stays above 1/2 (calc.), so at most 50% is saved — under its exponential-latency model
Speculate with memory, §2.1 and §4.3 (Li et al., 2026c, arXiv · Salesforce Research)	saving per hit $= \min(\ell_{\text{env}}, \ell_{\text{LLM}} - \ell_{\text{spec}})$ (guess the action) or $\min(\ell_{\text{LLM}}, \ell_{\text{env}} - \ell_{\text{spec}})$ (guess the observation); extra cost $= k C_S + (1 - \text{acc}) C_{\text{act}}$	ℓ latencies of the environment, the model and the speculator; k speculators of cost C_S ; acc hit rate; C_{act} cost of the pre-launched actor call a miss wastes (the extra-cost form holds for observation and chained guesses)	a hit can hide at most the idle window of the other side; its App. A uses slightly different per-type forms
Continuum, Eqs. 1–2 (Li et al., 2026a, arXiv · UC Berkeley)	$\tau^* = \arg \max_{\tau} \mathcal{P}(\tau, f) \text{Benefit}(r) - \text{Cost}(\tau, r)$	$\mathcal{P}(\tau, f)$ empirical probability that tool f returns within τ ; Benefit: reload and reordering cost avoided; Cost: memory held for τ	how long to keep a run’s cache through a tool wait: t^{queue} and re-prefill against GPU memory
InferCept, Eqs. 1–5 (Abhyankar et al., 2024)	$\text{Waste}_{\text{preserve}} = T_{\text{INT}} C_i^j M$; also discard, swap and chunked discard; $\text{Waste} = \min(\text{Waste}_{\text{preserve}}, \text{Waste}_{\text{chunkD}})$ (Eq. 5)	T_{INT} duration of the tool call; C context tokens; M memory per token; i request, j interception (not Part 1’s step and call)	the same trade inside the server: keep the cache during E_i , or pay t^{prefill} again
FocusAgent, App. H.1 (Kerboua et al., 2026)	$C_S o_i + C_L o_r \leq C_L o_i \Leftrightarrow \alpha \leq \frac{C_L - C_S}{C_L}$	C_S, C_L prices of the small reader and the large actor; $ o_i , o_r $ observation before and after reduction; $\alpha = o_r / o_i $	when a reader call pays in money; the paper notes it ignores API latency — the time can still grow (page 6)

B0 · Source formulas for money and set-up, and one closer source for Part 1

verbatim from each source · behind pages 12 and 14 · the last row is a finding for Part 1’s Appendix A0

SOURCE, LOCATION	THE FORMULA IN THE SOURCE’S OWN SYMBOLS	ITS SYMBOLS	WHAT IT BOUNDS, IN PART 1’S TERMS
TokenPilot, App. A.2, Eq. 11 (Xu et al., 2026a)	$\text{Cost} = C'_{\text{hit}} p_{\text{hit}} + C'_{\text{miss}} p_{\text{miss}} + H_{\text{out}} p_{\text{out}}$	$ C'_{\text{hit}} , C'_{\text{miss}} $ cached and uncached input tokens; H_{out} generated tokens; p prices	Part 1’s money sum over three classes; no cache-write term
SpeedRunner, App. A.3 (Huang et al., 2026c, arXiv · JHU)	$\text{cost} = p_{\text{in}} N_{\text{in}}^{\text{uncached}} + p_{\text{cache}} N_{\text{in}}^{\text{cached}} + p_{\text{out}} N_{\text{out}}$	N summed over all calls of an episode, the skill-inducer calls amortised over its rollouts	the same three classes; set-up calls spread over the runs they serve
LATM, Table 2 (Cai et al., 2024)	$O(nc + C)$ against $O(nC)$	n samples; C one call to the large tool-making model; c one call to the small tool-using model	one expensive set-up amortised over n cheap runs — the form of page 14’s C_{setup}/n
AI Agents That Matter, §3 (Kapoor et al., 2025)	in words: total cost = a fixed cost (one-time optimisation) + a variable cost (per run, from input and output tokens)	—	page 14’s $v_n(m, p)$ adds this fixed cost to Cost-of-Pass’s per-success cost (adapted)
Speculative tool calls, Eq. 4 (Nichols et al., 2025, arXiv · LLNL)	$T_{\text{vanilla}} = 2Ko + \phi \sum_{i=1}^K X_i + \delta \sum_i (R_i + t_i) + \sum_i T_i, \quad X_{i+1} = X_i + t_i + t_{o,i}$	K turns; o per-call overhead; ϕ prefill and δ decode seconds per token; X_i prompt tokens of turn i; R_i, t_i reasoning and tool-call tokens; $t_{o,i}$ tool output; T_i tool time	a published form of Part 1’s time sum with its growing prompt, found in these checks: per-call overhead, prefill of a prompt that grows each turn, decoding, tool wait. Part 1 does not cite it yet (open question)

B1 · Compile or replay: the research systems, with full conditions

behind page 3 · term $J_i \rightarrow 0$ on compiled or replayed steps · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
JIT-Planner (Winston et al., 2026)	Candidate code plans over cached site tools with pre/postcondition checks, drafted by parallel planning calls; a separate scheduler (not in the headline) can hedge by running up to 4 copies of the chosen plan	Wall-clock from submission to completion, planning included: 150.1 $\rightarrow$ 15.4 s (9.7 $\times$, its Table 1), success 61 $\rightarrow$ 90%. 37 tasks in 5 apps (18 written by the authors), 3 runs, GPT-4.1; Browser-Use v0.7.10 on the same model. Three-model average 10.4 $\times$	Tool synthesis 25–90 min + traces 25–45 min per app; no tokens or \$ reported
ActionEngine (Zhong et al., 2026, arXiv · Georgia Tech)	Offline state-machine graph of the site; one sketch-program call compiled by graph search	106 WebArena Reddit tasks: success 66 $\rightarrow$ 95%, end-to-end latency 237 $\rightarrow$ 118 s, model calls 10.2 $\rightarrow$ 1.8, input tokens 62.3k $\rightarrow$ 8.1k, \$0.71 $\rightarrow$ \$0.06 per task (tokens $\times$ list price). AgentOccam on GPT-4-Turbo vs ActionEngine on Claude Sonnet 4.5	About 2.5 $\times$ of the 11.8 $\times$ \$ gap is the cheaper model's price (calc.); crawl cost not reported
AutoDroid-V2 (Wen et al., 2025)	A local model writes a whole-task script from an offline app document	Model inference time 669.2 $\rightarrow$ 46.3 s per task (prompt to last token, screen actions excluded); uncached input 3,021 $\rightarrow$ 68 tokens; success 43.9 $\rightarrow$ 54.4%. 158 DroidTask tasks, fine-tuned Llama-3.1-8B on a phone, both sides	Offline GPT-4o \$4.11 + \$7.83 + \$70.48 = \$82.42 per app (calc.), ~2.5 GPU-hours of fine-tuning
EchoPath (Zhao et al., 2026, arXiv · JHU)	Validated trajectories kept as executable memories; a gate checks the start state	Second pass, medians: tokens 586,386 $\rightarrow$ 20,370 (–96.5%), time 315.7 $\rightarrow$ 127.5 s (–59.6%), success 91.8 $\rightarrow$ 91.2% (calc. for the %). OSWorld-Verified, 159 replayable tasks, Codex GPT-5.5, vs Synapse; the second pass changes the resolution	First pass $\approx$ 572k tokens, $\approx$ 4.5 min per task; the gate accepted 22% of incompatible starts (calc.)
MobileGPT (Lee et al., 2024)	App memory of learned sub-tasks; recall with slot filling, the model only for what memory lacks	Warm start (same task, new parameters) vs a GPT-4-Turbo baseline with the same prompts that derives every action: latency –62.5%, cost –68.8%; 70.6% (Gmail) to 98.6% (Telegram) of actions from memory. 80 tasks (160 instructions), 8 Android apps	Offline exploration 10–15 min per app, \$10.78 in all; slot filling on GPT-3.5 Turbo, so part of the cost cut is price
AppAgentX (Jiang et al., 2025, arXiv · Westlake U.)	Frequent action sequences evolved into shortcut actions	AndroidWorld, 116 tasks, GPT-4o: task time 147.17 $\rightarrow$ 59.74 s, tokens 18.9k $\rightarrow$ 6.2k, success 41.7 $\rightarrow$ 62.5%; time compared only on tasks every method completed	Evolution runs before the test
AutoTool (Jia & Li, 2026)	Repeated tool calls issued from a usage graph without a model call, capped at 30% of operations	Model calls 24.1 $\rightarrow$ 20.4 (ALFWorld), 23.3 $\rightarrow$ 17.8 (ScienceWorld); progress rate 0.394 $\rightarrow$ 0.531 and 0.716 $\rightarrow$ 0.708 — Llama4-Scout-17B	No wall-clock; calls used as the proxy
EAM (Qin et al., 2026)	Executable memory of sub-tasks for mobile agents	Per step 9.3 $\rightarrow$ 2.8 s and 50.8K $\rightarrow$ 8.3K tokens vs GPT-4o (benchmark not named); AndroidWorld success 34.5 $\rightarrow$ 52.6%, 3 runs	Per step, not per task
AgentReuse (Li et al., 2024)	Reuses stored plans for semantically similar requests	–93.12% latency is computed from an assumed 31.8 s per plan, not measured end to end	Modelled estimate; not used on the main page

B1 · Compile or replay in products: replay first, a model only to repair

behind page 3 · vendor documentation and pages, read 28 September 2026 · vendor figures are labelled as such; marketing is not evidence

PRODUCT	WHAT REPLAYS, AND WHEN A MODEL IS CALLED	PUBLISHED NUMBERS	WHAT IS MISSING
UiPath Healing Agent (UiPath, 2026)	Recorded automation; a model recommends or applies a fix when a step fails	Price only: one heal = 3 Platform Units; failed heals free; healing is not kept for the next run	Heal success rate, heal latency
Power Automate self-healing (Microsoft, 2026b)	Desktop flows repaired with Copilot	None; free on premium accounts	Any measured rate
Automation Anywhere Generative Recorder (Automation Anywhere, 2026)	Recorded bot with a generative fallback	“Nearly 50%” of failures fixed in customer previews: a vendor statement with no sample, period or definition (the product page’s “over 60%” is marketing, not cited)	A defined, sampled rate
Stagehand cache (Browserbase, 2026a)	An action’s model output is cached; the second run of the same action makes no call	None usable: the blog’s “as high as ~80%” (one action run twice) is vendor marketing, not cited	Whole-workflow numbers, sample, model
Skyvern code cache (Skyvern, 2026)	Generated code re-run without a model	None (“faster, cheaper”); plans imply about \$0.024 per action (calc.)	Any measured rate
HyperAgent action cache (Hyperbrowser, 2026)	Replay by XPath; the model only when the page has changed	None; the docs concede that a fallback adds latency and cost	Replay and fallback rates

Across all six: the model call is replaced by replay (**J_i** → **0**) and the environment actions stay (**E_i unchanged**); no vendor publishes a replay or repair success rate with a defined sample.

B2 · Fewer steps: more rows, with full conditions

behind page 4 · term N ↓ · steps are counted as each paper counts them · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
UF02, GUI + API (Zhang et al., 2026a)	Native API calls replace GUI sequences in Office apps	Model-involved steps 16.0 → 6.6 with o1, but 13.8 → 12.9 with GPT-4o, on tasks both settings completed; 12 hand-built APIs	No wall-clock
CUA-Verse (Shi et al., 2026b, arXiv · SJTU)	A command-line tool added to a fine-tuned 9B agent	OSWorld subset of 244 tasks, same model: success 23.4 → 40.2%, mean steps 39.6 → 28.6, tokens 325.7K → 286.5K	Training on 4,923 episodes
SPACE (Yang et al., 2026b)	RL-trained variable-length action chunks	ALFWorld unseen split, Qwen3-4B: model rounds 20.9 → 4.4, success 81.3 → 96.9% vs the best baseline (multi-action GRPO)	Text environments only; no time or tokens
SpeedRunner (Huang et al., 2026c, arXiv · JHU)	A coding agent refactors the skill library for cost	Crafter: model calls −94% over training; Gemini-3-Flash, 3 seeds	Text game; with Gemini-3-Flash success fell significantly below the OPO baseline on every benchmark
AWM (Wang et al., 2025c)	Induced workflows injected into the prompt	WebArena 812 tasks: success 15.0 → 35.5%, steps 7.9 → 5.9 vs the page-tree baseline; +10.8% computation for induction (its App. D). In ASI’s re-run: 5.9 steps vs 5.6 plain	Prompt grows
Beyond Browsing (Song et al., 2025)	API calls and API documentation in context, with browsing as fallback	WebArena, GPT-4o: success 14.8 → 38.9% (hybrid), steps 8.4 → 8.5, \$0.1 → \$1.4 per task	API documents lengthen the prompt
ComputerRL (Lai et al., 2026)	An API-GUI action space	OSWorld, prompted GPT-4o: success 11.2 → 26.2%	No step, time or \$ figures
CoAct-1 (Song et al., 2026)	An orchestrator hands sub-tasks to a coding agent or a GUI agent	OSWorld-Verified: success 53.07 → 59.93% vs GTA-1-7B with o3; 10.15 vs 15.22 steps per passed task	Step budget behind 10.15 ambiguous
OSWorld-Human (Abhyankar, Qi & Zhang, 2026)	Human reference trajectories, grouped by screenshot	Steps per trajectory, single → grouped: Chrome 5.8 → 4.3 (1.35×) to Calc 13.2 → 4.5 (2.93×) (calc.); 369 tasks	A human-reference headroom estimate, not a method or a strict bound

B2 · Capping the steps: stopping rules and budgets

behind page 4 · term N capped · a stop can end attempts that would have succeeded (R_m(p) ↓) · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
EET (Guo et al., 2026)	Stops a run early when past experience predicts no gain	SWE-bench Verified, GPT-5-mini, Agentless: resolved 33.2 → 41.0%, benchmark cost \$13.77 → \$6.18, API calls −26.4%; six-configuration average cost −31.8%	Offline experience base; one configuration −0.2 points
Runaway is ashamed (Lu et al., 2025b)	A verifier decides when to exit	ALFWorld 134 tasks, Llama-3.1-70B: steps 19.0 → 13.4, success 76.1 → 70.2%	Success falls
Budget Tracker (Liu et al., 2026a)	Tells the agent its remaining tool budget	BrowseComp, Gemini-2.5-Pro: 12.8% at budget 10 vs 12.6% at 100; cost per question 9.9¢ → 6.8¢ (−31.3%)	Tool calls priced at a flat \$0.001
Efficient Agents (Wang et al., 2025a, arXiv · OPPO)	A cheaper agent configuration	GAIA, 165 tasks (calc.): \$0.398 → \$0.285 per task, 53.33 → 51.52%; one run	Baseline configuration not stated
TTI (counterpoint) (Shen et al., 2025)	Trains the agent to take more steps (interaction scaling)	WebVoyager 427 tasks: 55.8 → 64.8% (zero-shot → TTI); WebArena 812: 18.3 → 26.1%; Gemma 3 12B. Longer thinking at equal compute gave under 3%	N grows by design; no cost figures

B3 · Agent-aware serving: more systems

behind page 5 · terms t^queue and t^prefill · self-hosted engines; all figures are throughput or latency under load, none reports task quality unless stated

SYSTEM	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	BASELINE
SGLang (Zheng et al., 2024)	Prefix sharing in a radix tree of KV caches	Up to 6.4× throughput, 3.7× lower latency; Llama-7B on one A10G; 12 workloads incl. ReAct and generative agents (replayed)	vLLM v0.2.5, Guidance, LMQL (which gives the maximum is not stated)
Preble (Srivatsa et al., 2025)	Prompt-sharing-aware scheduling across GPUs	1.5–14.5× lower mean and 2–10× lower p99 latency; tool use, embodied agent, program generation, video and long-document QA; Mistral 7B, Llama-3 70B	Data-parallel SGLang
Parrot (Lin et al., 2024)	Semantic variables expose a program’s structure to the server	Up to 11.7× lower end-to-end latency; MetaGPT multi-agent programming, LLaMA 13B, one A100	Latency-centric vLLM
KVFlow (Pan et al., 2025)	Workflow-aware cache eviction and prefetching	1.24× vs SGLang with hierarchical cache (camera-ready, Qwen2.5-32B, one H100); the 1.83× headline is from the v1 A10G set-up only	SGLang HiCache
Helium (Wadlom et al., 2026)	Workflow-level optimisation of batched agent queries	Up to 1.56× vs KVFlow (end-to-end latency of a batch, optimisation included)	KVFlow
KVCOMM (Ye et al., 2025)	Reuses KV caches across agents with offset correction	First-token time 7.82× lower at the fifth agent (428.6 → 54.8 ms); Llama-3.1-8B, one H100, HuggingFace	Full prefill
DroidSpeak (Liu et al., 2026b)	Shares KV caches between fine-tuned variants of one model	Prefill 1.7–3.1× faster (average 2.1×); eight model pairs, three datasets each	Full prefill
AsymCache (Shi et al., 2026a, arXiv · Peking U.)	Asymmetric cache eviction for multi-session serving	With Continuum, average job latency 4.4–18.1% below Continuum alone; alone 0.4–4.2% below vLLM-LRU (BFCL web search, no task count); its own headline ranges are inconsistent	Continuum; vLLM-LRU
Continuum (Li et al., 2026a, arXiv · UC Berkeley)	Keeps a run’s KV cache for a predicted tool-wait time	Job delay 1.12–3.66× lower — replayed SWE-bench, BFCL and OpenHands traces; Llama-3.1 8B/70B, Gemma-3 12B; no quality metric; GPU memory held during waits	not named for the range (evaluated against vLLM, LMCache, Autellix, InferCept)
CacheBlend (Yao et al., 2025)	Blends precomputed chunk caches, recomputing a few tokens	First-token time 2.2–3.3× lower, quality within 0.02 F1 — RAG, not agents	Full recompute

B4 · Prefill and context: more rows, with full conditions

behind pages 6 and 11 · terms |o_{a,i}|, g̃ and t^prefill · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
FocusAgent, other settings (Kerboua et al., 2026)	Reader model GPT-5-mini instead of GPT-4.1-mini	WorkArena L1: input cost −31.5% (calc.) at 53.2% success; WebArena 381 tasks: 36.5 → 39.6%, −21.7%. Input tokens only are priced	No latency for this reader
LineRetriever (Kerboua et al., 2025, arXiv · INSA Lyon)	A reader model keeps whole lines of the page tree	Observation −61% on WorkArena L1 (330 runs), success 52.7 → 44.8%	Success falls
Prune4Web (Zhang et al., 2026c)	A model writes a scoring program; the top 20 elements go to the grounder	25–50× fewer candidate elements; grounding accuracy 46.8 → 88.28% with ground-truth sub-tasks (1,101 steps)	Grounding, not task success; no latency measured
SimpAgent (Chen et al., 2025)	Masks irrelevant screenshot regions and compresses history	FLOPs per step 11.90 → 8.71 T; offline step success 69.0 → 71.3% (AITW), Qwen2-VL-2B	Offline steps
ScreenSeeker (counter) (Li et al., 2025)	A planner searches the screen recursively for the target	Grounding 18.9 → 48.1% on 1,581 ScreenSpot-Pro instructions, GPT-4o planner	More model calls per step
Observation masking, other models (Lindenbauer et al., 2025)	As on page 11	Gemini 2.5 Flash \$0.41 → \$0.18, 32.8 → 35.6%; with thinking 40.4 → 36.4% (significant); Qwen3-32B \$1.12 → \$0.55, 17.0 → 15.0%	Success can fall
AgentDiet, per task (Xiao et al., 2026)	As on page 11	\$0.535 → \$0.422 (Claude, SWE-bench Verified), \$0.701 → \$0.449 (Gemini, Multi-SWE-bench Flash); cached-input discount applied	Reducer 5.5–11.8% of the original cost
FireAct (Chen et al., 2023, arXiv · System2 Research)	A fine-tuned GPT-3.5 drops the few-shot examples from every prompt	HotpotQA 500 questions: 9.0 → 2.7 s per trial, \$2.6e-3 → \$2.2e-3, EM 31.4 → 39.2	Price per token ×8 (c_κ(μ) ↑)
TokenPilot, mixed sessions (Xu et al., 2026a)	As on page 12	123 tasks shuffled into 10 sessions: \$21.16 → \$4.83 (−77%, calc.), score 68.77 → 69.10	Partial-credit scores

B4 · Decoding: more rows, with full conditions

behind page 7 · terms n^out and TPOT · vendor = the vendor’s own statement

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
Think Twice, Click Once (Tang et al., 2025, arXiv · Zhejiang U.)	Switches between a fast and a slow grounding mode	Time per sample 3.2 → 2.6 s, accuracy 74.8 → 77.4%; slow mode only 5.4 s — 1,272 samples, Qwen2-VL-2B, same model with and without its switch threshold	Grounding only
Does CoT help mobile agents? (Zhang et al., 2026d)	Caps the reasoning tokens	AndroidControl, GLM-4.6V: action accuracy 71.55% at 128 tokens vs 70.41% unlimited and 70.32% without reasoning	Sample size not stated
TSDS (Farzaneh & Simeone, 2026, arXiv · Northeastern U. London)	Short edge thinking, deferral to the cloud when unsure	MBPP (257 problems, single step): edge thinking tokens 1,183 → 422, reward 0.719 → 0.714	Cloud deferral rate 0.346; cloud tokens not counted
GUI-KV (Huang et al., 2026a)	Keeps 40% of the KV cache per layer (5 screenshots in context, UI-TARS-1.5-7B)	Decoding MFLOPs per token –38.9%, offline step accuracy 17.5 → 21.6% (AgentNetBench); OSWorld 26.0 → 25.1%	No wall-clock
ToolSpec (Xia et al., 2026, arXiv · PolyU)	Schema-aware drafts of tool-call tokens, verified losslessly	Tool-call generation 3.5–4.2× faster — API-Bank, ToolAlpaca, BFCLv2; batch 1; 3–14B open models on GPUs	Decoding speed, not task time
Effort guidance (vendor) (Anthropic, 2026a)	Medium instead of high reasoning effort	“Roughly half the output tokens of high” with close to the best success, and the same success once retries are allowed — internal UI-automation suite, no counts	Vendor statement
Ultrafast (vendor) (OpenAI, 2026; Cerebras, 2026)	GPT-5.6 Sol on Cerebras hardware	Up to 14× faster than Standard, up to 750 output tokens per second; limited preview, no price	OpenAI vendor statements; a hardware partner’s 6-task chart is marketing, not cited

B5 · Environment and parallel calls: more rows, with full conditions

behind pages 8–9 · terms E_i and T_saving · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
Skim, details (Wong et al., 2026, arXiv · Princeton)	As on page 8; a local Qwen2.5-14B runs the fast path and the verifier	Accuracy vs the default agent: 40.6 vs 37.6 (WebVoyager agent), 52.0 vs 49.6 (AgentOccam), 45.6 vs 45.0 (BrowserUse); verifier precision 82.0%, recall 86.2%	Site profiling 6–24 s per site
ParaGUI (Yu et al., 2026, arXiv · BUPT)	A planner sends sub-tasks to up to 5 GUI workers on separate desktops	233 tasks: critical-path steps 38.7 vs 75.9, success 46.4 vs 33.5% vs Claude Sonnet 4.6; vs its own worker model 38.7 vs 36.7 steps	Desktops × workers; no wall-clock
AsyncLM (Gim et al., 2024, arXiv · Yale)	Asynchronous function calls with interrupts	Local fine-tuned Llama-3.2 on one RTX 4090: 1.6–2.4× lower latency than sequential calls, 1.23–1.5× than parallel calls (calc.; first to last generated token); cloud figures emulated	Emulated for GPT-4o
ThunderAgent (Kang et al., 2026)	Prepares the sandbox while the program waits for GPU memory (overlap, T_saving)	Environment time 4.8 → 0.3 s — OpenHands rollouts, self-hosted (its Fig. 6a)	Throughput system; no task success
TCclone · DeltaBox (Huang et al., 2026b, arXiv · UC San Diego; Dong et al., 2026, arXiv · SJTU)	Fast container clone; millisecond checkpoint and rollback	Clone for one OSWorld branch point 18.2 → 3.7 s vs KVM; rollback 1.86 ms on the fast path (9.29 ms slow path)	Cost of branching E_i
IdleSpec (counter) (Choi et al., 2026, arXiv · KAIST)	Drafts plans while tools run	GAIA 165 + FRAMES 50, Gemini-2.5-Flash: accuracy 50.5 → 55.6%, wall-clock about unchanged; output tokens +58% (calc., Qwen3.5-4B)	Tool-wait overlap spent on accuracy (R_m(p) ↑), not on time

B5 · Speculation and its safety: more rows, with full conditions

behind page 10 · term T_saving by guessing · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
Speculative tool calls (Nichols et al., 2025, arXiv · LLNL)	Client- and engine-side speculation of tool calls	6–21% of time saved with synthetic tool latencies and cached tool outputs; BFCL, 32 tasks per agent	Synthetic latencies
SPORK (Bai et al., 2026, arXiv · Tsinghua)	Probe calls from a fork of the model’s own cache	GAIA 165 tasks, Qwen3-32B: p50 34.7 → 31.2 s (–10%), p95 131.9 → 108.1 s (–18%), exact match within 1 point; baseline already speculative-decoding	Probe calls, wasted tool runs
Speculate with memory (Li et al., 2026c, arXiv · Salesforce Research)	Guesses from a memory of past runs	Analytical peak 1.128×; HotpotQA 1.05× — replay estimates	Not end-to-end
AgenticCache (Kim et al., 2026)	Cached plans for embodied agents	TDW-COOK, 18 episodes: 12.86 → 1.75 h, \$21.0 → \$4.4, success 94.44 → 100%; TDW-MAT 1.86× (calc.)	Embodied simulator
WebOperator (Dihan et al., 2025, arXiv · BUET)	A check before executing actions flagged as destructive	About 37% of pre-flagged actions were truly destructive — WebArena, gpt-4o	arXiv; rejected at two venues

B6 · Prices by billing class and tier, per vendor

behind pages 12–13 · vendor price pages and documentation, read 28 September 2026 · ratios are our arithmetic (calc.)

VENDOR, LATEST MODEL	UNCACHED → CACHE HIT (PER 1M INPUT TOKENS)	CACHE WRITE, STORAGE, LIFETIME	TIERS
DeepSeek-V4.1-Flash (DeepSeek, 2026)	\$0.30 → \$0.006 (0.02×), peak; off-peak half	none listed	—
Anthropic, Claude Opus 5.5 (Anthropic, 2026b)	\$4 → \$0.20 (0.05×); Fable 5.1 and Mythos 5.1 0.025×; other Claude models 0.1×	1.25× for 5 min, 2× for 1 h; refreshed free on each use; a change of speed, tools, thinking or effort invalidates parts of the cache	Fast mode 2× (\$8 / \$40), up to 2.5× output speed
OpenAI, GPT-6 Sol / Luna (OpenAI, 2026)	\$2 → \$0.20; \$0.10 → \$0.01 (0.1×)	1.25× from GPT-5.6 on; at least 30 min	Fast 2×; Flex and Batch 0.5×
Google, Gemini 3.8 Flash (Google, 2026)	\$0.75 → \$0.075 (0.1×), to 31 Dec 2026	storage \$0.50 per 1M tokens per hour	Priority 1.8× on every listed price (text says 75–100%), Preview
xAI, Grok 4.7 (xAI, 2026)	\$2 → \$0.50 (0.25×)	none listed	—
Browserbase (Browserbase, 2026b)	—	—	Cloud browser \$0.10–0.12 per hour above the plan allowance (smaller vendor)
Microsoft Foundry (Microsoft, 2026a)	—	—	Priority processing 1.75–2.5× (calc., from the pricing page’s HTML)

Worked example (OpenAI, GPT-5.6 and later): one write and one read cost 1.25 + 0.1 = 1.35× against 2× uncached; one write and nine reads cost 2.15× against 10× (calc.).

B7 · A cheaper model per call: more rows, with full conditions

behind page 13 · term $c_k(\mu_{ij})$ · calc. = our arithmetic

WORK	MECHANISM	MEASURED EFFECT, WITH ITS CONDITIONS	LEFT OUT, OR GROWS
StepWise, web (Wei et al., 2026)	gpt-oss-20b by default, GPT-5.2 when stuck	WebArena-Verified (~812 tasks, calc.): \$0.335 → \$0.211 per task, 60.1 → 57.8%, time per request 19.6 → 12.2 s; local 2×H100 timing	Large model on 66.9% of steps
BoPO (Zhang et al., 2026b, arXiv · Cambridge)	A router under a cap on large-model calls	AppWorld 168 tasks, GPT-4.1 / GPT-4.1 mini, 3 seeds: large calls per task 18.75 → 13.2 (−29.6%, calc.), success 66.7 → 66.5%	Router cost excluded
Agentic plan caching, per workload (Zhang et al., 2025)	As on page 13	FinanceBench \$4.03 → \$1.86, 91.0 → 85.5%; QASPER \$2.14 → \$0.78; GAIA \$69.02 → \$16.27, 37.58 → 36.97%	GAIA task count not stated
RouteLLM (Ong et al., 2025)	A learned router between a strong and a weak model	3.66× fewer GPT-4 calls than a random router at equal quality (MT-Bench); 1.41× (MMLU), 1.49× (GSM8K)	Single-turn; not against always-GPT-4
FrugalGPT (Chen, Zaharia & Zou, 2024)	A cascade of models with a stopping scorer	Up to 98% lower cost at equal accuracy on one single-query dataset (HEADLINES)	Single-turn

B8 · Cost per success: the inputs, the formula and the rounding

behind page 14 · v = cost per attempt ÷ success rate (Cost-of-Pass, Eq. 2) · all results are calc. unless stated

WORK	COST PER ATTEMPT (AS PRINTED)	SUCCESS RATE	COST PER SUCCESS (CALC.)	NOTE
AXIS vs UFO (Lu et al., 2025a)	\$0.4 → \$0.2	52.0 → 84.0%	\$0.77 → \$0.24	costs printed to one decimal
W&D (Lin et al., 2026)	\$1.025 → \$0.657 (\$102.5, \$65.7 per 100)	66 → 68%	\$1.55 → \$0.97	includes search and scraping fees
OSWorld 2.0, Opus 4.8 (XLANG Lab, 2026, arXiv · HKU)	~\$76.1 → ~\$72.4	18.5 → 20.6%	\$411 → \$351	Opus 4.7: \$258 → \$185
Beyond Browsing (Song et al., 2025)	\$0.1 · \$1.2 · \$1.4	14.8 · 29.2 · 38.9%	\$0.68 · \$4.11 · \$3.60	browsing · API · hybrid; one-decimal costs, so the ratio is 3–11×
BATS vs ReAct@100 (Liu et al., 2026a)	9.9¢ → \$1.1	12.6 → 24.6%	\$0.79 → \$4.47	BrowseComp, Gemini-2.5-Pro
ActionEngine (Zhong et al., 2026, arXiv · Georgia Tech)	\$0.71 → \$0.06	66 → 95%	\$1.08 → \$0.06	different models on the two sides
StepWise, OSWorld (Wei et al., 2026)	\$0.881 → \$0.224	58.1 → 55.4%	\$1.52 → \$0.40	
WebRouter (Li et al., 2026b)	\$0.98 → \$0.12	86.1 → 82.3%	\$1.14 → \$0.15	
Fara-7B vs computer-use-preview (Awadallah et al., 2025, arXiv · Microsoft Research)	\$0.913 → \$0.025	70.9 → 73.5%	\$1.29 → \$0.03	WebVoyager
AI Agents That Matter (Kapoor et al., 2025)	LATS \$134.50 vs warming \$2.45 (164-problem run totals)	88.0 vs 93.2%	stated totals, not per success	HumanEval, GPT-4

With a set-up cost (page 14): $v_n = C_{\text{setup}}/n + v$. Two retained works print both parts. EET’s experience base (Agentless, GPT-5-mini) cost \$4.3 once, against a benchmark run of \$13.77 → \$6.18 (500 tasks), so it pays off within the first run: \$10.48 with the set-up charged to it (calc.). SpeedRunner already spreads its skill-inducer calls over the runs they serve. The page-3 systems print their set-up in minutes, tokens or GPT-4o dollars, never next to a per-task dollar figure.

B9 · Works held back or dropped, and why

the source-credibility rule of slides/references.md · (iii) = numbers from a public benchmark with the task count and model stated

STATUS	WORKS	REASON
Held back: condition (iii) not met as written, pending Edwin’s decision	SpecHop (arXiv:2605.21965), PASTE (2603.18897), AOSpec (2608.00881), TAB, CacheScout (2608.14624), DualPath, PBKV, AgentKVShift, SpecBox (2607.23933), LLM-Tool Compiler, Cordon (2606.17573), GoClick, BAVT; DualSpec’s XBench row; Continuum’s real-run 8.18×	No task count, internal traces, or an unnamed model; their numbers stay off the pages
Shown with a flag, pending the same decision	Skim (“300+” tasks, page 8); ARES (~129 WebArena tasks, calc.), Overthinking (SWE-bench Verified), StepWise (OSWorld), Fara-7B (WebVoyager): task counts not printed; the author-built test suites of Atomix (leak test), Safe to Resume? (96 workflows) and Ghost Tool Calls (30 tasks)	Counts derived by us, implied by the benchmark, or from suites the authors built; labelled on the pages
Dropped: fail the rule	Octopus v2, GPA, SkillDroid (qualitative only), AgentServe, Stateful Inference (Norgren), Cost-Aware Speculative Execution (Fareed), Dynamic ReAct, AdaGUI-R1, a speculative-tools GitHub repository	In-house sets without counts, single authors without measurements, no institution, or unreadable
Could not be read or found	OS-Catalyst; ToolSEE; the Self-Guide “+22%”, VITA-VLA “–76%” and “2,000–8,000 schema tokens” figures	The figures are not in the sources
Off-topic	SEAR, DREAM-Chunk (robotics); VITA-VLA, m2mKD (not LLM agents); SRMT (not an LLM); MemRefine (offline storage); DARE (math reasoning); Self-Guide (no efficiency measure)	No Part 1 term
Qualitative only	ECLAIR, AgentRR, Signal-Driven Observation, the small-models position paper, the LATM / DiLogics / ALLOY / Voyager / WebAgent lineage, RAC, Revisable by Design, AAPT, Qwen-UI-Agent, Log2Plan, ConServe, TOPAS, SmoothAgent, AAFLOW+ (modelled), PEEK, PANDO (flagged), vendor cache marketing	No usable measured number, or internal inconsistencies
Numbers replaced by corrections	SkillDroid 2.4×; Hajimiri v1; KVFlow 1.83× (absent from the camera-ready body); AgentServe 2.8× (against llama.cpp); RouteLLM 3.66× “vs GPT-4”; UFO2 “51.5% lower inference cost”; WALT “1.3–1.4× on average”	Superseded by the source checks

B9 · Full references of the works named in B9 (1 of 3)

every work B9 names that is not in the Part 2 reference list (works also used as evidence are listed there) · status in brackets · checked 29 September 2026

1. Belcak, P., Heinrich, G., Diao, S., Fu, Y., Dong, X., Muralidharan, S., Lin, Y. C., & Molchanov, P. (2026). Small language models are the future of agentic AI. arXiv:2506.02153v3. NVIDIA Research; Georgia Institute of Technology. [\[qualitative only\]](#)

2. Chang, C., Zhou, Y., Fu, K., An, D., Feng, T., Lu, H., Yao, S., Guo, P., Yu, Y., Shan, Y., Li, B., Yuan, B., & Wang, W. (2026). From LLM inference to agentic workloads: Characterization and implications for serving systems (AgentSysBench). arXiv:2608.15127v1. Hong Kong University of Science and Technology; Alibaba Group; ByteDance. [\[could not be read, or the figure is not in it\]](#)

3. Chen, H. M., Guo, J., Luk, W., & Fan, H. (2026). AOSpec: Action and observation co-speculation for low-latency agent serving. arXiv:2608.00881v1. Imperial College London. [\[held back: condition \(iii\)\]](#)

4. Chen, J., Lv, R., Jing, H., Dang, Z., Fang, S., Ma, C., Zhao, L., & Teng, J. (2026). Difficulty-aware reasoning for mobile GUI automation via reinforcement fine-tuning (AdaGUI-R1) [Submission to ICLR 2026, rejected]. OpenReview. <https://openreview.net/forum?id=Ric2If6Xur> [\[dropped: credibility rule\]](#)

5. Chen, Q., Bellucci, A., Sun, Z., & Jacucci, G. (2026). SkillDroid: Compile once, reuse forever. arXiv:2604.14872v1. University of Helsinki; Universidad Carlos III de Madrid; Shenzhen University. [\[dropped: credibility rule\]](#)

6. Chen, W., & Li, Z. (2026). Octopus v2: On-device language model for super agent. arXiv:2404.01744v7. Stanford University. [\[dropped: credibility rule\]](#)

7. Chen, W., Zhang, K., Lin, C., Zhang, Z., She, Y., Liu, Y., Yeh, R. A., Mou, S., & Gu, Y. (2026). DREAM-Chunk: Reactive action chunking with latent world model. arXiv:2606.18589v1. Purdue University; Stanford University. [\[off-topic\]](#)

8. Chen, Z., Liu, H., Xu, D., Dong, D., Li, J., Pu, B., & Zhai, J. (2026). Cordon: Semantic transactions for tool-using LLM agents. arXiv:2606.17573v1. Tsinghua University; Shanghai Jiao Tong University; Renmin University of China; AetherHeart Tech Co., Ltd. [\[held back: condition \(iii\)\]](#)

9. Ding, J., Hosseini, R., Gholami, P. M., Xiang, M., & Hoffmann, H. (2026). Observation, not prediction: Conversation-level disaggregated scheduling for agentic serving (ConServe). arXiv:2606.01839v1. University of Chicago. [\[qualitative only\]](#)

10. Dong, S., Fu, C., Gao, H., Zhang, Y.-F., Yan, C., Wu, C., Liu, X., Shen, Y., Huo, J., Jiang, D., Cao, H., Gao, Y., Sun, X., He, R., & Shan, C. (2025). VITA-VLA: Efficiently teaching vision-language models to act via action expert distillation. arXiv:2510.09607v2. Nanjing University; Tencent Youtu Lab; Institute of Automation, Chinese Academy of Sciences. [\[off-topic\]](#)

11. Dong, Z., Qian, R., Zhan, Q., Peng, D., Li, K., & Li, Y. (2026). Why are GUI agents correct but late? Decode on the decision-time critical path, tested with pre-compiled policy trees (AAPT). arXiv:2607.28399v1. Georgia Institute of Technology; Fudan University; Marquette University; University of North Carolina at Chapel Hill; National University of Singapore; Southeast University. [\[qualitative only\]](#)

12. Fareed, F. (2026). Cost-aware speculative execution for LLM-agent workflows: An integrated five-dimension method. arXiv:2606.07846v1. Amazon Web Services. [\[dropped: credibility rule\]](#)

13. Feng, E., Zhou, W., Liu, Z., Chen, L., Dong, Y., Zhang, C., Zhao, Y., Du, D., Hua, Z., Xia, Y., & Chen, H. (2025). Get experience from practice: LLM agents with record & replay (AgentRR). arXiv:2505.17716v1. Shanghai Jiao Tong University (IPADS). [\[qualitative only\]](#)

14. Gaur, S., & Lane, I. (2026). Signal-driven observation for long-horizon web agents. *ICML 2026 Workshop on Failure Modes in Agentic AI (FAGEN)* [Non-archival workshop paper]. <https://icml.cc/virtual/2026/workshop/54094>. arXiv:2606.06708v2. University of California, Santa Cruz. [\[qualitative only\]](#)

15. Gaurav, N., Akarsh, A., Ranjan, A., & Bajaj, M. (2025). Dynamic ReAct: Scalable tool selection for large-scale MCP environments. arXiv:2509.20386v1. No institution printed (agentr.dev). [\[dropped: credibility rule\]](#)

16. Gu, Z., Zhang, Q., Khattab, O., & Madden, S. (2026). PEEK: Context map as an orientation cache for long-context LLM agents. arXiv:2605.19932v1. MIT CSAIL; Stanford University. [\[qualitative only\]](#)

17. Gur, I., Furuta, H., Huang, A., Safdari, M., Matsuo, Y., Eck, D., & Faust, A. (2024). A real-world WebAgent with planning, long context understanding, and program synthesis. *The Twelfth International Conference on Learning Representations (ICLR 2024)*, oral. <https://iclr.cc/virtual/2024/poster/19300>. arXiv:2307.12856v4. Google DeepMind; The University of Tokyo. [\[qualitative only\]](#)

18. Jali, N., Nayak, A., & Joshi, G. (2026). Not all turns are equally hard: Adaptive thinking budgets for efficient multi-turn reasoning in agents (TAB). arXiv:2604.05164v3. Carnegie Mellon University. [\[held back: condition \(iii\)\]](#)

B9 · Full references of the works named in B9 (2 of 3)

every work B9 names that is not in the Part 2 reference list (works also used as evidence are listed there) · status in brackets · checked 29 September 2026

19. joelvarun. (2026). *speculative-tools* [Source code]. GitHub. <https://github.com/joelvarun/speculative-tools> [dropped: credibility rule]

20. Kim, M., Baek, J., Jeong, S., & Hwang, S. J. (2026). MemRefine: LLM-guided compression for long-term agent memory. arXiv:2606.13177v1. Korea University; KAIST; DeepAuto.ai. [off-topic]

21. Lee, S., Yoon, S., Lee, S., Kim, H., & Sim, J. Y. (2025). Log2Plan: An adaptive GUI automation framework integrated with task mining approach. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*, 1–13. <https://doi.org/10.1145/3746059.3747663>. arXiv:2509.22137v1. Sookmyung Women's University. [qualitative only]

22. Li, H., Chen, Y., & Zhang, Z. (2026). GoClick: Lightweight element grounding model for autonomous GUI interaction [Technical report]. arXiv:2604.23941v1. Institute of Automation, Chinese Academy of Sciences; University of Chinese Academy of Sciences; Hong Kong Institute of Science & Innovation, Chinese Academy of Sciences. [held back: condition (iii)]

23. Li, J., Ning, Z., Tian, Y., & Li, T. J.-J. (2025). ALLOY: Generating reusable agent workflows from user demonstration. arXiv:2510.10049v1. University of Michigan; University of Notre Dame; Purdue University. [qualitative only]

24. Li, Y., Deng, W., Li, J., & Li, X. (2026). Spend less, reason better: Budget-aware value tree search for LLM agents (BAVT). arXiv:2603.12634v1. University of British Columbia; Vector Institute. [held back: condition (iii)]

25. Li, Y., Miao, Y., Shen, Y., & Liu, Y. (2026). PANDO: Efficient multimodal AI agents via online skill distillation. arXiv:2605.24785v2. Carnegie Mellon University (from e-mail domain; no affiliation line printed). [qualitative only]

26. Lo, K. M., Liang, Y., Du, W., Fan, Y., Wang, Z., Huang, W., Ma, L., & Fu, J. (2024). m2mKD: Module-to-module knowledge distillation for modular transformers. arXiv:2402.16918v3. University of Macau; Institute of Automation, Chinese Academy of Sciences; The University of Hong Kong; Beijing University of Posts and Telecommunications; 01.AI; Peking University; Hong Kong University of Science and Technology. [off-topic]

27. Nagy, C. F. M., Celik, O., Gospodinov, E., Seligmann, F., Liao, W., Kaushik, A., & Neumann, G. (2026). SEAR: Sample efficient action chunking reinforcement learning. arXiv:2603.01891v2. Karlsruhe Institute of Technology; FZI Research Center for Information Technology. [off-topic]

28. Ni, H., Tian, H., Zhang, C., Li, G., & Tan, H. (2026). TOPAS: Workflow-aware prefix-state scheduling for multi-agent LLM serving. arXiv:2608.25523v1. University of Science and Technology of China; Hefei University of Technology. [qualitative only]

29. Norgren, V. (2026). Stateful inference for low-latency multi-agent tool calling. arXiv:2605.26289v1. LayerScale, Inc. [dropped: credibility rule]

30. Pan, Z., Wang, Q., Hu, Z., Chen, C., Guan, Y., Zhou, Y., Swanson, S., & Ding, Y. (2026). SmoothAgent: Efficient long-horizon LLM-based agent serving with lookahead context engineering. arXiv:2607.00151v1. University of California, San Diego. [qualitative only]

31. Pandey, N. P., Kong, J., Hu, L., Zhao, Q., Zhao, Y., Gungor, O., Zhang, H., & Rosing, T. (2026). AgentKVShift: Efficient KV cache reuse for agentic memory systems. arXiv:2607.21604v1. University of California, San Diego. [held back: condition (iii)]

32. Perera, S., Hapuarachchi, K., Leymann, F., & Khalaf, R. (2026). Robust Agent Compensation (RAC): Teaching AI agents to compensate. In *Proceedings of the ACM Conference on AI and Agentic Systems (CAIS '26)*, 253–262. <https://doi.org/10.1145/3786335.3813141>. arXiv:2605.03409v2. WSO2; University of Stuttgart. [qualitative only]

33. Pu, K., Yang, J., Yuan, A., Ma, M., Dong, R., Wang, X., Chen, Y., & Grossman, T. (2023). DiLogics: Creating web automation programs with diverse logics. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*, 1–15. <https://doi.org/10.1145/3586183.3606822>. University of Toronto; University of Michigan; Virginia Tech. [qualitative only]

34. Saberi, M., Rezaei, K., & Feizi, S. (2026). SpecHop: Continuous speculation for accelerating multi-hop retrieval agents. arXiv:2605.21965v1. University of Maryland, College Park. [held back: condition (iii)]

35. Sagirova, A., Kuratov, Y., & Burtsev, M. (2026). SRMT: Shared memory for multi-agent lifelong pathfinding. arXiv:2501.13200v2. AIRI; Moscow Institute of Physics and Technology; London Institute for Mathematical Sciences. [off-topic]

36. Sarker, A. K., Halpern, A. J., Staylor, M., Alsaadi, A., von Laszewski, G., Cheng, Y., Jha, S., & Fox, G. (2026). [AAFLOW+] Stateful operator abstraction with zero-copy distributed KV cache orchestration for multi-agent workflows. arXiv:2607.10987v1. University of Virginia; Rutgers University; Princeton Plasma Physics Laboratory. [qualitative only]

B9 · Full references of the works named in B9 (3 of 3)

every work B9 names that is not in the Part 2 reference list (works also used as evidence are listed there) · status in brackets · checked 29 September 2026

37. Singh, S., Karatzas, A., Fore, M., Anagnostopoulos, I., & Stamoulis, D. (2024). An LLM-tool compiler for fused parallel function calling. arXiv:2405.17438v1. Microsoft; Southern Illinois University. [\[held back: condition \(iii\)\]](#)

38. Sui, Y., Zhao, H., Ma, R., He, Z., Wang, H., Li, J., Xu, K., Chen, K., & Yang, Y. (2026). Parallelizing tool execution and LLM generation for low-latency agent serving (PASTE). arXiv:2603.18897v3. Shanghai Jiao Tong University; Microsoft Research; Stevens Institute of Technology; Hong Kong University of Science and Technology; Google. [\[held back: condition \(iii\)\]](#)

39. Vadlapati, P. (2025). ToolSEE: Agent tool search engine for efficient and scalable tool discovery using retrieval [Preprint]. Preprints.org. <https://doi.org/10.20944/preprints202512.1744.v1> [\[could not be read, or the figure is not in it\]](#)

40. Wang, G., Xie, Y., Jiang, Y., Mandlkar, A., Xiao, C., Zhu, Y., Fan, L., & Anandkumar, A. (2024). Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*. <https://openreview.net/forum?id=ehfRiFOR3a>. arXiv:2305.16291v2. NVIDIA; California Institute of Technology; The University of Texas at Austin; Stanford University; University of Wisconsin–Madison. [\[qualitative only\]](#)

41. Wang, X., Wu, H., Song, J., Zhang, S., Zhang, J., Kong, F., Kwok, T. S. T., Chang, X.-W., Luo, Y., Wu, C., & Liu, B. (2026). Co-evolution of policy and internal reward for language agents (Self-Guide). arXiv:2604.03098v1. McGill University; McMaster University; The University of Hong Kong; HKUST (Guangzhou); Peking University; University of California, Los Angeles; DeepWisdom; Université de Montréal; Mila. [\[off-topic\]](#)

42. Wornow, M., Narayan, A., Opsahl-Ong, K., McIntyre, Q., Shah, N. H., & Ré, C. (2024). Automating the enterprise with foundation models (ECLAIR). *Proceedings of the VLDB Endowment*, 17(11), 2805–2812. <https://doi.org/10.14778/3681954.3681964>. arXiv:2405.03710. Stanford University. [\[qualitative only\]](#)

43. Wu, Y., Chen, S., Zhong, Y., Huang, R., Tan, Y., Zhang, W., Zhang, L., Zhou, S., Liu, Y., Zhou, S., Zhang, M., Jin, X., & Huang, P. (2026). DualPath: Breaking the storage bandwidth bottleneck in agentic LLM inference. arXiv:2602.21548v2. Peking University; Tsinghua University; DeepSeek-AI. [\[held back: condition \(iii\)\]](#)

44. Yuan, X., Sun, Q., Chen, Y., Li, R., Chen, X., Yuan, S., Wang, X., Ding, Z., Li, Z., Qi, B., & Yang, D. (2026). OS-Catalyst: Advancing computer-using agents efficiency through adaptive action compression [Withdrawn submission to ICLR 2026]. OpenReview. <https://openreview.net/forum?id=QpKXNYtF3x> [\[could not be read, or the figure is not in it\]](#)

45. Zhai, Z., Li, M., & Wang, X. (2026). Revisable by design: A theory of streaming LLM agent execution. arXiv:2604.23283v1. Fudan University; Guangming Lab. [\[qualitative only\]](#)

46. Zhang, R., Kim, C., Feng, S., Du, K., Liu, Y., Zhong, Y., Ching, C.-W., Jiang, J., & Hu, L. (2026). Learning agent execution for KV-cache management in agentic serving (CacheScout). arXiv:2608.14624v1. University of California, Santa Cruz; University of Washington; University of Chicago. [\[held back: condition \(iii\)\]](#)

47. Zhang, Y., Wo, T., Wang, J., Sun, X., Zhang, M., Yuan, C., Li, L., Hu, C., Zomaya, A. Y., & Yang, R. (2026). SpecBox: Speculative sandbox scheduling for efficient LLM agent serving. arXiv:2607.23933v2. Beihang University; University of Leeds; The University of Sydney. [\[held back: condition \(iii\)\]](#)

48. Zhang, Y., Yan, Y., Yang, N., & Yuan, D. (2026). AgentServe: Algorithm-system co-design for efficient agentic AI serving on a consumer-grade GPU. arXiv:2603.10342v1. The University of Sydney. [\[dropped: credibility rule\]](#)

49. Zhao, Z., Liew, J. H., Yang, Y., Yang, W., Luo, Z., Sahoo, D., Savarese, S., & Li, J. (2026). GPA: Learning GUI process automation from demonstrations. arXiv:2604.01676v2. Salesforce AI Research. [\[dropped: credibility rule\]](#)

50. Zheng, H., Fu, F., Wu, J., Yuan, B., Zhang, Y., Wang, H., Zhu, Y., Yan, X., & Jiang, J. (2026). Efficient serving for dynamic agent workflows with prediction-based KV-cache management (PBKV). arXiv:2605.06472v1. Wuhan University; Shanghai Jiao Tong University; Macquarie University; Hong Kong University of Science and Technology; Dameng Database. [\[held back: condition \(iii\)\]](#)

51. Zhong, S., Lu, B., Chen, Q., Liu, C., Yang, F., & Li, M. (2026). DualSpec: Accelerating deep research agents via dual-process action speculation. arXiv:2603.07416v1. Peking University; Microsoft Research; Microsoft. [\[held back: condition \(iii\)\]](#)

52. Zhou, H., Tong, P., Zhang, X., Kong, Q., Cai, C., Xia, T., Zhang, G., Zhang, J., Li, L., Chen, L., Wang, L., Dai, G., Li, P., Chen, L., Wang, Y., & Hoi, S. (2026). Qwen-UI-Agent technical report: Toward next-generation real-world centric foundation GUI agents. arXiv:2607.28227v1. Alibaba Group. [\[qualitative only\]](#)

53. Zhou, Y., Jin, C., Dong, Z., Wang, Z., Yang, Y., Zhao, S., Li, L., Bao, R., Xie, Y., & Metaxas, D. N. (2026). DARE: Difficulty-adaptive reinforcement learning with co-evolved difficulty estimation. arXiv:2605.09188v1. Rutgers University; Amazon; Washington University; Google. [\[off-topic\]](#)

B10 · What no retained work measures yet, by term

gaps in the evidence, not claims of a research gap · the one-line versions are in each page’s foot

TERM	NOT YET MEASURED
J_i (compile, replay)	Replay under live application change; a vendor replay or repair rate with a sample; cost per success with set-up included; a multi-page enterprise form with a final submit
t^{queue}	Its share of task time for API agents; measured latency of API priority tiers (only classes or targets are published)
$t^{\text{prefill}}, o_{a,i} $	Task time with caching on; how pruning interacts with prefix caching; the load time of cache-hit tokens, which Part 1 has no term for
n^{out} , TPOT	The net effect of per-step effort changes that invalidate the message cache; independent task-level timing of fast tiers
E_i	Page-load and wait time in write-heavy web workflows (Skim is read-only); wall-clock of batched actions
T_{saving}	Speculation on a live web GUI with side effects held until commit; cost per success of any speculation method; several speed figures are analytical or replayed
$\bar{g}, N^2$	The N^2 token volume together with the cache hit rate in a GUI agent; structural removal is shown only by compiled plans; eviction (TokenPilot) may change the shape but is not measured as such
$n^{\kappa}, x_{\text{env}}$	A GUI or browser agent’s tokens split by billing class, cache writes included (Don’t Break the Cache covers a web-search agent); environment hours priced beyond a vendor list price (coding agents only in TraceLab)
$c_{\kappa}(\mu_{ij})$	The router’s own cost; small-model losses on enterprise forms; routing with cached prices
$R_m(p)$, set-up	Cost per success reported by the papers themselves; time, money and success, set-up included, on one web or GUI benchmark